

Utilities Process Flows — Common Information Model (CIM) from EPRI with IEC 61970/IEC 61850/IEC 62351/IEC 61968, OpenADR 2.0b and Green Button (NAESB REQ.21 ESPI) for Smart Grid and Metering, EPRI Utility Business Capability Model (UCBM) and TM Forum eTOM for Multi-Utilities

Generated on 2026-07-07 | Domain: Utilities | Framework: Common Information Model (CIM) from EPRI with IEC 61970/IEC 61850/IEC 62351/IEC 61968, OpenADR 2.0b and Green Button (NAESB REQ.21 ESPI) for Smart Grid and Metering, EPRI Utility Business Capability Model (UCBM) and TM Forum eTOM for Multi-Utilities

EPRI Utility Business Capability Model Reference

The EPRI Utility Business Capability Model (UCBM) defines standard business capabilities for electric, gas, and water utilities. The table below lists all Level 2 capabilities grouped by Level 1 domain. Each process flow in this document is mapped to its relevant UCBM capability codes.

 EPRI Utility Business Capability Model overview - Customer, Grid Operations, Generation, Water, Enterprise, and Technology domains

Source: EPRI Utility Business Capability Model (2024) — used under EPRI public version license

Customer & Energy Services

Capability	Code	Description
Customer Information Management	UT - CIM	Manage customer accounts, contacts, and relationship data across utility services
Customer Service & Support	UT - CSS	Handle customer inquiries, service requests, complaints, and field service coordination
Marketing & Sales	UT - MKT	Plan and execute marketing campaigns, lead management, and customer acquisition
Credit & Collections	UT - CCL	Manage credit risk assessment, collections, and disconnection/reconnection processes

Metering & Smart Grid

Capability	Code	Description
Advanced Metering Infrastructure	UT - AMI	Operate AMI head-end systems, meter communications, and smart meter network
Meter Data Management	UT - MDM	Validate, estimate, and edit (VEE) meter data; manage data quality and storage
Smart Grid Operations	UT - SGO	Monitor and control smart grid devices, DERMS integration, and grid-edge intelligence
Green Button Data Access	UT - GBD	Provide customer energy usage data via Download My Data and Connect My Data standards

Billing & Revenue Management

Capability	Code	Description
Rate & Tariff Management	UT-RTM	Define and manage rate structures, tariffs, riders, and pricing schedules
Billing Calculation & Presentment	UT-BIL	Calculate charges, apply rates, generate bills, and manage presentment
Payment Processing	UT-PAY	Process payments via multiple channels, manage payment plans and deposits
Revenue Assurance	UT-REV	Detect revenue leakage, audit billing accuracy, and manage write-offs

Generation Operations

Capability	Code	Description
Thermal Generation Control	UT-TGC	Monitor and control thermal power plant operations including boiler, turbine, and generator
Renewable Generation Management	UT-RGM	Manage solar, wind, and hydro generation assets including variability and forecasting
Generator Dispatch & AGC	UT-GDA	Coordinate generator dispatch, automatic generation control, and load following
Fuel Management	UT-FMG	Manage fuel procurement, inventory, quality, and consumption for thermal plants

Transmission Operations

Capability	Code	Description
EMS Real-time Operations	UT-EMS	Monitor and control transmission network via energy management system SCADA
State Estimation & Contingency	UT-SEC	Perform network state estimation, contingency analysis, and security assessment
Wide Area Monitoring	UT-WAM	Monitor grid stability using synchrophasors, PMUs, and wide-area situational awareness
Transmission Switching & Clearance	UT-TSC	Manage transmission switching orders, clearance requests, and tagging procedures

Substation Automation

Capability	Code	Description
RTU & SCADA Control	UT-RTU	Operate remote terminal units and substation SCADA for data acquisition and control
Protection & IED Management	UT-PIM	Manage protection relays, intelligent electronic devices, and fault recording
IEC 61850 Communications	UT-IEC	Implement IEC 61850 GOOSE, MMS, and sampled value communications for substation automation
Substation Security & Access	UT-SSA	Manage substation physical security, cyber access, and electronic security perimeters

Distribution Grid

Capability	Code	Description
Distribution SCADA / ADMS	UT-DSC	Monitor and control distribution network via ADMS, DMS, and distribution SCADA
DERMS & Distributed Energy	UT-DER	Manage distributed energy resource interconnection, dispatch, and optimization
Fault Location & Restoration	UT-FLR	Detect faults, locate via FLISR, and automatically restore distribution service
Distribution Planning & Optimization	UT-DPO	Plan distribution network capacity, volt/VAR optimization, and loss reduction

Water Operations

Capability	Code	Description
Water Treatment Control	UT-WTC	Monitor and control water treatment plant processes including chemical dosing and filtration
Water Distribution Management	UT-WDM	Manage water distribution network pressure zones, flow, and reservoir levels
Wastewater Treatment Control	UT-WWC	Monitor and control wastewater treatment processes and effluent quality
Water Quality Management	UT-WQM	Monitor water quality parameters, regulatory compliance, and sampling programs

Outage & Workforce Management

Capability	Code	Description
Outage Detection & IVR	UT-ODI	Detect outages via AMI, SCADA, and customer calls; manage IVR outage reporting
Damage Assessment & Switching	UT-DAS	Assess storm damage, perform switching orders, and isolate faulted sections
Crew Dispatch & Restoration	UT-CDR	Dispatch field crews, manage mobile workforce, and track restoration progress
Outage Analytics & Reporting	UT-OAR	Analyze outage duration, frequency, SAIDI/SAIFI metrics, and regulatory reporting

Asset Management

Capability	Code	Description
Asset Registry & Hierarchy	UT-ARH	Maintain asset register, hierarchy, criticality ratings, and lifecycle status
GIS Network Model	UT-GIS	Manage geographic information system network model and asset geolocation
Condition Monitoring & IoT	UT-CMI	Monitor asset condition via IoT sensors, online monitoring, and predictive analytics
Predictive Maintenance & CBM	UT-PMC	Plan condition-based and predictive maintenance activities for critical assets

Enterprise Resources

Capability	Code	Description
Financial Management	UT-FIN	Manage general ledger, accounts payable/receivable, budgeting, and financial reporting
Human Resources & Payroll	UT-HRP	Manage workforce planning, recruitment, time tracking, payroll, and benefits
Procurement & Supply Chain	UT-PSC	Manage procurement, vendor contracts, purchase orders, and logistics
Inventory & Warehousing	UT-INV	Manage materials inventory, warehouse operations, and stock replenishment

Supply Chain & Materials

Capability	Code	Description
Vendor & Supplier Management	UT-VSM	Manage vendor qualifications, performance, contracts, and compliance
Materials & Equipment Management	UT-MEM	Manage materials catalog, equipment specifications, and standardization
Contract Lifecycle Management	UT-CLM	Manage contract creation, negotiation, approval, and renewal processes
Logistics & Warehousing	UT-LGW	Manage inbound/outbound logistics, warehouse operations, and fleet management

Cybersecurity & Compliance

Capability	Code	Description
Identity & Access Management	UT-IAM	Manage user identities, role-based access control, and privileged access for OT/IT
Security Monitoring & SIEM	UT-SSM	Monitor security events, analyze threats, and manage security incident response
Vulnerability & Patch Management	UT-VPM	Manage vulnerability scanning, patch deployment, and configuration hardening
NERC CIP Compliance	UT-NCC	Manage NERC Critical Infrastructure Protection compliance, audits, and evidence

Data & Analytics

Capability	Code	Description
Operational Data Lake	UT-ODL	Ingest and store OT/IT data via PI System, data lake, and historian platforms
Grid Analytics & Forecasting	UT-GAF	Analyze grid data for load forecasting, renewable integration, and optimization
Customer Analytics	UT-CAN	Analyze customer usage patterns, segmentation, and energy efficiency program targeting
Asset Health & Predictive Analytics	UT-AHP	Predict asset failures, calculate remaining useful life, and optimize maintenance

Regulatory & Market Operations

Capability	Code	Description
Regulatory Reporting	UT-RRP	Prepare and submit regulatory reports to FERC, NERC, state PUCs, and environmental agencies
ISO/RTO Market Operations	UT-ISO	Participate in wholesale electricity markets, bid management, and settlement
Environmental Compliance & Emissions	UT-ENV	Monitor emissions, manage environmental permits, and report compliance data
Environmental Health & Safety	UT-EHS	Manage safety programs, incident reporting, industrial hygiene, and OSHA compliance

Demand Response & Energy Efficiency

Capability	Code	Description
Demand Response Program Management	UT-DRP	Design, market, and manage demand response programs across customer segments
OpenADR VEN/VTN Operations	UT-OAD	Operate OpenADR 2.0b virtual top node for automated DR signaling and reporting
Energy Efficiency Programs	UT-EEP	Manage energy efficiency audits, rebates, incentives, and measurement/verification
DER Customer Enrollment	UT-DCE	Enroll customer-owned DER (solar, battery, EV) in utility programs and markets

Engineering & Planning

Capability	Code	Description
T&D Planning & Studies	UT-TDP	Plan transmission and distribution capacity, reliability, and expansion projects
Generation Planning	UT-GNP	Plan generation capacity, fuel mix, retirement, and new resource acquisition
Standards & Specifications	UT-STS	Develop and maintain technical standards, specifications, and design criteria
System Protection & Coordination	UT-SPC	Design and coordinate protection schemes, relay settings, and arc flash studies

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Test Data

The following test data is used consistently across all sequence diagrams:

Entity	Field	Value
Customer	customer_id	CUST-2026-001
	name	Jane Smith
	address	123 Oak Street, Springfield
	service_type	Electric + Water
	status	Active
Account	account_id	ACCT-2026-001
	customer_id	CUST-2026-001
	billing_cycle	Monthly
	balance	\$245.80
	payment_method	Auto-Pay
Meter	meter_id	MTR-2026-001
	type	Smart Meter (Itron OpenWay)
	installation_date	2024-03-15
	status	Commissioned
	interval	15-min
MeterReading	reading_id	RDG-2026-001
	meter_id	MTR-2026-001
	timestamp	2026-07-07T10:00:00Z
	kwh	12.45
	quality	Validated
ServicePoint	sp_id	SP-2026-001
	address	123 Oak Street, Springfield
	transformer_id	TX-2026-001
	feeder_id	FDR-2026-001
	voltage	240V
Transformer	transformer_id	TX-2026-001
	type	Padmount 75kVA
	location	123 Oak St pad
	status	In Service
	capacity	75
Feeder	feeder_id	FDR-2026-001
	substation_id	SUB-MAIN-001
	voltage	12.47kV
	rating	600A
	length_km	4.2
Substation	substation_id	SUB-MAIN-001
	name	Springfield Main Substation
	voltage	138kV/12.47kV

Entity	Field	Value
Breaker	type	Distribution
	status	Operational
	breaker_id	BRK-2026-001
	substation_id	SUB-MAIN-001
	status	Closed
WorkOrder	rating	1200A
	type	SF6
	wo_id	WO-2026-001
	type	Emergency Repair
	status	Assigned
Crew	priority	High
	crew_id	CREW-2026-001
	crew_id	CREW-2026-001
	name	Line Crew Alpha
	members	4
Plant	location	Springfield Depot
	status	Available
	plant_id	PLT-GEN-001
	name	Springfield Thermal Power Station
	type	Combined Cycle Gas Turbine
Generator	capacity_mw	500
	status	Operating
	generator_id	GEN-2026-001
	plant_id	PLT-GEN-001
	type	Gas Turbine
Boiler	capacity_mw	250
	status	Online
	boiler_id	BLR-2026-001
	plant_id	PLT-GEN-001
	type	Heat Recovery Steam Generator
Turbine	steam_pressure	1800 psi
	status	Operating
	turbine_id	TRB-2026-001
	plant_id	PLT-GEN-001
	type	Steam Turbine
WaterTreatmentPlant	rating_mw	250
	speed_rpm	3600
	wtp_id	WTP-2026-001

Entity	Field	Value
	name	Springfield Water Treatment Plant
	capacity_mgd	50
	source	Springfield Reservoir
	status	Operating
	sample_id	WQ-2026-001
WaterQualitySample	sample_id	WQ-2026-001
	wtm_id	WTP-2026-001
	ph	7.2
	turbidity_ntu	0.15
	chlorine_residual	1.2 mg/L
	status	Compliant
Pump	pump_id	PUMP-2026-001
	station	Springfield Booster Station
	type	Centrifugal
	flow_gpm	2500
	status	Running
Valve	valve_id	VALV-2026-001
	type	Butterfly
	diameter_inches	16
	position	85% Open
	status	Operational
Pipe	pipe_id	PIPE-2026-001
	material	Ductile Iron
	diameter_inches	24
	length_ft	3200
	zone	Pressure Zone 3
LiftStation	ls_id	LS-2026-001
	name	Oak Street Lift Station
	type	Wastewater
	pump_count	3
	status	Auto
SCADA_Point	point_id	PT-2026-001
	description	Turbine Speed
	value	3600
	unit	RPM
	quality	Good
Alarm	alarm_id	ALM-2026-001
	source	Boiler BLR-2026-001
	type	High Pressure

Entity	Field	Value
	severity	Critical
	timestamp	2026-07-07T09:45:00Z
	status	Acknowledged
DER	der_id	DER-2026-001
	customer_id	CUST-2026-001
	type	Solar PV
	capacity_kw	10
	status	Interconnected
Battery	battery_id	BAT-2026-001
	customer_id	CUST-2026-001
	capacity_kwh	13.5
	type	Lithium-Ion
	status	Charging
EV_Charger	ev_id	EV-2026-001
	customer_id	CUST-2026-001
	type	Level 2
	rating_kw	7.2
	status	Idle
DR_Event	event_id	DR-2026-001
	program	Peak Rewards
	status	Active
	start_time	2026-07-07T14:00:00Z
	end_time	2026-07-07T18:00:00Z
	reduction_target_kw	500
Invoice	invoice_id	INV-2026-007
	account_id	ACCT-2026-001
	amount	\$245.80
	due_date	2026-07-25
	status	Pending
Payment	payment_id	PAY-2026-042
	invoice_id	INV-2026-007
	amount	\$245.80
	method	Credit Card
	reference	CC-4242
Outage	outage_id	OUT-2026-001
	feeder_id	FDR-2026-001
	cause	Tree Contact
	start_time	2026-07-07T08:30:00Z

Entity	Field	Value
Switchgear	customers_affected	450
	status	Active
	switch_id	SW-2026-001
	substation_id	SUB-MAIN-001
	type	Load Break Switch
	status	Open
Relay	location	Pole #47
	relay_id	REL-2026-001
	substation_id	SUB-MAIN-001
	type	Distance Protection (SEL-421)
	setting_group	1
PMU	status	Enabled
	pmu_id	PMU-2026-001
	substation_id	SUB-MAIN-001
	phasor	V1
	magnitude	138.2 kV
	angle	-2.4 deg
EnvironmentalReport	frequency	60.02 Hz
	report_id	ENV-2026-001
	plant_id	PLT-GEN-001
	co2_tons	1250
	nox_lbs	320
	so2_lbs	15
Permit	reporting_period	2026-Q2
	permit_id	PER-2026-001
	type	NPDES Discharge
	facility	WTP-2026-001
	issue_date	2026-01-01
	expiry_date	2030-12-31
Contract	status	Active
	contract_id	CON-2026-001
	vendor	GE Vernova
	type	Turbine Maintenance
	value	\$2.5M
	start_date	2026-01-01
Vendor	end_date	2028-12-31
	vendor_id	VEN-2026-001
	name	ACME Electrical Supply

Entity	Field	Value
ComplianceReport	category	Switchgear
	status	Approved
	rating	4.5/5
	comp_id	COMP-2026-001
	standard	NERC CIP-005
	status	Compliant
	audit_date	2026-04-15
	findings	0

Power Generation & SCADA Operations

Monitor and control power generation assets including thermal (combined cycle gas turbine), hydro, and renewable generation. Manages real-time DCS/SCADA operations, automatic generation control (AGC), fuel management, and emissions monitoring across the generation fleet.

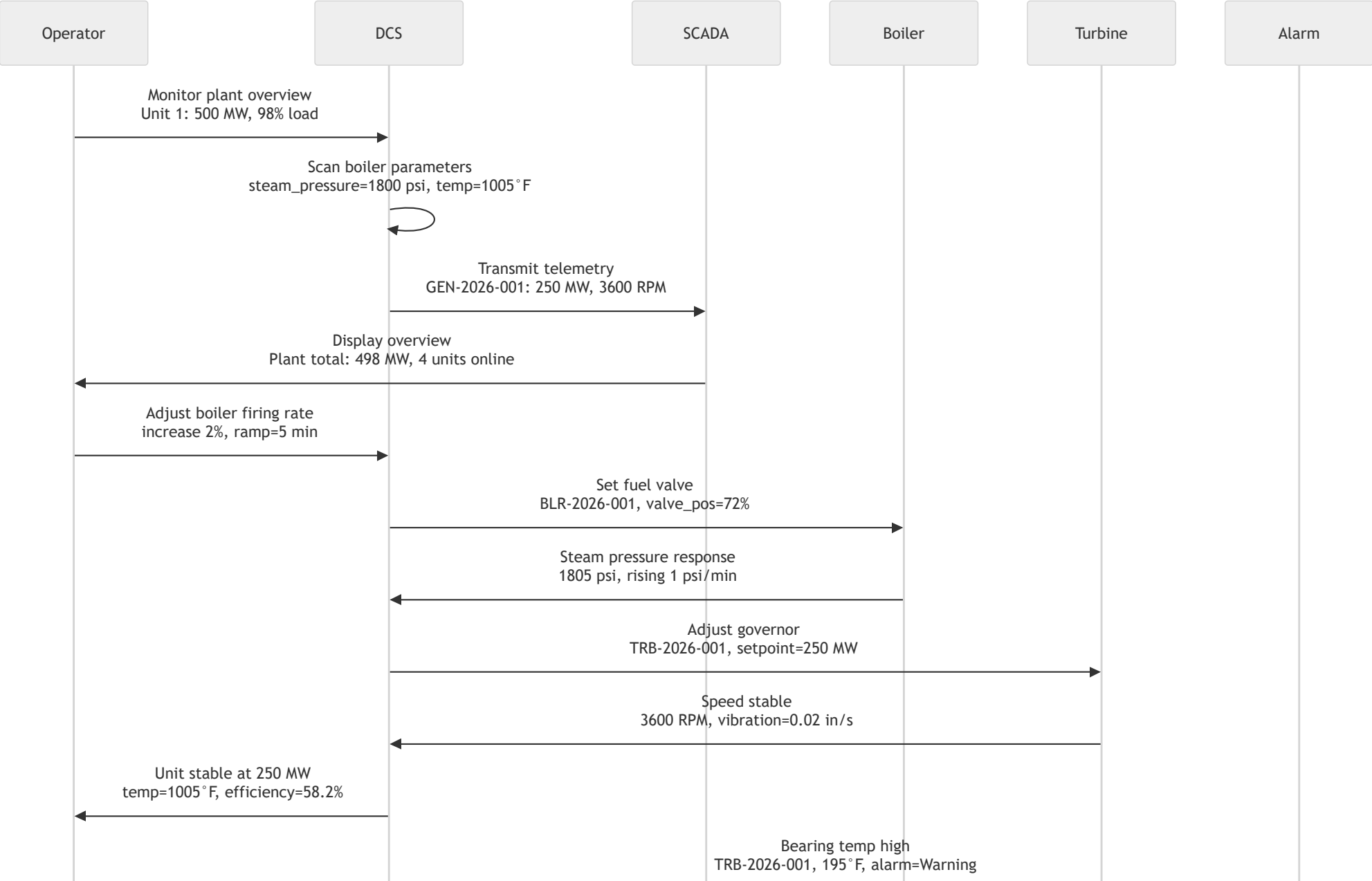
1.1 Thermal Power Plant DCS/SCADA Operations

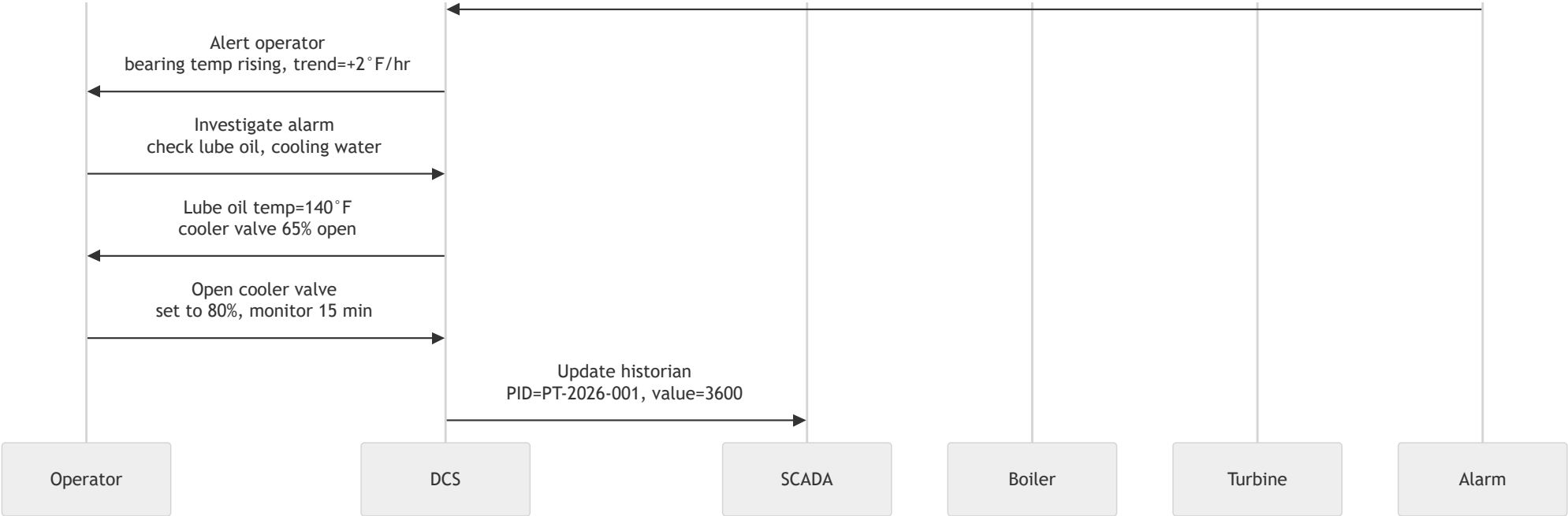
Description: Control room operators monitor boiler, turbine, and generator parameters via distributed control system (DCS). SCADA system provides real-time data acquisition, alarm management, and historical trending for the thermal power plant.

Actors: Control Room Operator, Shift Supervisor, Plant Engineer

Systems: DCS (ABB 800xA), SCADA (Siemens Spectrum Power), Plant Historian (OSIsoft PI), Alarm Management System

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Thermal Plant	Boiler	MonitorBoilerParameters	DCS (ABB 800xA)	SCADA (Siemens Spectrum Power)	DCS OPC-UA	SCADA OPC-DA	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Thermal Plant	Turbine	AdjustGovernor	DCS (ABB 800xA)	Turbine Governor (Woodward)	DCS Modbus TCP	Governor Modbus RTU	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Thermal Plant	Generator	TransmitTelemetry	SCADA (Siemens Spectrum Power)	Plant Historian (OSIsoft PI)	SCADA API (ICCP)	PI Interface (OPC-DA)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Thermal Plant	Alarm	AlertOperator	DCS (ABB 800xA)	Alarm Management System	DCS OPC-A&E	Alarm API (REST)	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
Thermal Plant	Setpoint	AdjustBoilerFiringRate	Operator HMI	DCS (ABB 800xA)	HMI API (OAuth2)	DCS OPC-UA	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Thermal Plant	Trend	LogTrendData	Plant Historian (OSIsoft PI)	Reporting Database	PI API (REST)	SQL JDBC	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints
Thermal Plant	Performance	CalculateEfficiency	DCS (ABB 800xA)	Plant Historian (OSIsoft PI)	DCS OPC-UA	PI API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Thermal Plant	Vibration	MonitorVibration	DCS (ABB 800xA)	Condition Monitoring System	DCS OPC-UA	CMS API (Modbus TCP)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-TP-01	Validate real-time telemetry accuracy within ±0.5% of field values	MonitorBoilerParameters	Telemetry values match field instruments within tolerance	High
AC-TP-02	Verify alarm response time under 30 seconds from event occurrence	AlertOperator	Alarms are displayed on HMI within 30s of trigger	High
AC-TP-03	Confirm DCS-to-SCADA data latency below 1 second for critical points	TransmitTelemetry	PI historian shows data timestamp within 1s of DCS scan	Medium
AC-TP-04	Ensure all operator setpoint changes are logged with user identity and timestamp	AdjustBoilerFiringRate	Audit trail shows user, value, time for every setpoint change	Medium

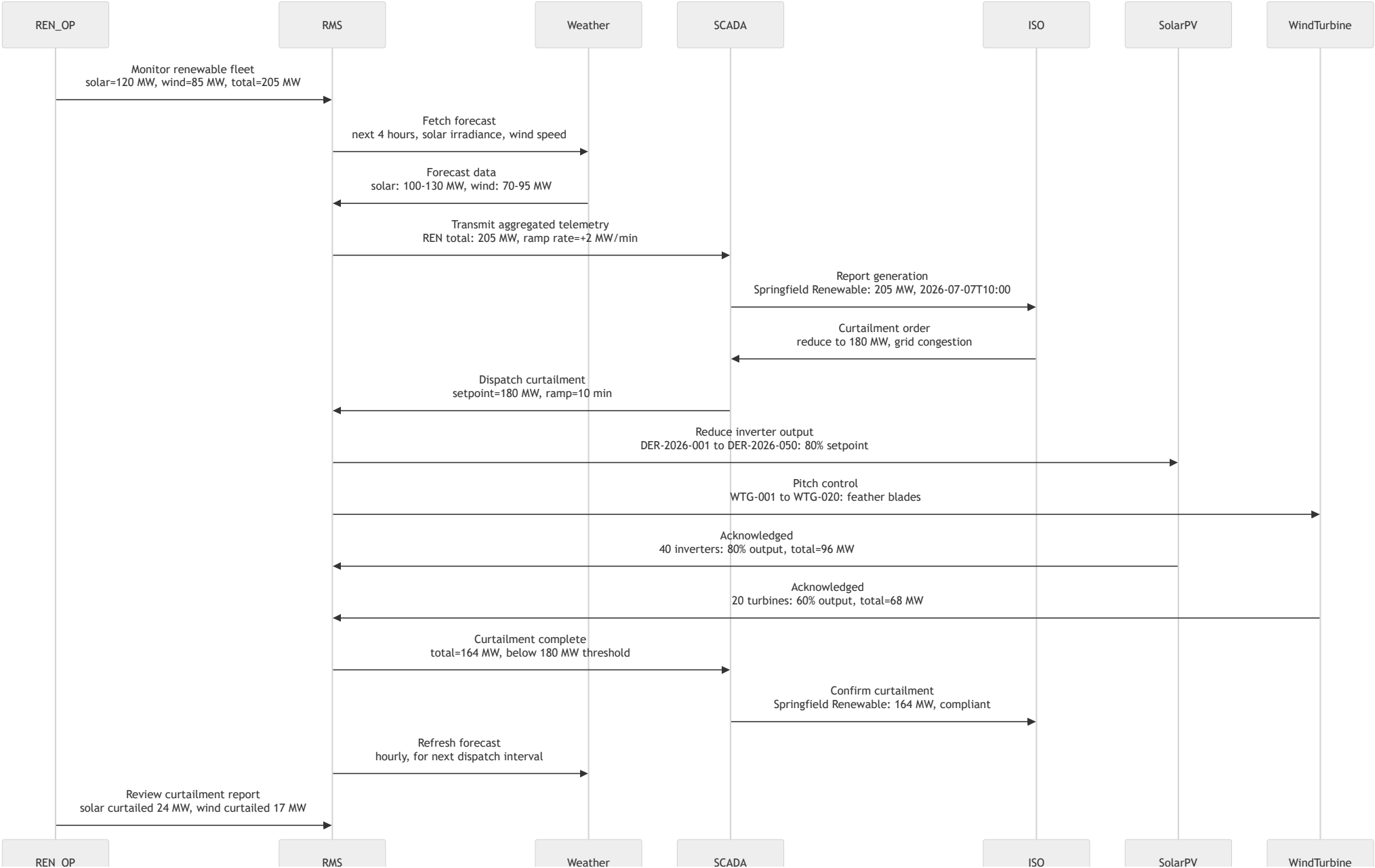
1.2 Renewable Generation & Solar/Wind Integration

Description: Operators monitor and control renewable generation assets including solar PV farms and wind turbines. The renewable management system handles variability forecasting, curtailment commands, and power quality compliance.

Actors: Renewable Operator, Grid Integration Engineer, ISO Scheduler

Systems: Renewable Management System (GE Renewable), SCADA (Siemens Spectrum Power), Weather Forecasting Service, ISO Market Portal

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Renewable	Forecast	FetchWeatherForecast	RMS (GE Renewable)	Weather Forecasting Service	RMS API (REST)	Weather API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Renewable	Curtailment	ReceiveCurtailmentOrder	ISO Market Portal	SCADA (Siemens Spectrum Power)	ISO API (ICCP)	SCADA API (ICCP)	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
Renewable	Dispatch	DispatchCurtailment	SCADA (Siemens Spectrum Power)	RMS (GE Renewable)	SCADA API (ICCP)	RMS API (Modbus TCP)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Renewable	Solar	ReduceInverterOutput	RMS (GE Renewable)	Solar PV Inverters	RMS API (Modbus TCP)	Inverter API (SunSpec Modbus)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Renewable	Wind	ActivatePitchControl	RMS (GE Renewable)	Wind Turbine Controllers	RMS API (OPC-UA)	Turbine API (IEC 61400-25)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Renewable	Compliance	ConfirmCurtailment	SCADA (Siemens Spectrum Power)	ISO Market Portal	SCADA API (ICCP)	ISO API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Renewable	Report	LogCurtailmentEvent	RMS (GE Renewable)	Plant Historian (OSIsoft PI)	RMS API (REST)	PI API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-RG-01	Verify curtailment order execution within mandated ramp time (10 min)	ReceiveCurtailmentOrder	Curtailment target achieved within ramp window	High
AC-RG-02	Validate forecast accuracy within ±15% for 4-hour horizon	FetchWeatherForecast	Weather forecast accuracy meets threshold	Medium
AC-RG-03	Confirm telemetry latency under 2 seconds from inverter to SCADA	DispatchCurtailment	SCADA reflects inverter changes within 2s	Medium
AC-RG-04	Ensure curtailment events are logged with timestamp, setpoint, and actual	ReduceInverterOutput	Complete audit trail for every curtailment event	Medium

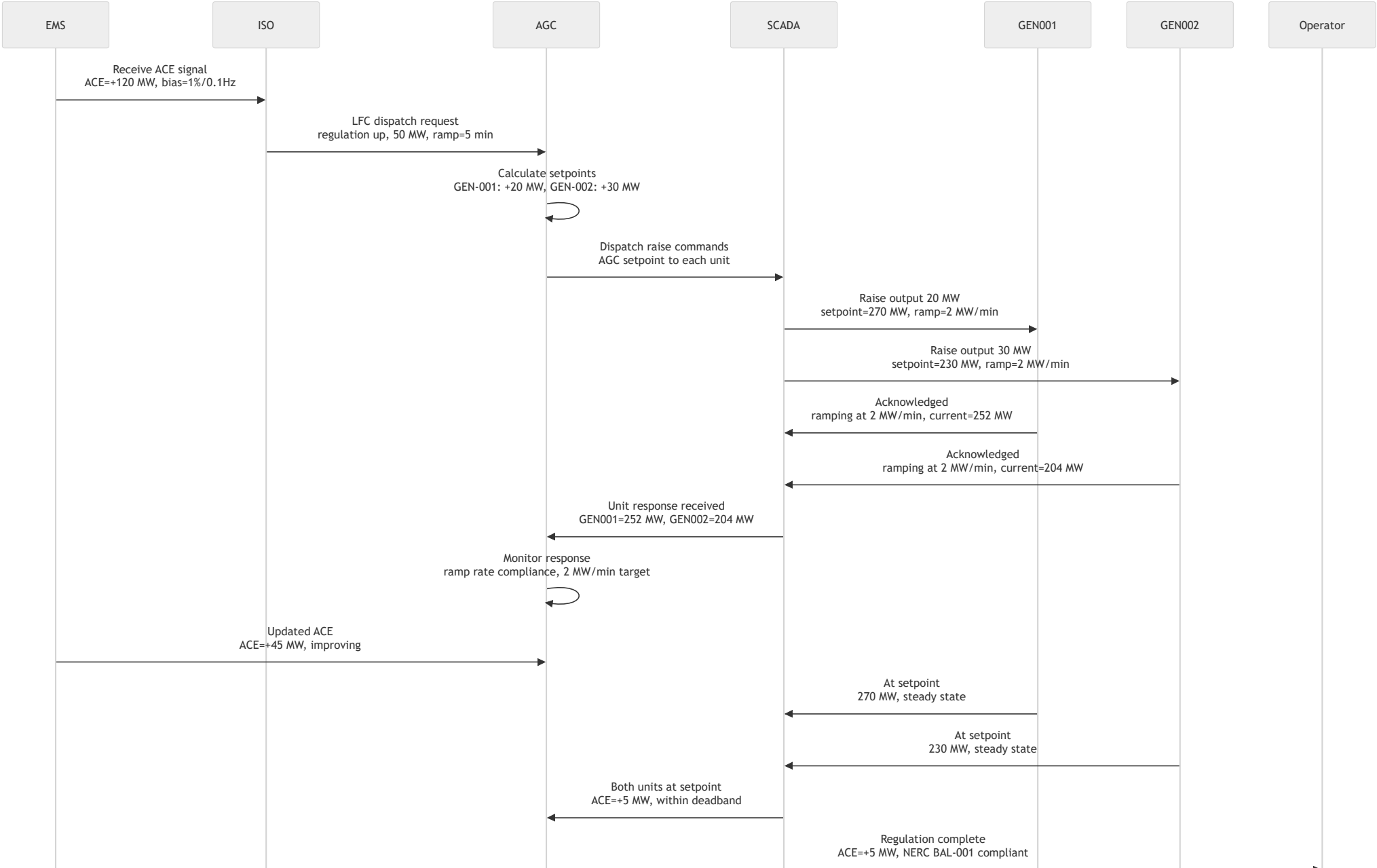
1.3 Automatic Generation Control & Dispatch

Description: The AGC system receives load frequency control signals from the ISO, calculates generator setpoints, and dispatches them to participating generators. Operators monitor ACE (Area Control Error) and ensure regulatory compliance.

Actors: System Operator, ISO Operator, Generator Dispatcher

Systems: AGC (GE/Alstom), EMS (Siemens Spectrum Power), ISO RTO System, SCADA, Plant DCS

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
AGC	ACE	ReceiveACESignal	EMS (GE/Alstom)	AGC System	EMS API (ICCP)	AGC Internal Bus	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
AGC	Dispatch	CalculateGeneratorSetpoints	AGC System	SCADA (Siemens Spectrum Power)	AGC Internal Calc	SCADA API (ICCP)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
AGC	Setpoint	DispatchRaiseCommand	SCADA (Siemens Spectrum Power)	Plant DCS (ABB 800xA)	SCADA API (ICCP)	DCS Modbus TCP	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
AGC	Response	MonitorRampResponse	AGC System	EMS (GE/Alstom)	AGC Internal	EMS API (ICCP)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
AGC	Compliance	VerifyACENERC	AGC System	EMS (GE/Alstom)	AGC Internal	EMS API (ICCP)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
AGC	Confirmation	SendLFCResponse	EMS (GE/Alstom)	ISO RTO System	EMS API (ICCP)	ISO API (ICCP)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
AGC	Log	LogDispatchEvent	AGC System	Plant Historian (OSIsoft PI)	AGC API (REST)	PI API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

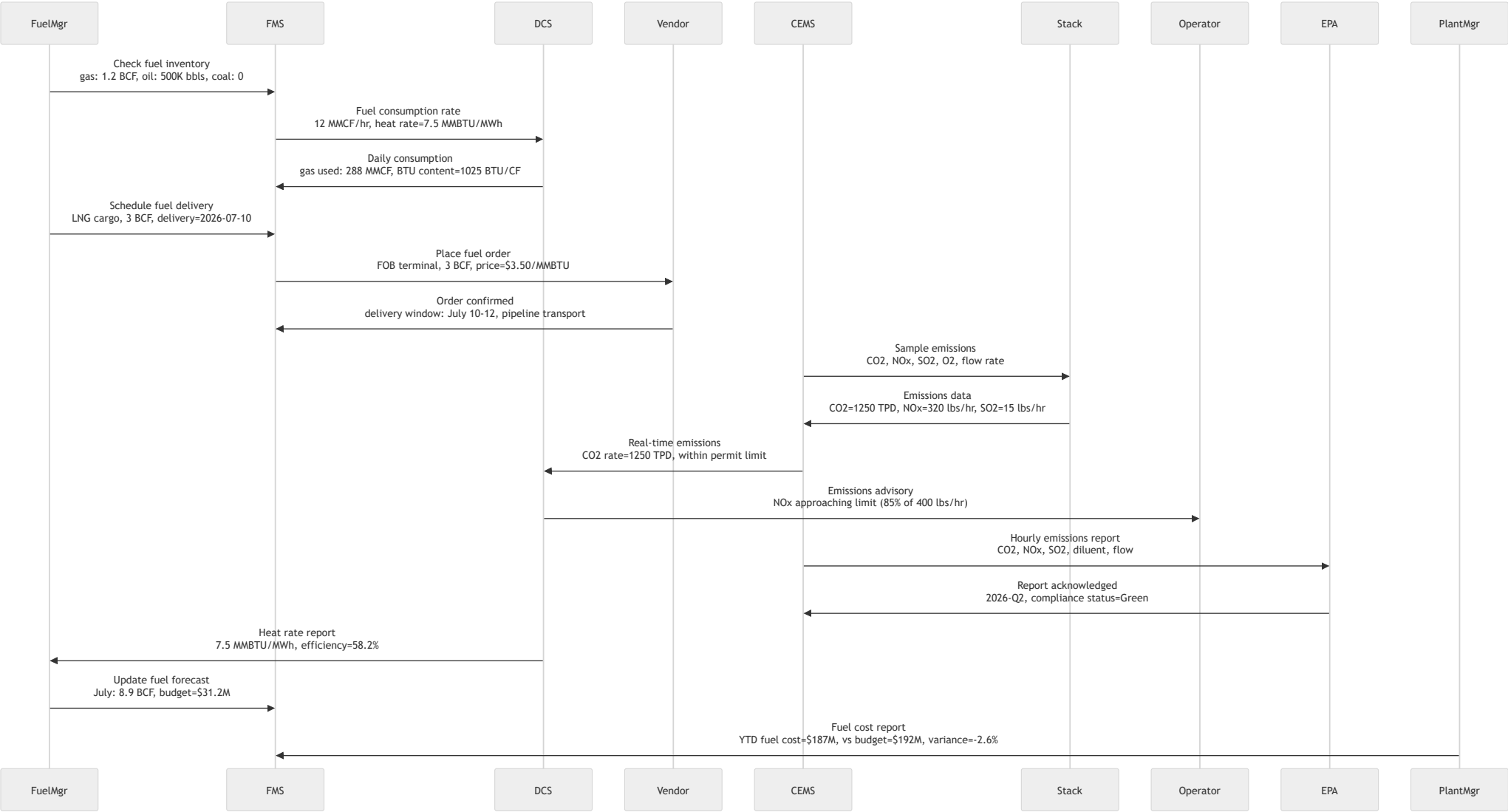
AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-AGC-01	Verify ACE returns to within ±10 MW deadband within 10 minutes of LFC request	ReceiveACESignal	ACE within deadband within 10 min of dispatch	High
AC-AGC-02	Validate generator ramp rate compliance with NERC BAL-001 standards	CalculateGeneratorSetpoints	Each generator ramps within ±0.5 MW/min of target	High
AC-AGC-03	Confirm ISO communication latency under 5 seconds for dispatch signals	ReceiveACESignal	ICCP data transfer completes in under 5s	Medium
AC-AGC-04	Ensure all AGC events logged with ACE value, setpoint, and unit response	LogDispatchEvent	Complete audit trail for NERC compliance	Medium

1.4 Fuel Management & Emissions Monitoring

Description: Fuel management system tracks fuel inventory, quality, and consumption for thermal plants. Emissions monitoring system (CEMS) continuously measures stack emissions and reports data to environmental regulators.

Actors: Fuel Manager, Environmental Engineer, Plant Manager
Systems: Fuel Management System, CEMS (Continuous Emissions Monitoring), DCS (ABB 800xA), EPA CAMD Portal, PI Historian

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Fuel	Inventory	CheckFuelInventory	Fuel Management System	DCS (ABB 800xA)	FMS API (REST)	DCS OPC-UA	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Fuel	Consumption	RecordDailyConsumption	DCS (ABB 800xA)	Fuel Management System	DCS OPC-UA	FMS API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
Fuel	Order	PlaceFuelOrder	Fuel Management System	Vendor Portal	FMS API (REST)	Vendor EDI	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Emissions	CEMS	SampleStackEmissions	CEMS	Stack Analyzer	CEMS Serial (Modbus)	Analyzer 4-20mA	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Emissions	DCS	TransferEmissionsData	CEMS	DCS (ABB 800xA)	CEMS API (Modbus TCP)	DCS OPC-UA	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Emissions	EPA	ReportHourlyEmissions	CEMS	EPA CAMD Portal	CEMS API (REST)	EPA CAMD Web Services (SOAP)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Emissions	Compliance	LogEmissionsReport	CEMS	Plant Historian (OSIsoft PI)	CEMS API (REST)	PI API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-FM-01	Verify fuel inventory accuracy within ±2% of physical measurement	CheckFuelInventory	FMS inventory matches tank gauging within 2%	High
AC-FM-02	Validate CEMS data completeness >95% for EPA reporting	SampleStackEmissions	No gaps > 1 hour in CEMS data stream	High
AC-FM-03	Confirm EPA report submission within regulatory deadline (hourly)	ReportHourlyEmissions	EPA acknowledges receipt within 1 hour of period end	High
AC-FM-04	Ensure heat rate calculation accuracy within ±1% of turbine heat balance	RecordDailyConsumption	Heat rate matches turbine manufacturer curve within 1%	Medium

AC ID	Description	Related Info Flow	Expected Outcome	Priority
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Transmission Operations & EMS

Real-time monitoring and control of the high-voltage transmission network via Energy Management System (EMS). Includes state estimation, contingency analysis, wide-area monitoring with synchrophasors, and transmission switching/clearance management.

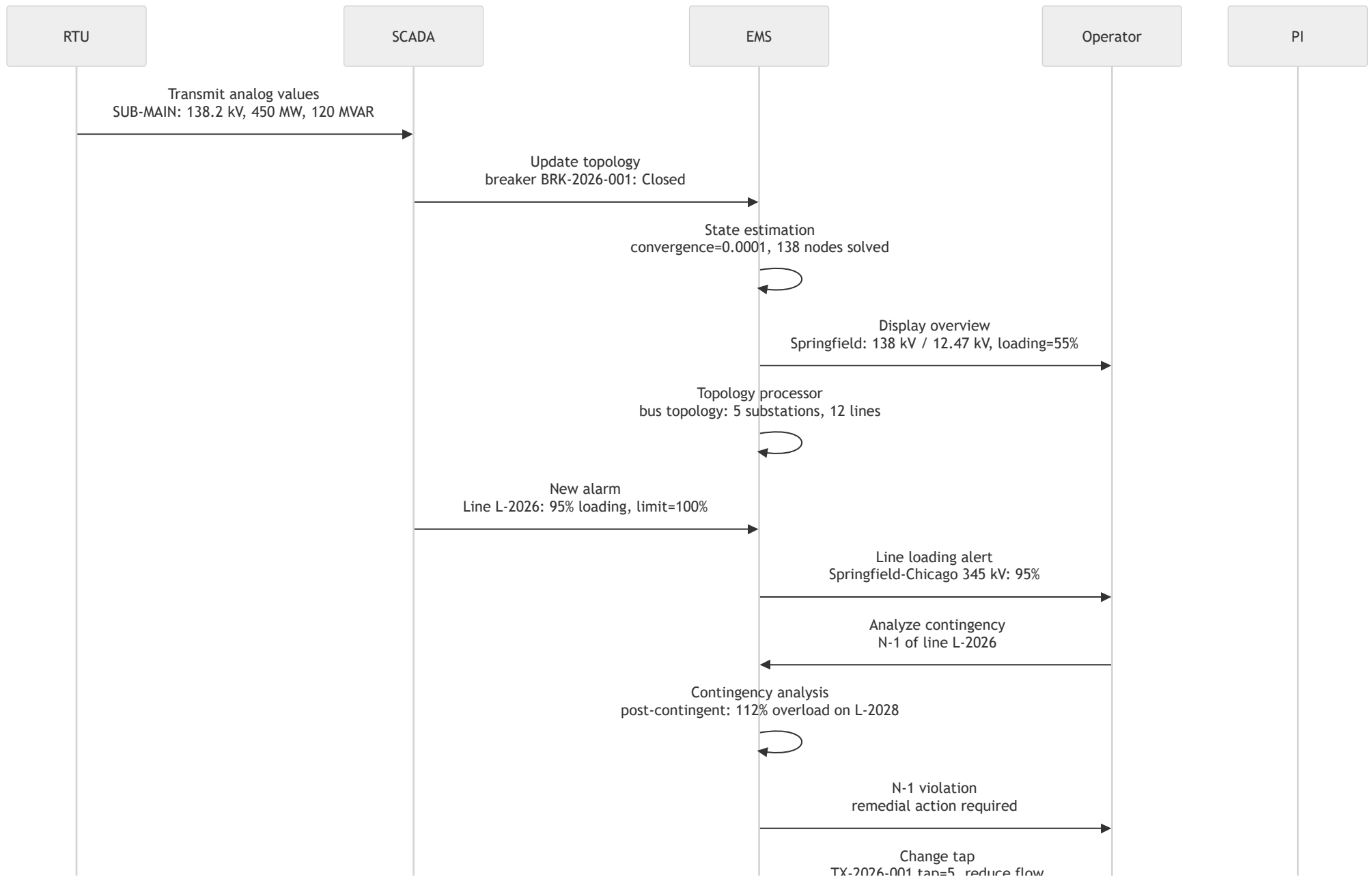
2.1 EMS Real-time Network Monitoring

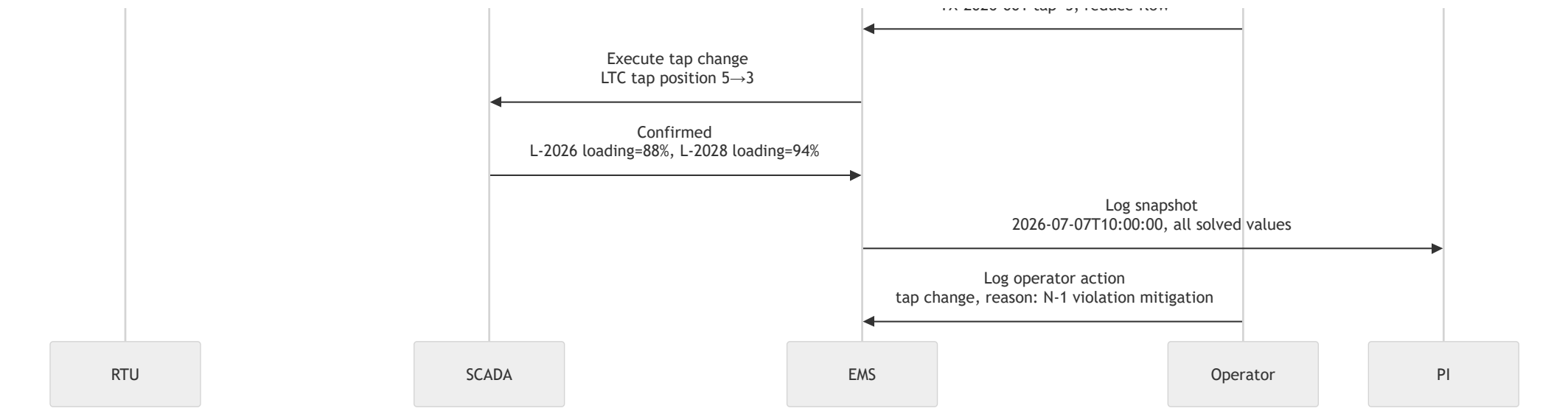
Description: System operators monitor the transmission network in real-time via EMS displays showing voltage profiles, line loadings, and system frequency. SCADA data from substations is processed for situational awareness.

Actors: Transmission Operator, EMS Engineer, Reliability Coordinator

Systems: EMS (Siemens Spectrum Power / OSI Monarch), SCADA RTUs, PI Historian, Substation IEDs

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
EMS	Telemetry	ReceiveAnalogValues	RTU (SUB-MAIN-001)	SCADA (Siemens Spectrum Power)	RTU DNP3	SCADA DNP3	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
EMS	StateEstimation	RunStateEstimation	EMS (Siemens Spectrum Power)	EMS (Siemens Spectrum Power)	Internal	Internal	Batch (Real-time)	High	High-frequency near-real-time batch processing with minimal latency tolerance
EMS	Alarm	ProcessLineAlarm	SCADA (Siemens Spectrum Power)	EMS (Siemens Spectrum Power)	SCADA API (ICCP)	EMS Internal	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
EMS	Contingency	RunContingencyAnalysis	EMS (Siemens Spectrum Power)	EMS (Siemens Spectrum Power)	Internal	Internal	Batch (Real-time)	High	High-frequency near-real-time batch processing with minimal latency tolerance
EMS	Control	ExecuteTapChange	EMS (Siemens Spectrum Power)	SCADA (Siemens Spectrum Power)	EMS API (ICCP)	SCADA DNP3	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
EMS	Historian	LogSnapshot	EMS (Siemens Spectrum Power)	Plant Historian (OSIsoft PI)	EMS API (REST)	PI API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints
EMS	Audit	LogOperatorAction	EMS (Siemens Spectrum Power)	Compliance Database	EMS API (REST)	SQL JDBC	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-EMS-01	Validate state estimation convergence within 30 seconds of topology change	RunStateEstimation	State estimation converges with residual <0.001 within 30s	High
AC-EMS-02	Confirm alarm processing latency under 5 seconds from RTU scan to HMI display	ProcessLineAlarm	Alarm visible on operator console within 5s of RTU scan	High
AC-EMS-03	Verify contingency analysis completes within 60 seconds for full network model	RunContingencyAnalysis	All N-1 contingencies evaluated within 60s	Medium
AC-EMS-04	Ensure all operator control actions are logged with user, target, and timestamp	LogOperatorAction	Complete audit trail for NERC compliance	Medium

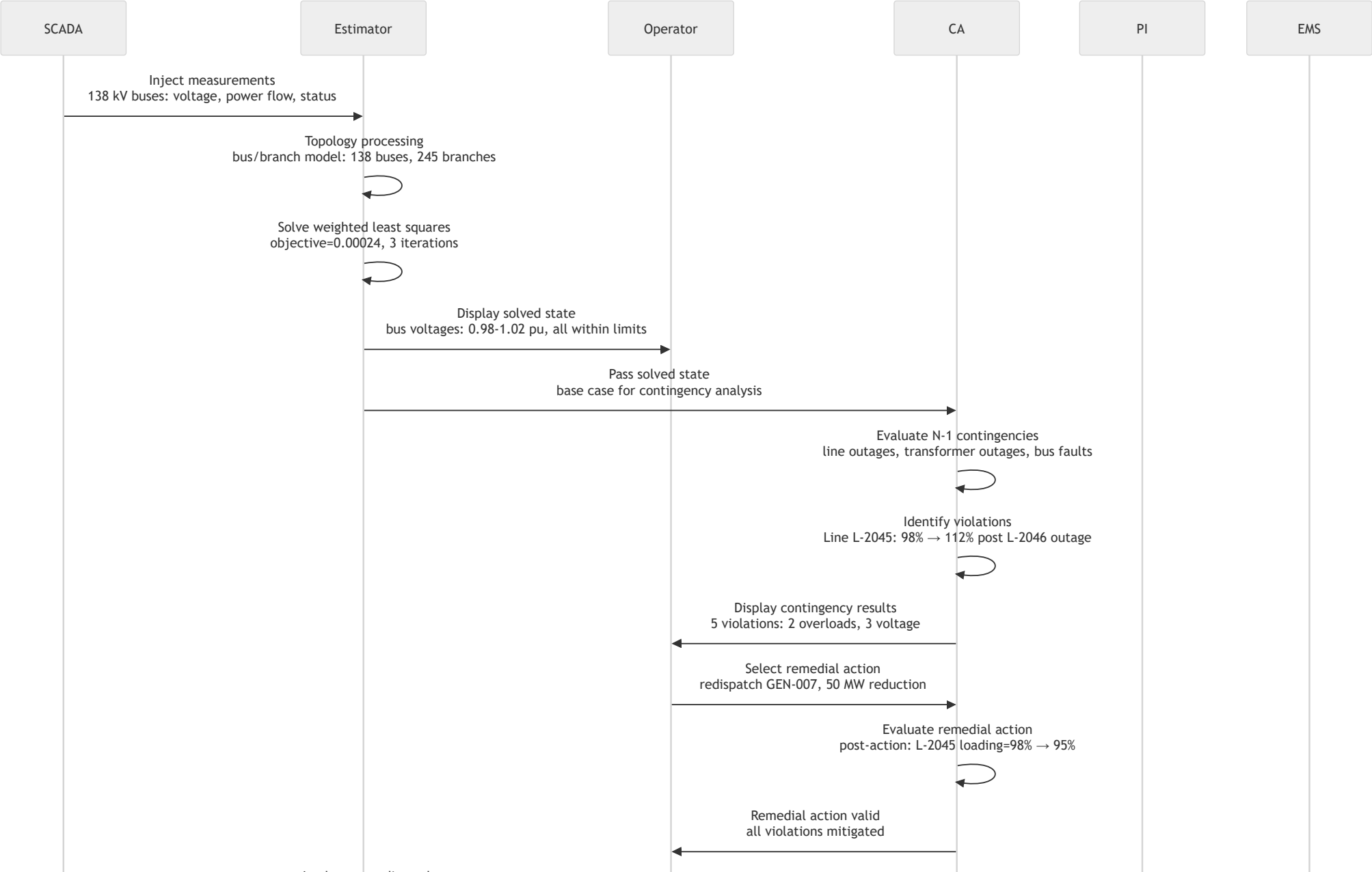
2.2 State Estimation & Contingency Analysis

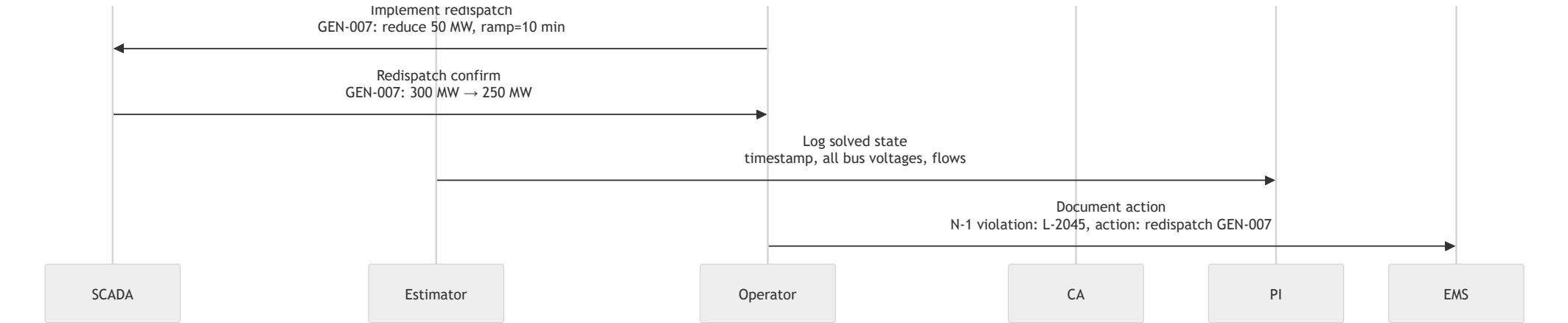
Description: State estimator processes real-time measurements to compute the best estimate of power system state. Contingency analysis evaluates N-1 reliability criteria and identifies potential violations requiring remedial action.

Actors: EMS Engineer, Transmission Operator, Planning Engineer

Systems: EMS State Estimator (OSI Monarch), SCADA, Contingency Analysis Module, Remedial Action System

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
EMS	Estimation	InjectMeasurements	SCADA	EMS State Estimator	SCADA API (ICCP)	Estimator API (Internal)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
EMS	Topology	ProcessTopology	EMS State Estimator	EMS Network Model	Internal	Internal	Batch (Real-time)	High	High-frequency near-real-time batch processing with minimal latency tolerance
EMS	Contingency	EvaluateN-1Contingencies	EMS State Estimator	Contingency Analysis Module	Internal	Internal	Batch (Real-time)	High	High-frequency near-real-time batch processing with minimal latency tolerance
EMS	Violation	IdentifyViolations	Contingency Analysis Module	Operator HMI	Internal	HMI API (REST)	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
EMS	Action	SelectRemedialAction	Operator HMI	Contingency Analysis Module	HMI API (REST)	Internal	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
EMS	Dispatch	ImplementRedispatch	Operator HMI	SCADA	HMI API (REST)	SCADA API (ICCP)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
EMS	Log	LogSolvedState	EMS State Estimator	Plant Historian (OSIsoft PI)	EMS API (REST)	PI API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-SE-01	Validate state estimation solves within 5-minute SCADA scan cycle	InjectMeasurements	State estimator converges before next SCADA scan	High
AC-SE-02	Confirm measurement residual <0.001 pu for all bus voltages	ProcessTopology	All bus voltage residuals within tolerance	High
AC-SE-03	Verify contingency analysis evaluates 100% of defined N-1 contingencies	EvaluateN-1Contingencies	All contingencies evaluated with results available	Medium
AC-SE-04	Ensure remedial action effectiveness validated before implementation	SelectRemedialAction	CA confirms all violations mitigated post-action	Medium

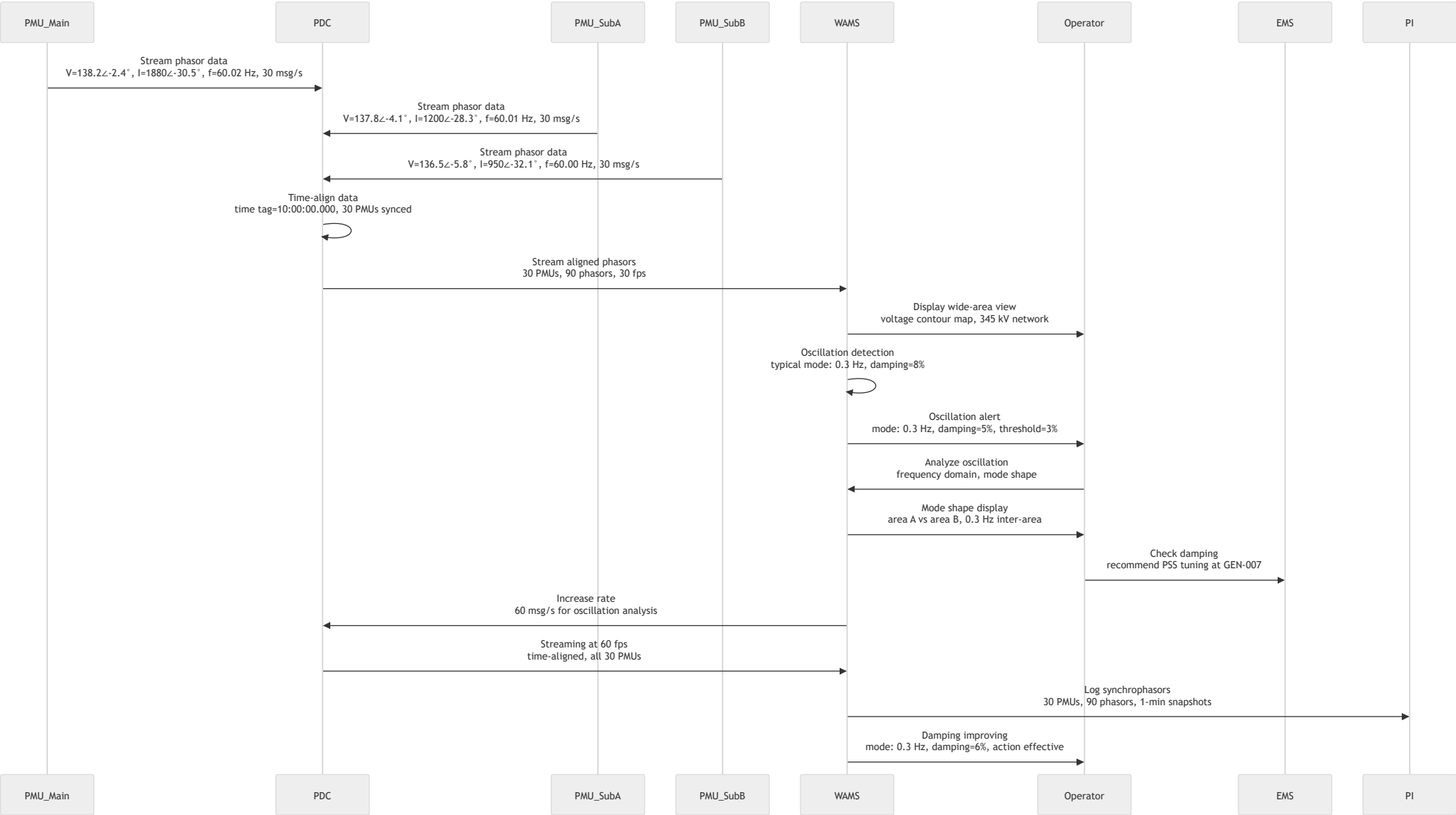
2.3 Wide Area Monitoring & Synchrophasors

Description: Wide Area Monitoring System (WAMS) collects synchrophasor data from PMUs deployed across the transmission network. Phasor data concentrators (PDCs) time-align and stream data for real-time visualization, oscillation detection, and post-event analysis.

Actors: System Operator, Protection Engineer, Grid Stability Analyst

Systems: WAMS (GE PhasorPoint / SEL), PDC (Phasor Data Concentrator), PMUs (SEL-421 / GE D60), EMS, PI Historian

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
WAMS	PMU	StreamPhasorData	PMU (SEL-421)	PDC (Phasor Data Concentrator)	PMU C37.118	PDC C37.118	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
WAMS	Alignment	TimeAlignPhasors	PDC (Phasor Data Concentrator)	PDC (Phasor Data Concentrator)	Internal	Internal	Batch (Real-time)	High	High-frequency near-real-time batch processing with minimal latency tolerance
WAMS	Stream	StreamAlignedPhasors	PDC (Phasor Data Concentrator)	WAMS (GE PhasorPoint)	PDC C37.118	WAMS C37.118	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
WAMS	Oscillation	DetectOscillations	WAMS (GE PhasorPoint)	WAMS (GE PhasorPoint)	Internal	Internal	Batch (Real-time)	Medium	Periodic near-real-time batch with data validation and transformation needs
WAMS	Alert	AlertOperatorOscillation	WAMS (GE PhasorPoint)	Operator HMI	WAMS API (REST)	HMI API (REST)	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
WAMS	Historian	LogSynchrophasors	WAMS (GE PhasorPoint)	Plant Historian (OSIsoft PI)	WAMS API (REST)	PI API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
WAMS	EMS	TransferWAMSData	WAMS (GE PhasorPoint)	EMS	WAMS API (ICCP)	EMS API (ICCP)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-WAM-01	Validate PMU data timestamp accuracy within ±50 μs of UTC via GPS	StreamPhasorData	All PMU timestamps synchronized to GPS within tolerance	High
AC-WAM-02	Confirm PDC time-alignment latency under 10 ms for 30 PMU streams	TimeAlignPhasors	Time-aligned data available within 10 ms of PMU frame	High
AC-WAM-03	Verify oscillation detection identifies modes with damping <10% within 5 cycles	DetectOscillations	Oscillation detection triggers within 5 cycles of onset	Medium
AC-WAM-04	Ensure synchrophasor data archival at minimum 1 snapshot per minute	LogSynchrophasors	PI historian contains 1-min snapshots for all 30 PMUs	Medium

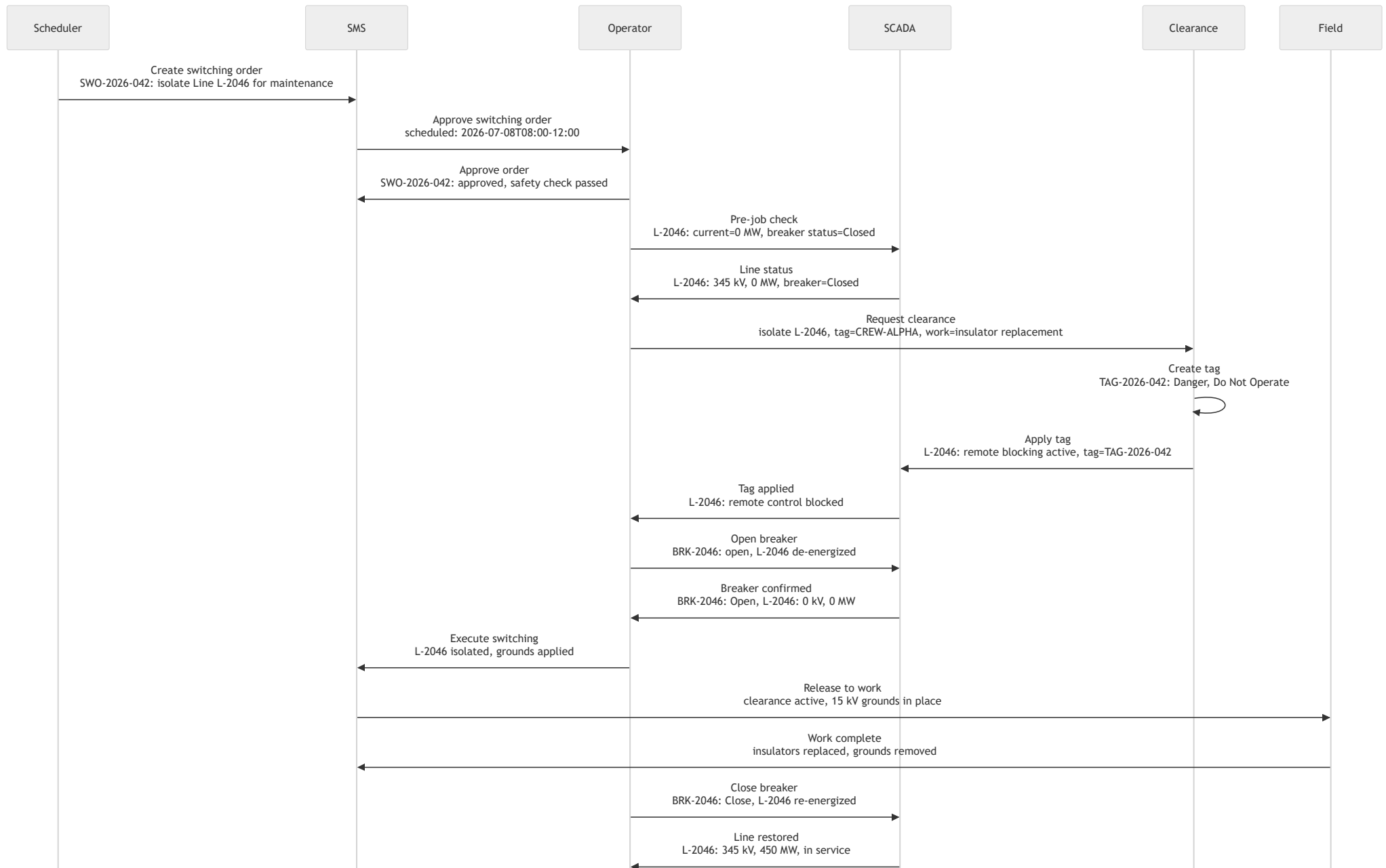
2.4 Transmission Switching & Clearance Management

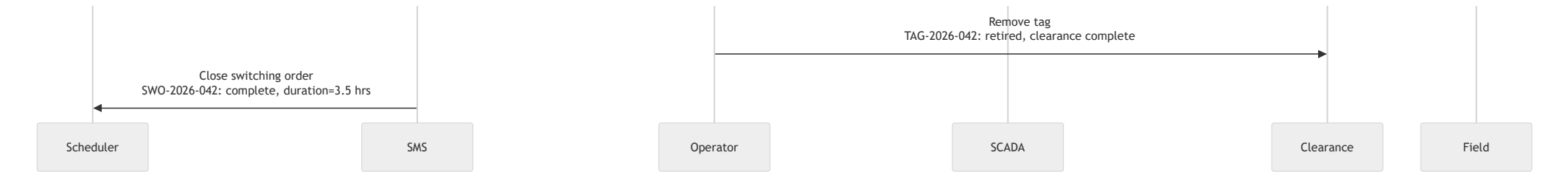
Description: Operators manage transmission switching orders and clearance requests to ensure safe isolation of equipment for maintenance. The clearance management system tracks tags, holds, and switching sequences with interlocking logic.

Actors: Transmission Operator, Switching Scheduler, Field Crew

Systems: Switching Management System, EMS, SCADA, Clearance & Tagging System, Mobile Field App

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Switching	Order	CreateSwitchingOrder	Switching Scheduler	Switching Management System	Scheduler UI	SMS API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Switching	Approval	ApproveSwitchingOrder	Switching Management System	Operator HMI	SMS API (REST)	HMI API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Switching	Clearance	RequestClearance	Operator HMI	Clearance & Tagging System	HMI API (REST)	Clearance API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Switching	Tag	ApplyTag	Clearance & Tagging System	SCADA	Clearance API (REST)	SCADA API (ICCP)	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
Switching	Breaker	OpenBreaker	Operator HMI	SCADA	HMI API (REST)	SCADA DNP3	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Switching	Field	ReleaseToWork	Switching Management System	Mobile Field App	SMS API (REST)	Field App API (OAuth2)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Switching	Restore	RestoreLine	Operator HMI	SCADA	HMI API (REST)	SCADA DNP3	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Switching	Close	CloseSwitchingOrder	Switching Management System	Switching Scheduler	SMS API (REST)	Scheduler UI	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-TS-01	Verify tag/clearance blocks remote SCADA control before breaker operation	ApplyTag	SCADA remote control blocked when tag is active	High
AC-TS-02	Confirm switching order approval workflow is complete before field release	ApproveSwitchingOrder	Switching order fully approved before field work begins	High
AC-TS-03	Validate clearance-to-work authorization is communicated to field crew	ReleaseToWork	Field crew acknowledges clearance within 5 min	Medium
AC-TS-04	Ensure all switching events are logged with user, equipment, and timestamp	CloseSwitchingOrder	Complete audit trail	Medium

AC ID	Description	Related Info Flow	Expected Outcome	Priority
			I for NERC compliance	

Substation Automation & Protection

Automated control and monitoring of substation equipment via RTUs, IEDs, and IEC 61850 communications. Includes protection relay management, fault recording, and substation security perimeters.

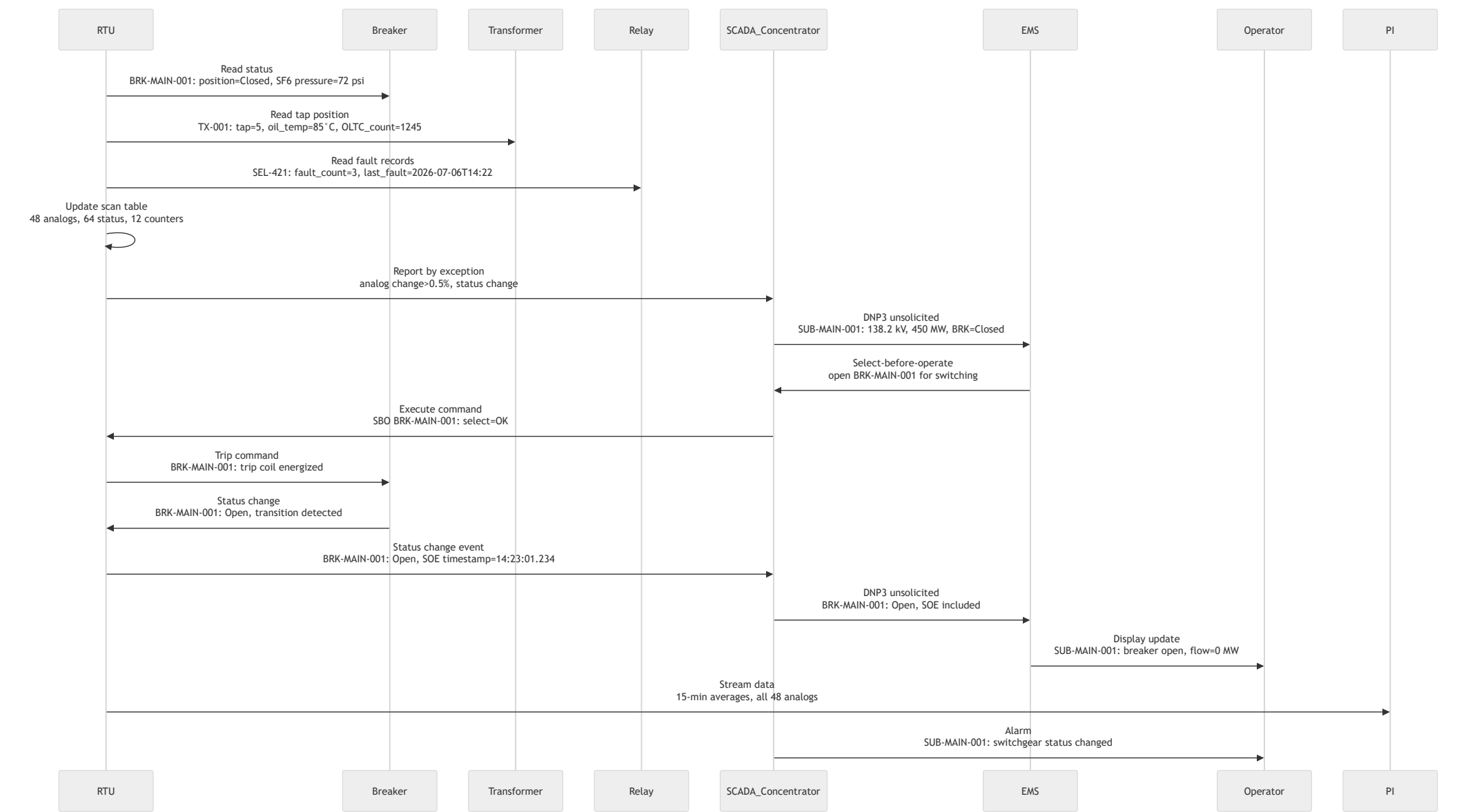
3.1 RTU & Substation SCADA Control

Description: Remote Terminal Units at substations acquire analog and status data from switchgear, transformers, and breakers. The substation SCADA concentrator aggregates RTU data and communicates with the central EMS via ICCP/DNP3.

Actors: Substation Technician, System Operator, SCADA Engineer

Systems: RTU (SEL-3530 / GE D20), Substation SCADA Concentrator, EMS, IEDs (SEL-421 Relays)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Substation	Status	ReadBreakerStatus	RTU (SEL-3530)	Breaker Control	RTU I/O	Breaker Aux Contact	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Substation	Analog	ReadAnalogValues	RTU (SEL-3530)	CT/PT	RTU Analog Input	CT/PT Secondary	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Substation	SCADA	ReportByException	RTU (SEL-3530)	Substation SCADA Concentrator	RTU DNP3	SCADA DNP3	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Substation	EMS	TransmitToEMS	Substation SCADA Concentrator	EMS (Siemens Spectrum Power)	SCADA API (ICCP)	EMS API (ICCP)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Substation	Control	SBOCommand	EMS (Siemens Spectrum Power)	Substation SCADA Concentrator	EMS API (ICCP)	SCADA DNP3	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Substation	Event	LogSOEEvent	RTU (SEL-3530)	Substation SCADA Concentrator	RTU DNP3	SCADA DNP3	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
Substation	Historian	StreamAnalogData	RTU (SEL-3530)	Plant Historian (OSIsoft PI)	RTU DNP3	PI Interface (DNP3)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-RTU-01	Validate RTU scan rate of 2 seconds for all analog points	ReadAnalogValues	All analogs scanned within 2s cycle	High
AC-RTU-02	Confirm SOE timestamp accuracy within ±1 ms of event occurrence	LogSOEEvent	SOE timestamps accurate to ±1 ms	High
AC-RTU-03	Verify select-before-operate sequence completes in under 5 seconds	SBOCommand	Control operation sequence completes within 5s	Medium
AC-RTU-04	Ensure data quality flags (good/suspect/override) are preserved in transmission	ReportByException	Data quality flags correctly mapped to DNP3 quality field	Medium

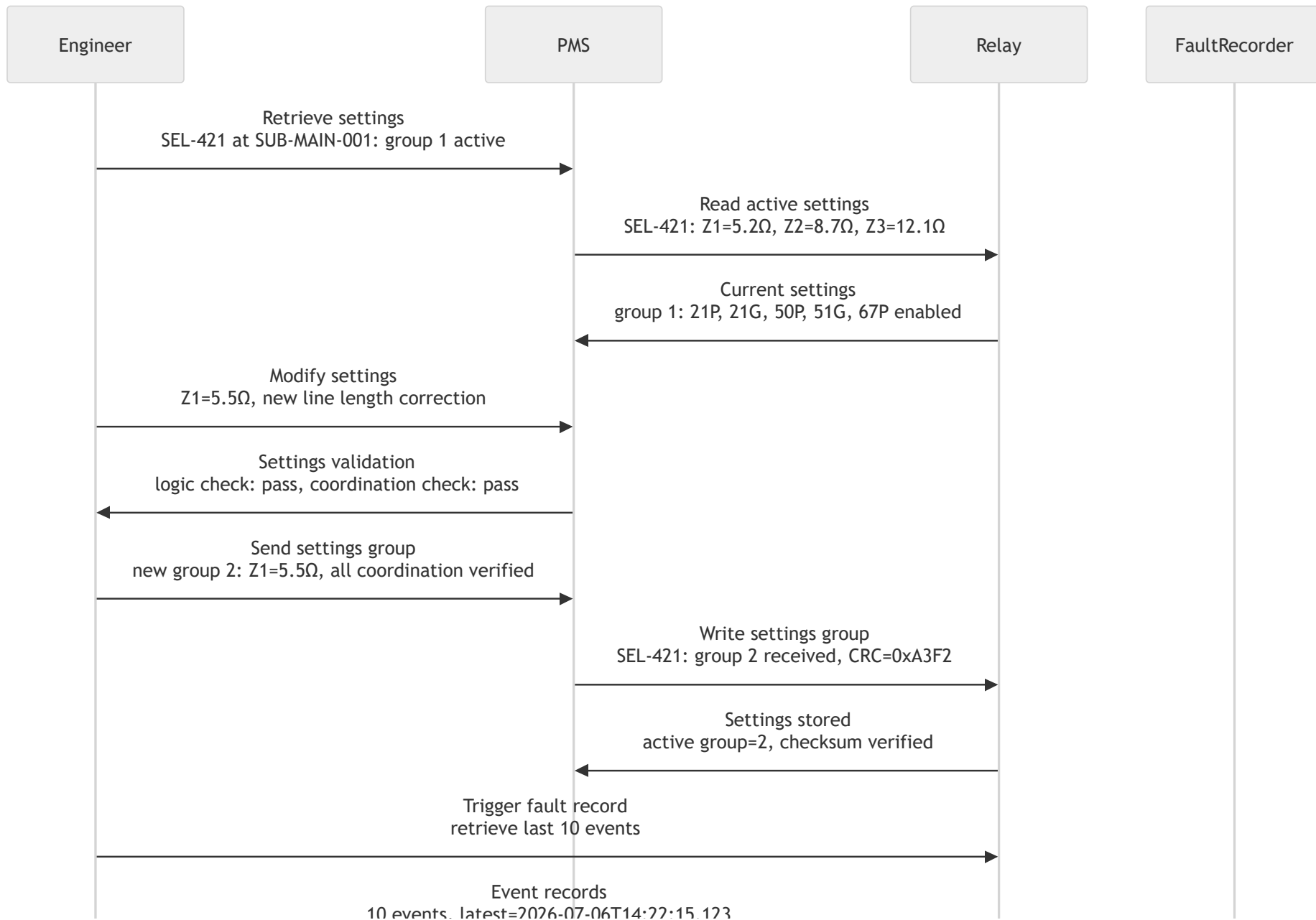
3.2 Protection Relay & IED Management

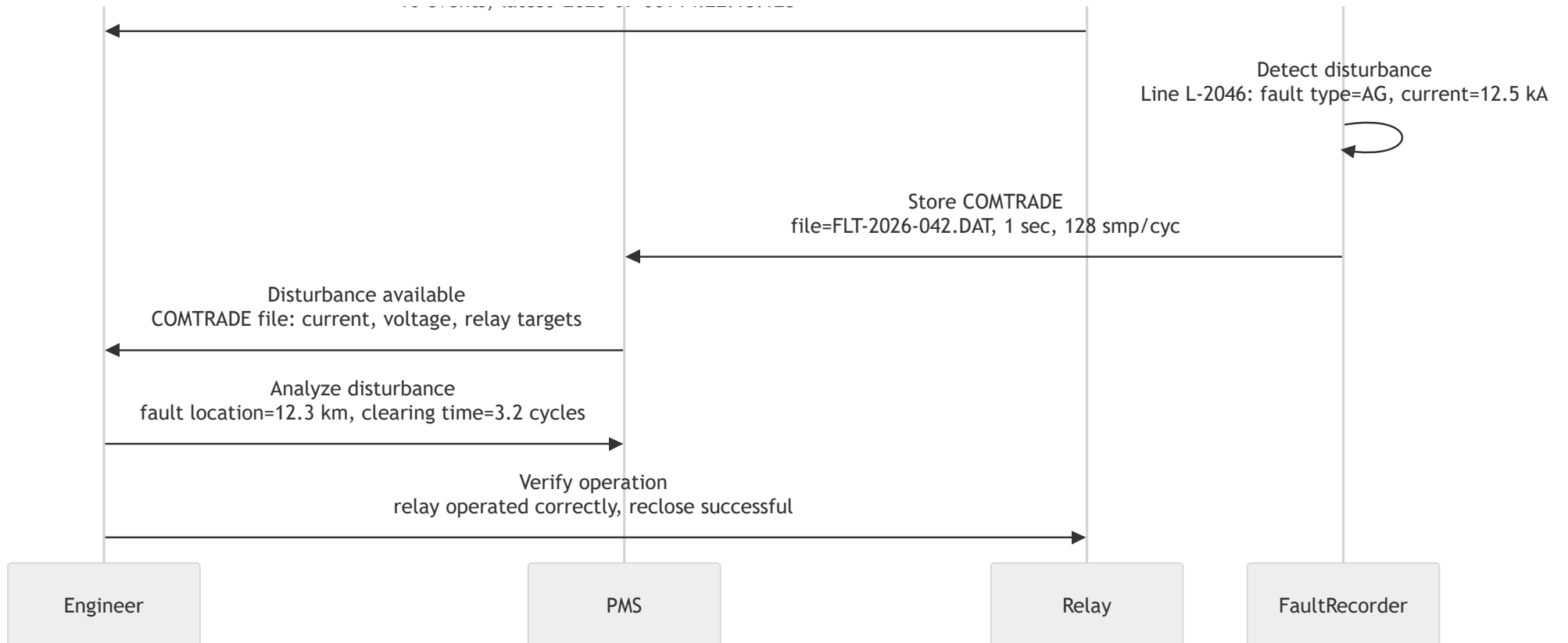
Description: Protection engineers manage relay settings, fault records, and event reports for microprocessor relays. The IED management system handles settings file versioning, group changes, and disturbance record retrieval.

Actors: Protection Engineer, Substation Technician, Fault Analyst

Systems: Protection Management System (SEL AcSElerator / GE Enervista), Relays (SEL-421, SEL-311, GE D60), Fault Recorder System, Time Sync (GPS / IRIG-B)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Protection	Settings	ReadRelaySettings	Protection Management System	Relay (SEL-421)	PMS API (REST)	Relay Serial (SEL Protocol)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Protection	Validation	ValidateRelaySettings	Protection Management System	Protection Management System	Internal	Internal	Batch (Real-time)	High	High-frequency near-real-time batch processing with minimal latency tolerance
Protection	Write	WriteSettingsGroup	Protection Management System	Relay (SEL-421)	PMS API (REST)	Relay Serial (SEL Protocol)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Protection	Events	RetrieveFaultRecords	Engineer HMI	Relay (SEL-421)	HMI API (REST)	Relay Serial (SEL Protocol)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Protection	Disturbance	StoreCOMTRADE	Fault Recorder	Protection Management System	Fault Recorder API	PMS API (REST)	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
Protection	Analysis	AnalyzeDisturbance	Protection Management System	Engineer HMI	PMS API (REST)	HMI API (REST)	Batch (Real-time)	Simple	Minimal near-real-time batch with basic data pass-through
Protection	TimeSync	SyncRelayTime	Time Sync (GPS)	Relay (SEL-421)	GPS IRIG-B	Relay IRIG-B	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-PR-01	Validate relay settings file CRC integrity after write operation	WriteSettingsGroup	CRC checksum matches between PMS and relay	High
AC-PR-02	Confirm settings group change does not affect other protection elements	ValidateRelaySettings	All existing protection elements remain enabled/disabled as configured	High
AC-PR-03	Verify COMTRADE file captures 60 cycles pre-fault and 180 cycles post-fault	StoreCOMTRADE	Disturbance record meets duration requirements	Medium
AC-PR-04	Ensure relay time drift remains under ±1 ms relative to GPS	SyncRelayTime	Relay time sync status shows IRIG-B locked	Medium

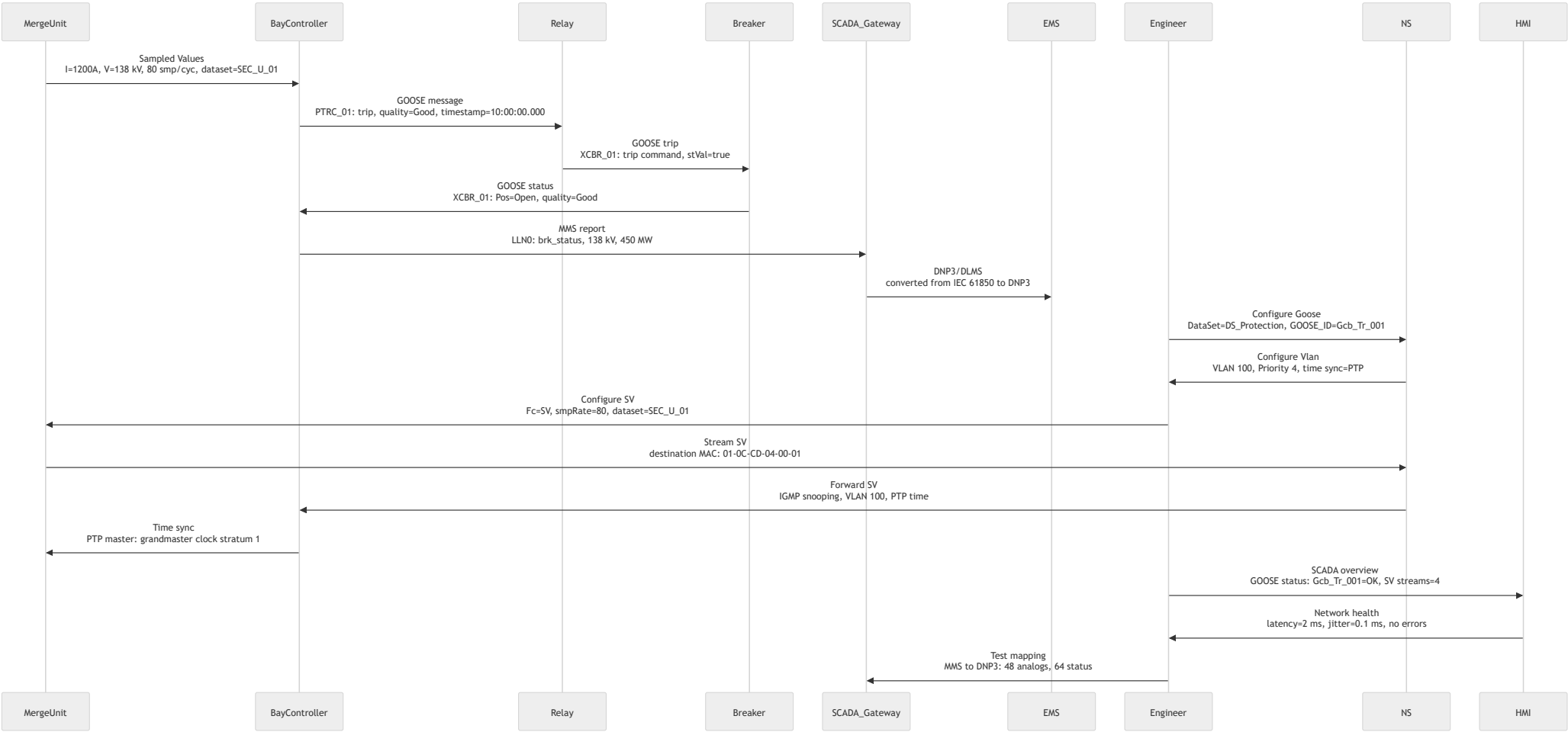
3.3 IEC 61850 Substation Communications

Description: Substation automation devices communicate using IEC 61850 protocols including GOOSE for high-speed peer-to-peer messaging, MMS for client-server monitoring and control, and Sampled Values for instrument transformer data.

Actors: Substation Automation Engineer, Protection Engineer, Network Engineer

Systems: IEC 61850 IEDs (SEL, GE, ABB), Substation HMI, SCADA Gateway (IEC 61850 to DNP3), Network Switch (Managed), Time Sync (PTP/IEEE 1588)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
IEC61850	SV	StreamSampledValues	Merge Unit	Bay Controller	SV (IEC 61850-9-2)	SV (IEC 61850-9-2)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
IEC61850	GOOSE	TripGOOSEMessage	Bay Controller	Relay	GOOSE (IEC 61850-8-1)	GOOSE (IEC 61850-8-1)	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
IEC61850	MMS	ReportMMSData	Bay Controller	SCADA Gateway	MMS (IEC 61850-8-1)	MMS (IEC 61850-8-1)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
IEC61850	Gateway	ConvertToDNP3	SCADA Gateway	EMS	Gateway Internal	DNP3 (IEC 61850 to DNP3)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
IEC61850	Config	ConfigureGOOSE	Engineer Workstation	Network Switch	Engineering Tool	Switch CLI/SNMP	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
IEC61850	PTP	SyncPTPTime	PTP Grandmaster	Merge Unit	PTP (IEEE 1588)	PTP (IEEE 1588)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
IEC61850	Health	MonitorNetworkHealth	Network Switch	Substation HMI	SNMP	HMI API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-IEC-01	Validate GOOSE message delivery time under 3 ms for protection messages	TripGOOSEMessage	GOOSE trip received by relay within 3 ms	High
AC-IEC-02	Confirm SV stream synchronization within ±1 μs via PTP	StreamSampledValues	All SV streams synchronized to PTP grandmaster	High
AC-IEC-03	Verify MMS report latency under 100 ms from data change to SCADA gateway	ReportMMSData	MMMS report buffered report delivered within 100 ms	Medium
AC-IEC-04	Ensure GOOSE dataset integrity with sequence number continuity	ConfigureGOOSE	No missing GOOSE sequence numbers over 24-hour test	Medium

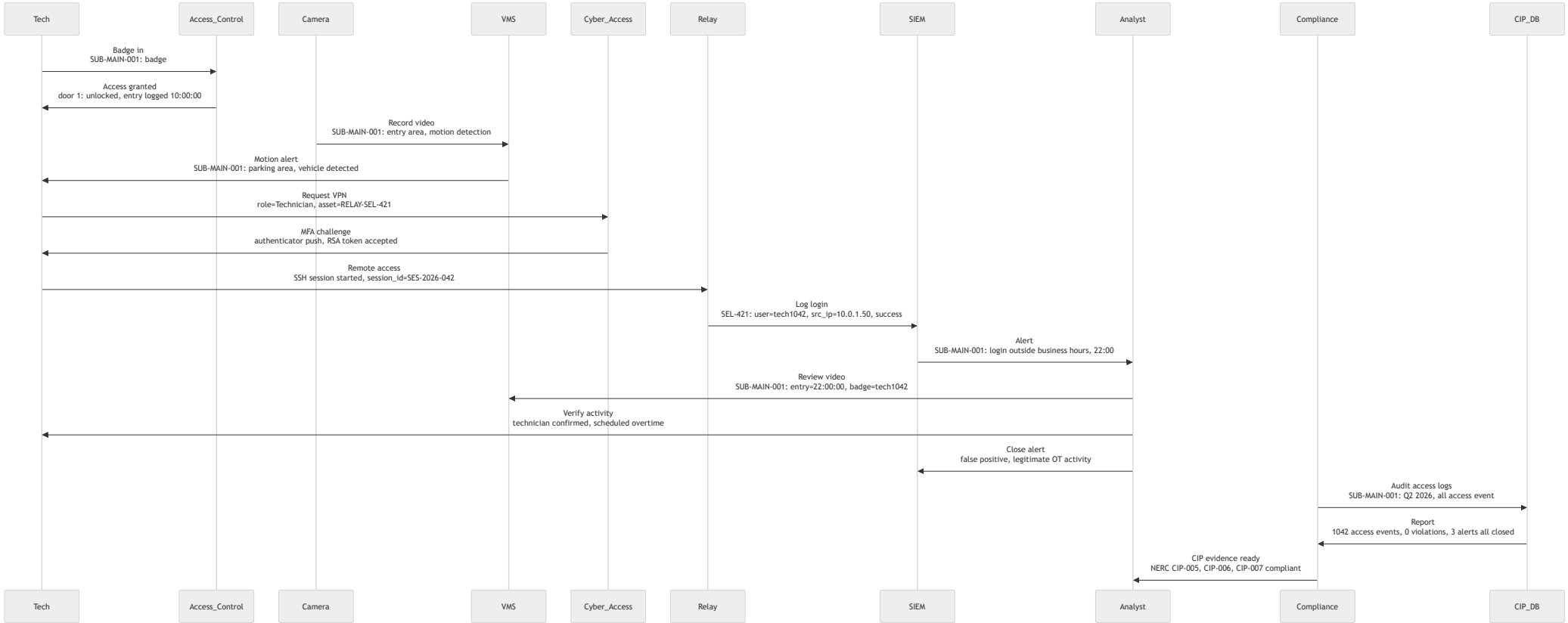
3.4 Substation Security & Access Management

Description: Physical and cyber security management for substations including electronic security perimeters (ESP), access control systems, video surveillance, and NERC CIP compliance for cyber assets.

Actors: Security Analyst, Substation Technician, NERC CIP Compliance Lead

Systems: Access Control System (Lenel / Honeywell), Video Management System, Substation HMI, CIP Compliance Database, SIEM (Splunk)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Security	Physical	BadgeAccess	Badge Reader	Access Control System	Badge Reader Wiegand	ACS API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Security	Video	RecordMotionEvent	Camera (IP)	Video Management System	Camera RTSP	VMS API (ONVIF)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Security	Cyber	RequestVPNAccess	Remote Technician	Cyber Access Gateway	VPN Client	Gateway API (RADIUS)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Security	MFA	ChallengeMFA	Cyber Access Gateway	MFA Server (RSA)	Gateway API (RADIUS)	MFA API (RADIUS/HTTP)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Security	SIEM	LogAccessEvent	Relay (SEL-421)	SIEM (Splunk)	Syslog	SIEM API (Syslog)	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
Security	CIP	AuditAccessLogs	Compliance Lead	CIP Compliance Database	CIP UI	CIP DB API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
Security	Report	GenerateCIPReport	CIP Compliance Database	Compliance Lead	CIP DB API (REST)	CIP UI	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

d>CIP compliance database contains Q2 2026 review evidence

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-SEC-01	Validate badge access response time under 3 seconds from badge swipe to door unlock	BadgeAccess	Door unlocks within 3s of badge presentation	High
AC-SEC-02	Confirm MFA enforcement for all remote cyber access to BES Cyber Assets	ChallengeMFA	All remote access requires MFA before session establishment	High
AC-SEC-03	Verify SIEM receives all access events within 60 seconds of occurrence	LogAccessEvent	SIEM shows event timestamp within 60s of actual event	Medium
AC-SEC-04	Ensure quarterly CIP access log reviews are completed with evidence archived	AuditAccessLogs	Medium	

Distribution Grid & ADMS

Real-time monitoring, control, and optimization of the distribution network via Advanced Distribution Management System (ADMS). Includes DER management, fault location/restoration, and distribution planning with volt/VAR optimization.

4.1 Distribution SCADA & ADMS Operations

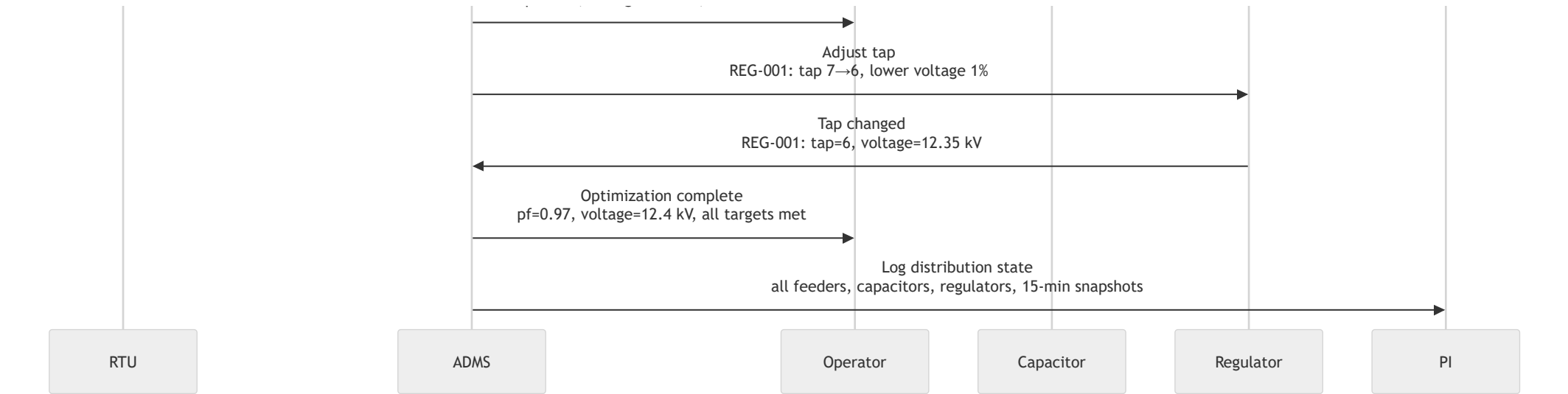
Description: Distribution operators monitor feeders, regulators, and capacitor banks via ADMS. The system provides real-time visibility, volt/VAR control, and switching management for the distribution grid.

Actors: Distribution Operator, Field Technician, Distribution Engineer

Systems: ADMS (GE Grid Solutions / OSI), Distribution SCADA RTUs, Capacitor Bank Controllers, Voltage Regulators, PI Historian

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Distribution	Telemetry	ReportFeederValues	RTU (Feeder)	ADMS (GE Grid Solutions)	RTU DNP3	ADMS DNP3	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Distribution	SCADA	UpdateTopology	ADMS (GE Grid Solutions)	ADMS (GE Grid Solutions)	Internal	Internal	Batch (Real-time)	High	High-frequency near-real-time batch processing with minimal latency tolerance
Distribution	Capacitor	MonitorCapacitorBank	Capacitor Bank Controller	ADMS (GE Grid Solutions)	Capacitor DNP3	ADMS DNP3	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Distribution	Regulator	ReadRegulatorTap	Voltage Regulator Control	ADMS (GE Grid Solutions)	Regulator DNP3	ADMS DNP3	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Distribution	VoltVAR	ExecuteVoltVARControl	ADMS (GE Grid Solutions)	Capacitor Bank Controller	ADMS DNP3	Capacitor DNP3	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Distribution	Optimize	OptimizePowerFactor	ADMS (GE Grid Solutions)	ADMS (GE Grid Solutions)	Internal	Internal	Batch (Real-time)	Medium	Periodic near-real-time batch with data validation and transformation needs
Distribution	Historian	LogDistributionState	ADMS (GE Grid Solutions)	Plant Historian (OSIsoft PI)	ADMS API (REST)	PI API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-DS-01	Validate feeder telemetry accuracy within ±1% of field meter values	ReportFeederValues	ADMS feeder values match field meters within 1%	High
AC-DS-02	Confirm capacitor bank switching response under 10 seconds from command	MonitorCapacitorBank	Capacitor status reflects command within 10s	Medium
AC-DS-03	Verify volt/VAR optimization converges to target pf within ±0.02	OptimizePowerFactor	Power factor within 0.02 of target post-optimization	Medium
AC-DS-04	Ensure all ADMS control actions are logged with operator identity and timestamp	LogDistributionState	Complete audit trail for every control action	Medium

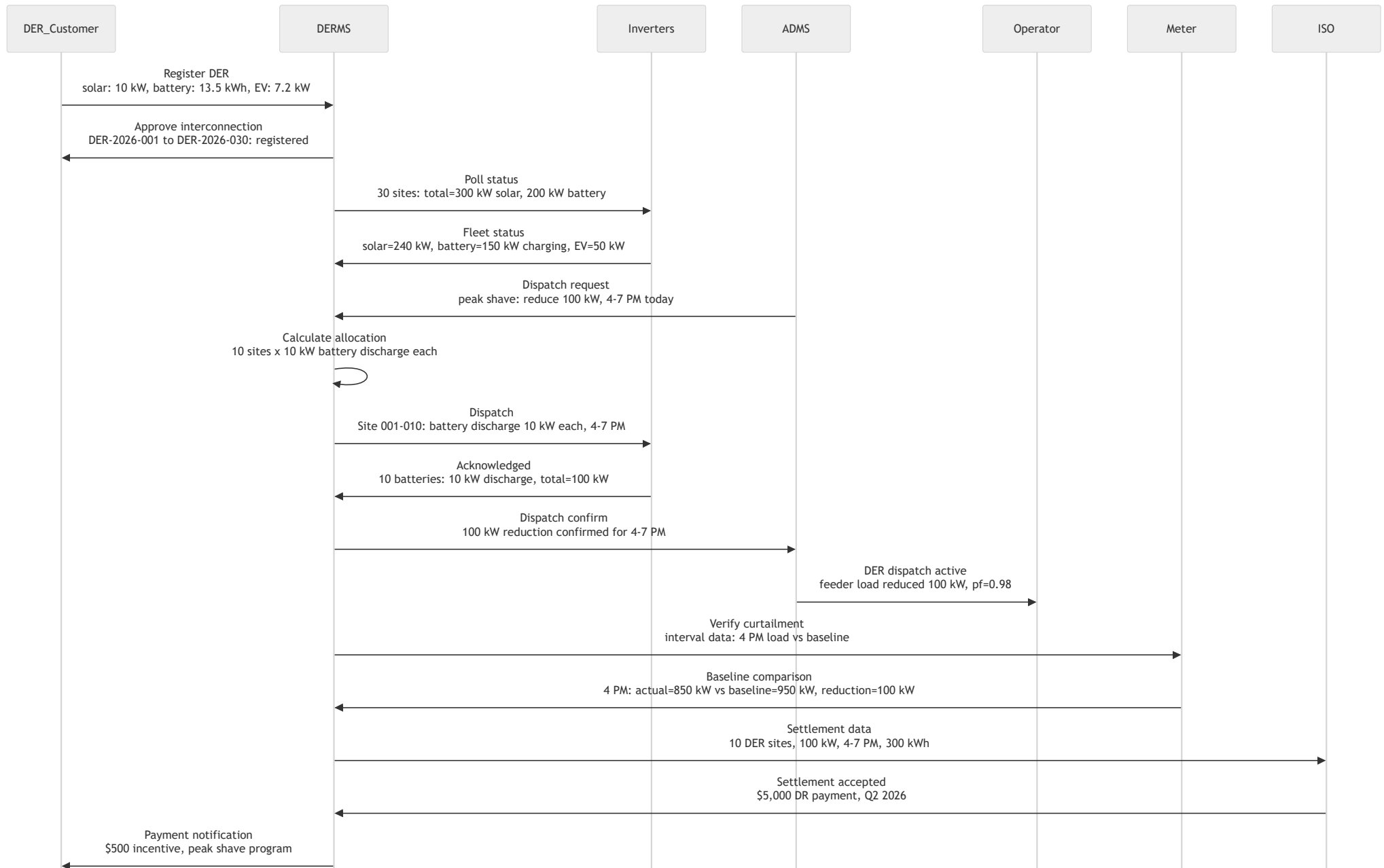
4.2 DER Management & Distributed Energy Resources

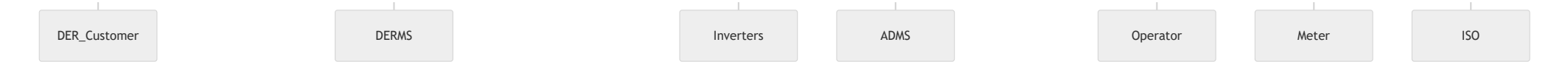
Description: Distribution operators manage interconnection of customer-owned DERs including solar PV, battery storage, and EV chargers. The DERMS system aggregates capacity, dispatches for peak shaving, and monitors power quality at the grid edge.

Actors: DER Operator, DER Customer, Grid Integration Engineer

Systems: DERMS (Generac / Enbala), ADMS (GE Grid Solutions), Customer DER Inverters, Meter Data System, ISO Market Portal

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
DER	Register	RegisterDERSite	DER Customer Portal	DERMS (Generac)	Customer UI	DERMS API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
DER	Poll	PollInverterFleet	DERMS (Generac)	DER Inverters	DERMS API (Modbus TCP)	Inverter API (SunSpec)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
DER	Dispatch	ReceiveDispatchRequest	ADMS (GE Grid Solutions)	DERMS (Generac)	ADMS API (ICCP)	DERMS API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
DER	Battery	DischargeBattery	DERMS (Generac)	Battery Inverters	DERMS API (Modbus TCP)	Inverter API (SunSpec)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
DER	Meter	VerifyLoadReduction	DERMS (Generac)	Meter Data System	DERMS API (REST)	Meter API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
DER	Settlement	SubmitSettlementData	DERMS (Generac)	ISO Market Portal	DERMS API (REST)	ISO API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
DER	Payment	NotifyDERPayment	DERMS (Generac)	DER Customer Portal	DERMS API (REST)	Customer UI	Event-driven	Simple	Simple event notification with fire-and-forget delivery semantics

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-DER-01	Validate DER dispatch command reaches inverter within 30 seconds	DischargeBattery	Inverter acknowledges dispatch within 30s	High
AC-DER-02	Confirm load reduction measurement accuracy within ±5% of baseline interval metering	VerifyLoadReduction	Meter data confirms reduction within 5% of DERMS estimate	Medium
AC-DER-03	Verify DERMS aggregates all registered DERs into dispatch calculation	ReceiveDispatchRequest	All 30 registered sites included in dispatch allocation	High
AC-DER-04	Ensure settlement data submitted to ISO within 1 hour of event end	SubmitSettlementData	Settlement submission timestamp within 1 hr of event close	Medium

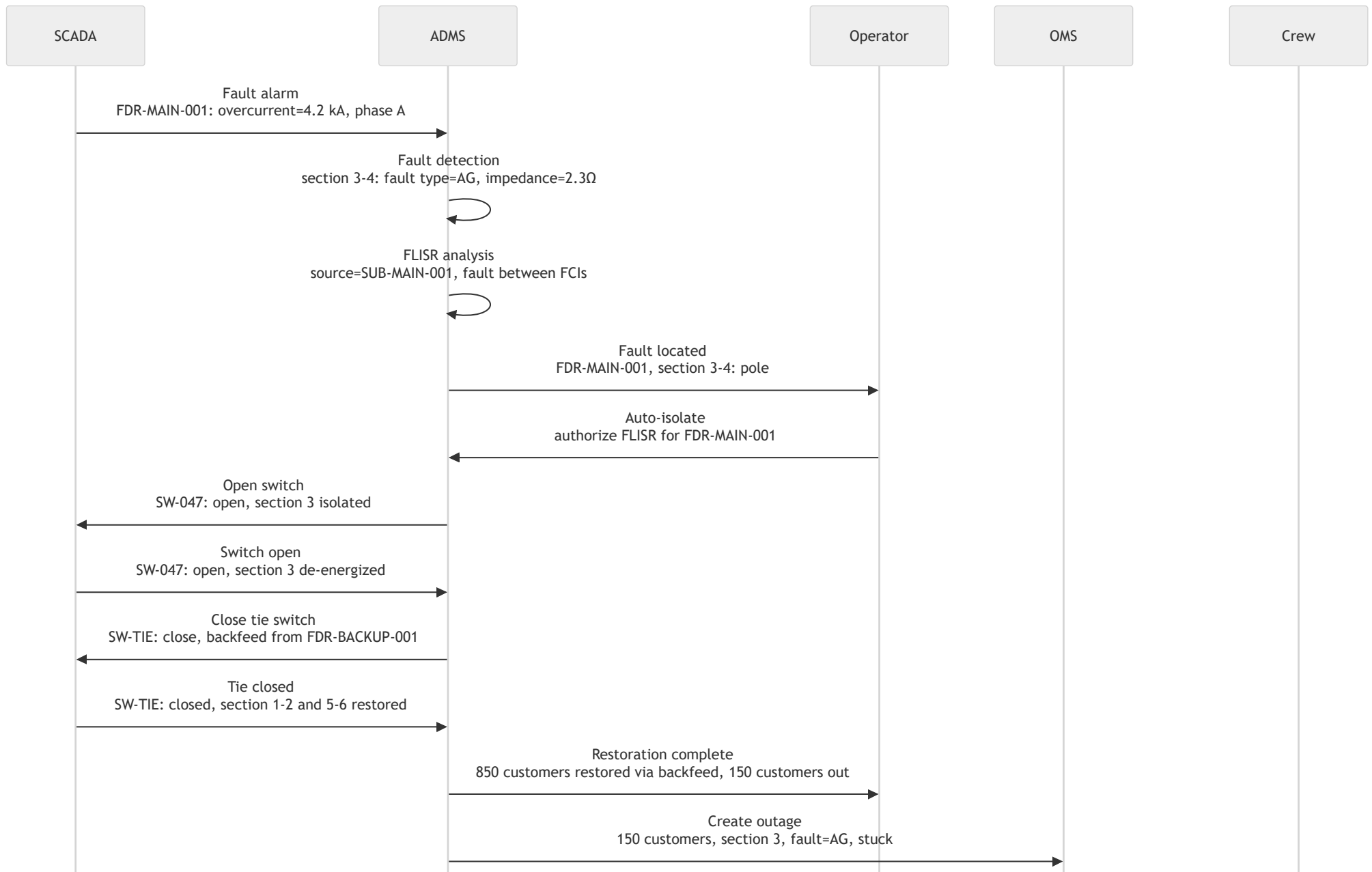
4.3 Fault Location, Isolation & Restoration (FLISR)

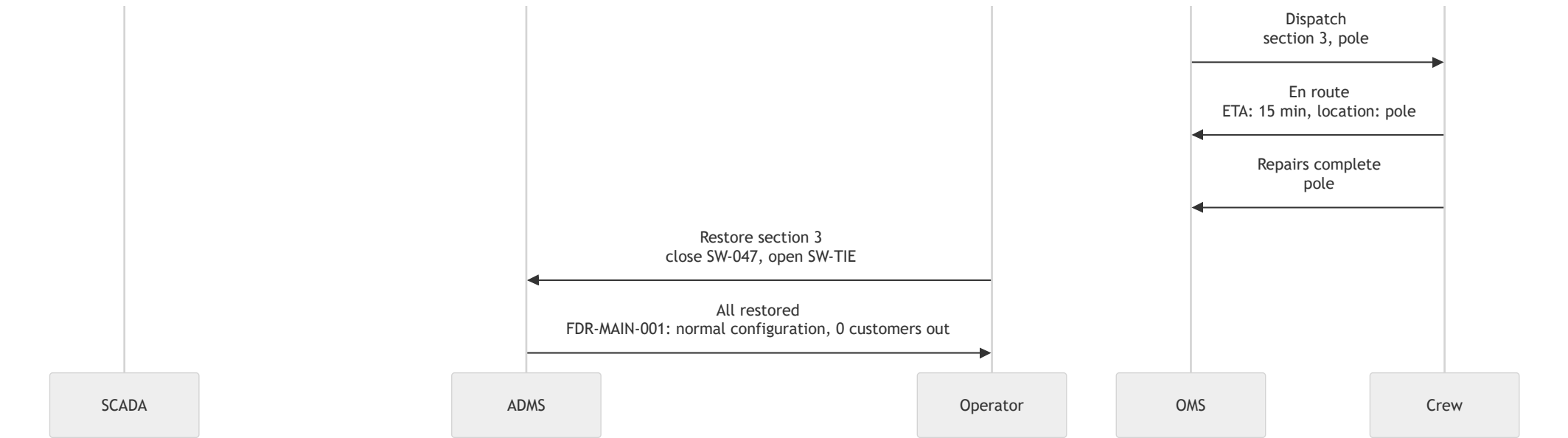
Description: ADMS detects distribution faults via SCADA alarms and AMI voltage complaints, locates the faulted segment using fault current indicators, and automatically isolates the fault while restoring service to unfaulted sections.

Actors: Distribution Operator, Outage Coordinator, Field Crew

Systems: ADMS (GE Grid Solutions), Fault Current Indicators, SCADA RTU, AMI Head-End, Outage Management System (OMS)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
FLISR	Fault	DetectFaultAlarm	SCADA RTU	ADMS (GE Grid Solutions)	RTU DNP3	ADMS DNP3	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
FLISR	Analysis	RunFLISRAnalysis	ADMS (GE Grid Solutions)	ADMS (GE Grid Solutions)	Internal	Internal	Batch (Real-time)	High	High-frequency near-real-time batch processing with minimal latency tolerance
FLISR	Isolate	OpenIsolationSwitch	ADMS (GE Grid Solutions)	SCADA RTU	ADMS DNP3	RTU DNP3	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
FLISR	Restore	CloseTieSwitch	ADMS (GE Grid Solutions)	SCADA RTU	ADMS DNP3	RTU DNP3	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
FLISR	OMS	CreateOutageEvent	ADMS (GE Grid Solutions)	Outage Management System	ADMS API (REST)	OMS API (REST)	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
FLISR	Dispatch	DispatchCrew	Outage Management System	Mobile Field Crew App	OMS API (REST)	Field App API (OAuth2)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
FLISR	Restore	RestoreNormalConfig	Operator HMI	ADMS (GE Grid Solutions)	HMI API (REST)	ADMS DNP3	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-FL-01	Validate fault detection and location within 30 seconds of alarm receipt	DetectFaultAlarm	FLISR identifies fault location within 30s of SCADA alarm	High
AC-FL-02	Confirm automatic isolation and restoration completes under 2 minutes	RunFLISRAnalysis	Unfaulted sections restored within 2 min of fault detection	High
AC-FL-03	Verify OMS outage record created automatically with fault details	CreateOutageEvent	OMS record created within 30s of FLISR completion	Medium
AC-FL-04	Ensure backfeed tie switch rating is not exceeded during restoration	CloseTieSwitch	Backfeed loading remains within 90% of tie switch rating	High

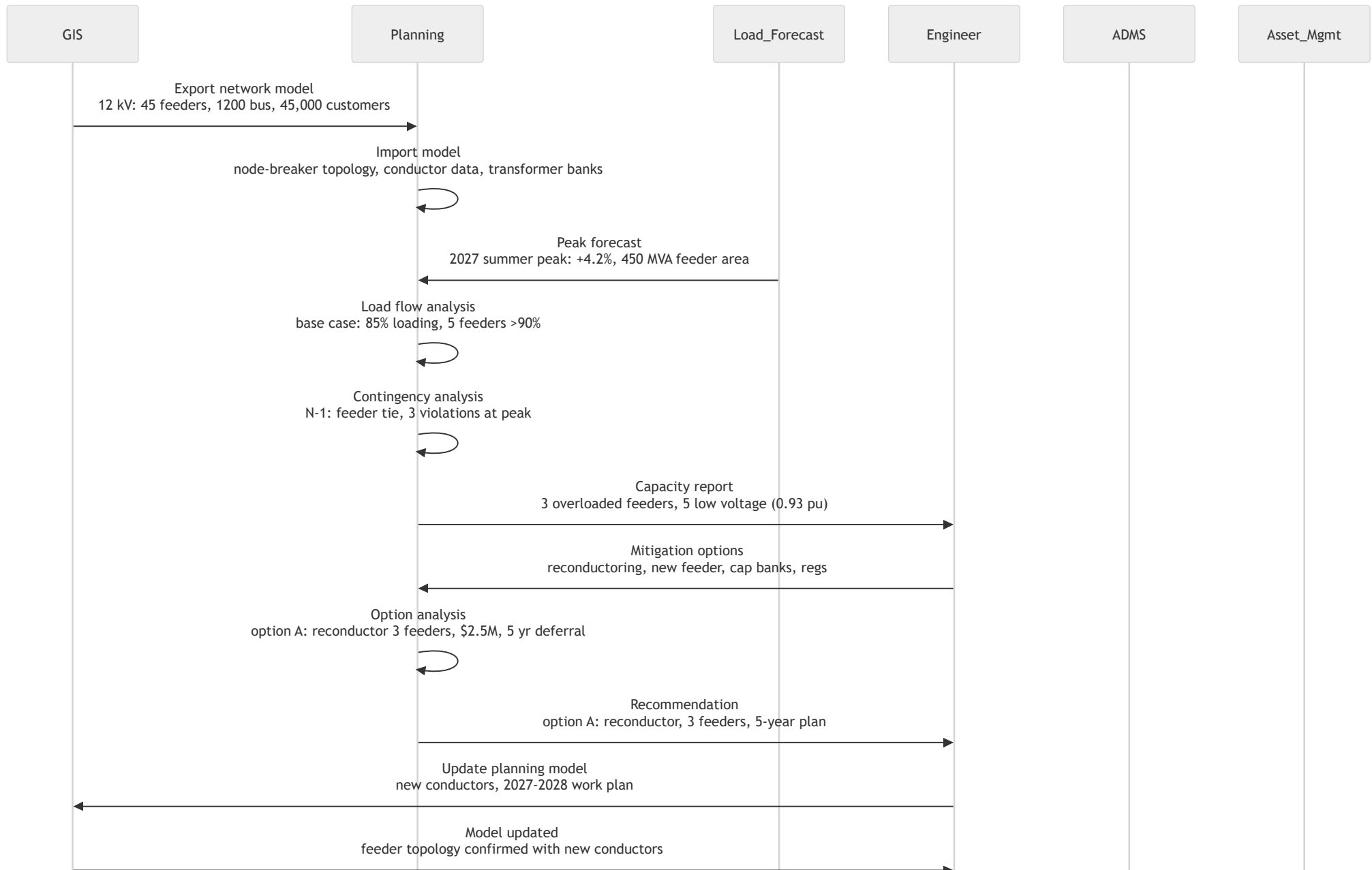
4.4 Distribution Planning & Network Optimization

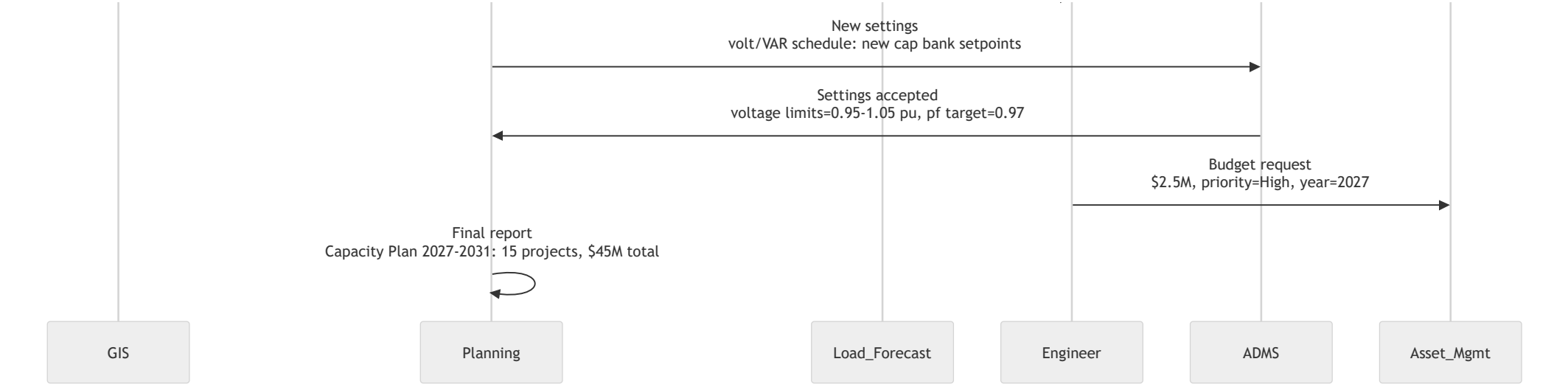
Description: Planning engineers analyze distribution network capacity, load growth, and voltage profiles to plan system upgrades. The optimization module performs volt/VAR control, loss minimization, and capacity forecasting.

Actors: Distribution Planning Engineer, System Planner, GIS Engineer

Systems: Distribution Planning System (CYME / Synergi), GIS (ArcGIS Utility Network), ADMS, Load Forecasting System, Asset Management System

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Planning	Model	ExportNetworkModel	GIS (ArcGIS Utility Network)	Distribution Planning System	GIS API (REST)	Planning API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
Planning	Forecast	ImportPeakForecast	Load Forecasting System	Distribution Planning System	Forecast API (REST)	Planning API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
Planning	Flow	RunLoadFlowAnalysis	Distribution Planning System	Distribution Planning System	Internal	Internal	Batch (Real-time)	Medium	Periodic near-real-time batch with data validation and transformation needs
Planning	Options	AnalyzeMitigationOptions	Distribution Planning System	Distribution Planning System	Internal	Internal	Batch (Real-time)	Medium	Periodic near-real-time batch with data validation and transformation needs
Planning	GIS	UpdatePlanningModel	Distribution Planning System	GIS (ArcGIS Utility Network)	Planning API (REST)	GIS API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints
Planning	ADMS	DeployVoltVARSettings	Distribution Planning System	ADMS (GE Grid Solutions)	Planning API (REST)	ADMS API (ICCP)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Planning	Budget	SubmitBudgetRequest	Planning Engineer	Asset Management System	Planning UI	Asset API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-DP-01	Validate load flow analysis matches field measurements within ±2% for all feeders	RunLoadFlowAnalysis	Load flow results within 2% of SCADA readings	High
AC-DP-02	Confirm planning model GIS topology completeness (100% of assets mapped)	ExportNetworkModel	All distribution assets in GIS are present in planning model	Medium
AC-DP-03	Verify mitigation options include at least feeder reconductor, cap bank, and regulator alternatives	AnalyzeMitigationOptions	At least 3 distinct mitigation options provided per violation	Medium
AC-DP-04	Ensure volt/VAR settings deployment does not cause temporary control disr	DeployVoltVARSettings	No ADMS alarms during settings deployment window	Medium

AC ID	Description	Related Info Flow	Expected Outcome	Priority
	ption			

Water Treatment Plant Operations (WTP) - UT-WTC UT-WQM

Monitor and control water treatment processes including raw water intake, chemical dosing, coagulation/flocculation, filtration, disinfection, and clearwell storage. SCADA and DCS systems manage plant operations with real-time quality monitoring.

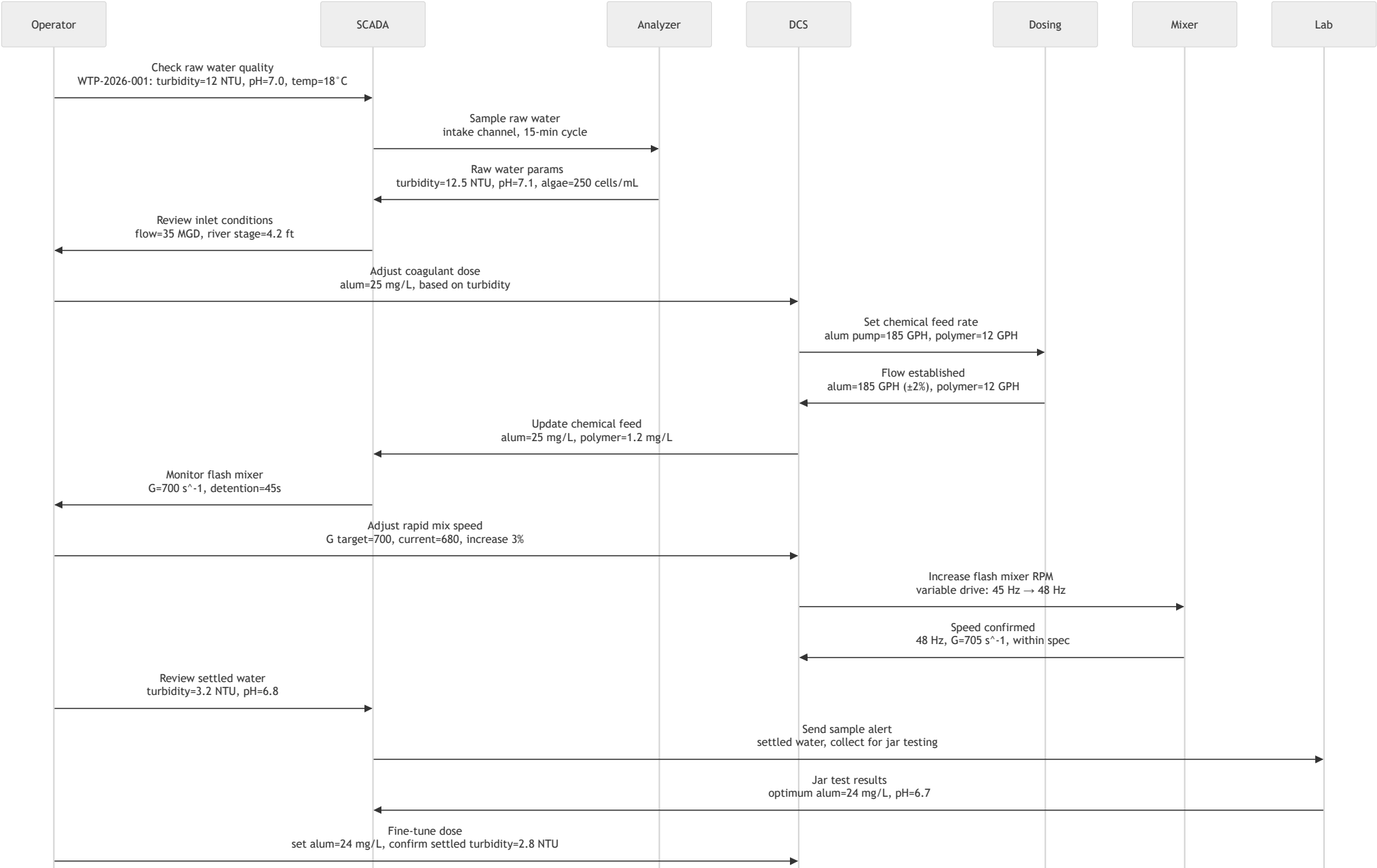
5.1 Raw Water Intake & Preliminary Treatment

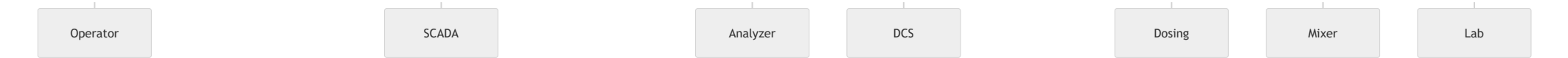
Description: Operators monitor raw water quality from the source reservoir, control intake gates and screens, manage chemical pre-treatment dosing, and oversee the settling basin operations.

Actors: Treatment Plant Operator, Plant Manager, Lab Technician

Systems: SCADA (Schneider Electric EcoStruxure), DCS (Rockwell PlantPAx), Water Quality Analyzers, Chemical Dosing System

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Water Treatment	Raw Water	SampleRawWater	Water Quality Analyzer	SCADA (EcoStruxure)	Analyzer Modbus RTU	SCADA Modbus TCP	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Water Treatment	Dosing	SetChemicalDose	DCS (Rockwell PlantPax)	Chemical Dosing System	DCS OPC-UA	Dosing Pump PLC	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Water Treatment	Mixing	AdjustFlashMixerSpeed	DCS (Rockwell PlantPax)	Flash Mixer VFD	DCS OPC-UA	VFD Modbus RTU	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Water Treatment	Settled	MonitorSettledWater	SCADA (EcoStruxure)	DCS (Rockwell PlantPax)	SCADA OPC-DA	DCS OPC-UA	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Water Treatment	Lab	AnalyzeJarTest	Lab Information System	SCADA (EcoStruxure)	LIS API (REST)	SCADA API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Water Treatment	Chemical	LogChemicalUsage	DCS (Rockwell PlantPax)	Plant Historian (OSIsoft PI)	DCS OPC-UA	PI API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints
Water Treatment	Compliance	RecordRawWaterQuality	SCADA (EcoStruxure)	Compliance Database	SCADA API (REST)	SQL JDBC	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-WT-01	Verify coagulant dose optimization achieves settled turbidity <5 NTU	SetChemicalDose	Settled turbidity below 5 NTU within 30 min of dose change	High
AC-WT-02	Validate raw water quality sample frequency meets regulatory minimum (every 15 min)	SampleRawWater	Analyzer logs show 15-min sampling interval maintained	High
AC-WT-03	Confirm flash mixer G value within ±10% of design target	AdjustFlashMixerSpeed	G value between 630-770 s^-1	Medium
AC-WT-04	Ensure chemical feed rate accuracy within ±5% of setpoint	SetChemicalDose	Flow meter verifies dosing within 5% of commanded rate	Medium

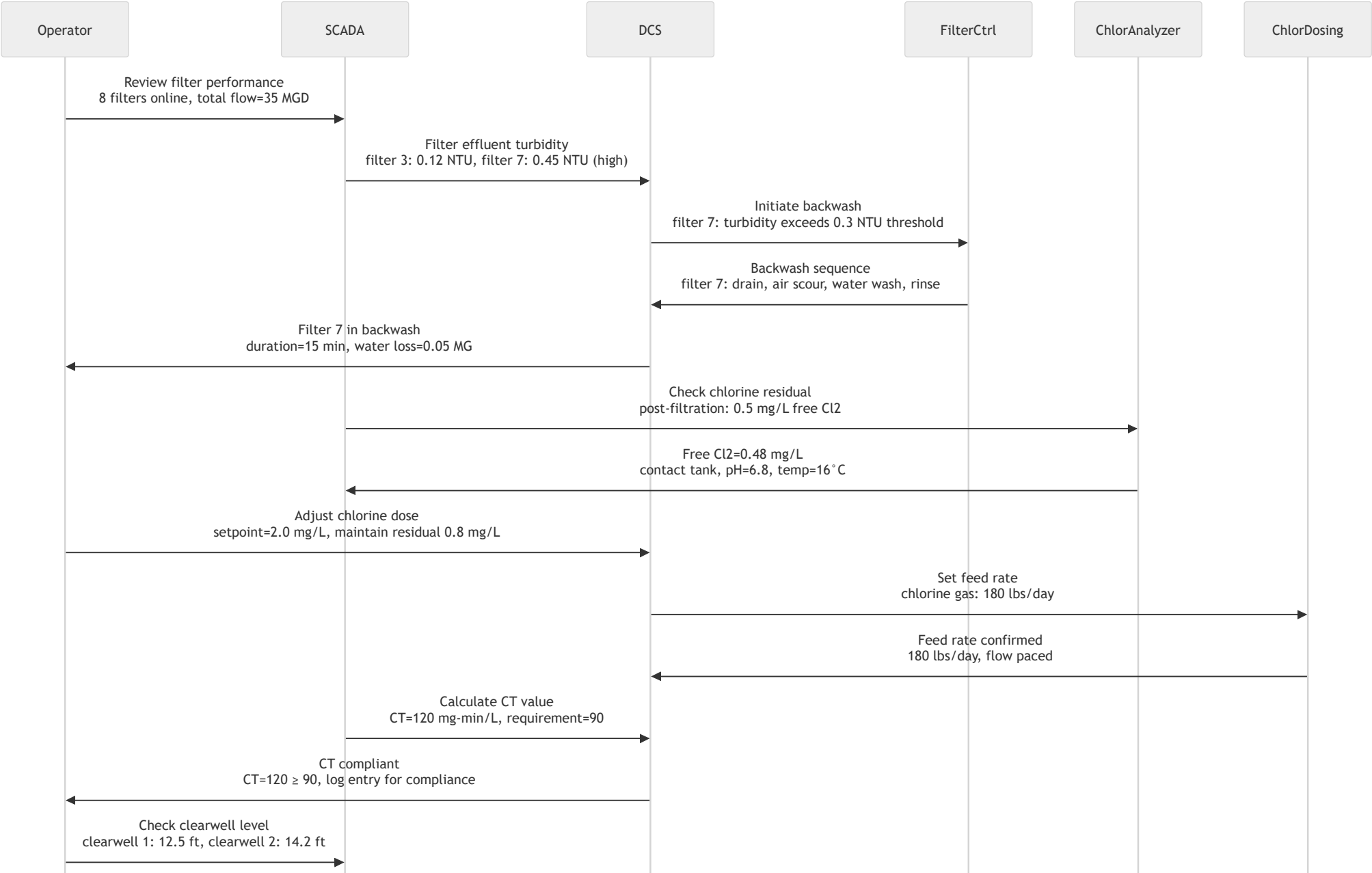
5.2 Filtration & Disinfection Control

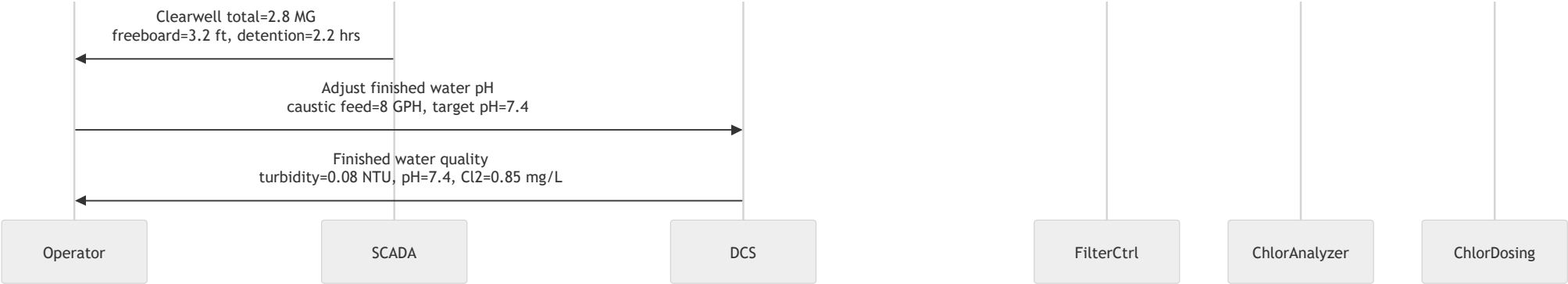
Description: Operators manage dual-media filtration process, backwash cycles, chlorination disinfection, and clearwell storage levels. CT (concentration × time) values are calculated for regulatory compliance.

Actors: Treatment Plant Operator, Lab Technician, Compliance Officer

Systems: DCS (Rockwell PlantPax), SCADA (EcoStruxure), Chlorine Analyzers, Filter Control System, Clearwell Level Sensors

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Filtration	Filter	MonitorFilterPerformance	SCADA (EcoStruxure)	DCS (Rockwell PlantPAx)	SCADA OPC-DA	DCS OPC-UA	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Filtration	Backwash	InitiateBackwashSequence	DCS (Rockwell PlantPAx)	Filter Control System	DCS OPC-UA	Filter PLC	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Filtration	Chlorine	AnalyzeChlorineResidual	Chlorine Analyzer	SCADA (EcoStruxure)	Analyzer 4-20mA	SCADA Modbus TCP	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Filtration	CT	CalculateCTValue	SCADA (EcoStruxure)	DCS (Rockwell PlantPAx)	SCADA API (REST)	DCS OPC-UA	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Filtration	Clearwell	MonitorClearwellLevel	Clearwell Level Sensor	SCADA (EcoStruxure)	Sensor 4-20mA	SCADA Modbus RTU	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Filtration	pH	AdjustFinishedWaterpH	DCS (Rockwell PlantPAx)	Chemical Dosing System	DCS OPC-UA	Dosing Pump PLC	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Filtration	Compliance	LogFinishedWaterQuality	SCADA (EcoStruxure)	Compliance Database	SCADA API (REST)	SQL JDBC	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-FL-01	Verify filter effluent turbidity <0.3 NTU for all operating filters	MonitorFilterPerformance	All filters below 0.3 NTU except those in backwash	High
AC-FL-02	Validate CT value meets regulatory minimum for Giardia inactivation (CT≥90)	CalculateCTValue	CT value meets or exceeds regulatory requirement	High
AC-FL-03	Confirm backwash sequence completes within 20 minutes without operator intervention	InitiateBackwashSequence	Backwash completes automatically in ≤20 min	Medium
AC-FL-04	Ensure clearwell detention time >30 minutes at peak flow	MonitorClearwellLevel	Clearwell volume provides ≥30 min detention at max flow	Medium

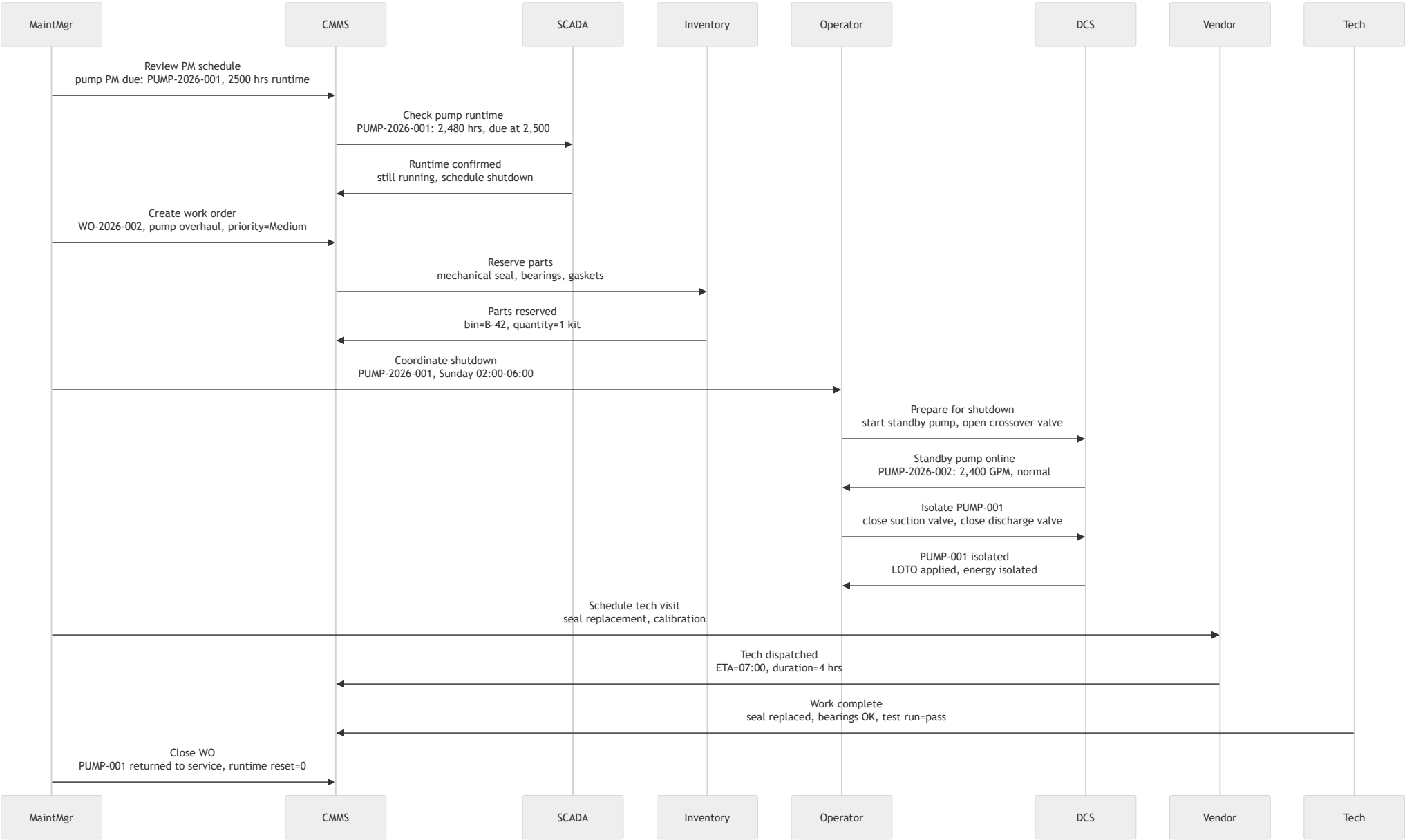
5.3 Plant Maintenance & CIP (Clean-in-Place)

Description: Maintenance team plans and executes preventive maintenance on plant equipment including pumps, valves, chemical systems, and instrumentation. Clean-in-Place (CIP) operations are managed for membrane systems.

Actors: Maintenance Manager, Plant Operator, Vendor Technician

Systems: CMMS (IBM Maximo), SCADA (EcoStruxure), DCS (Rockwell PlantPAx), Vendor Portal, Inventory System

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Maintenance	PM	CheckPMPriority	CMMS (IBM Maximo)	SCADA (EcoStruxure)	Maximo API (REST)	SCADA API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Maintenance	WO	CreateWorkOrder	CMMS (IBM Maximo)	CMMS (IBM Maximo)	Internal	Internal	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Maintenance	Parts	ReserveSpareParts	CMMS (IBM Maximo)	Inventory System	Maximo API (REST)	Inventory API (SOAP)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Maintenance	Shutdown	CoordinatePumpShutdown	CMMS (IBM Maximo)	DCS (Rockwell PlantPAx)	Maximo API (REST)	DCS OPC-UA	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
Maintenance	Isolation	IsolateEquipment	Operator HMI	DCS (Rockwell PlantPAx)	HMI API	DCS OPC-UA	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Maintenance	Vendor	DispatchTech	CMMS (IBM Maximo)	Vendor Portal	Maximo API (REST)	Vendor API (REST)	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
Maintenance	Close	CloseWorkOrder	Tech Mobile App	CMMS (IBM Maximo)	Mobile API (OAuth2)	Maximo API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-MT-01	Verify PM completion rate >90% of scheduled tasks within window	CheckPMPriority	At least 90% of PMs completed within scheduled period	Medium
AC-MT-02	Validate LOTO (Lock-Out/Tag-Out) compliance before equipment isolation	IsolateEquipment	Zero energy verified before maintenance begins	High
AC-MT-03	Confirm spare parts availability >95% for critical equipment	ReserveSpareParts	Critical parts in stock or on order with ETA	Medium
AC-MT-04	Ensure work order closure includes test run results and parts consumed	CloseWorkOrder	WO complete with test results and parts used	Medium

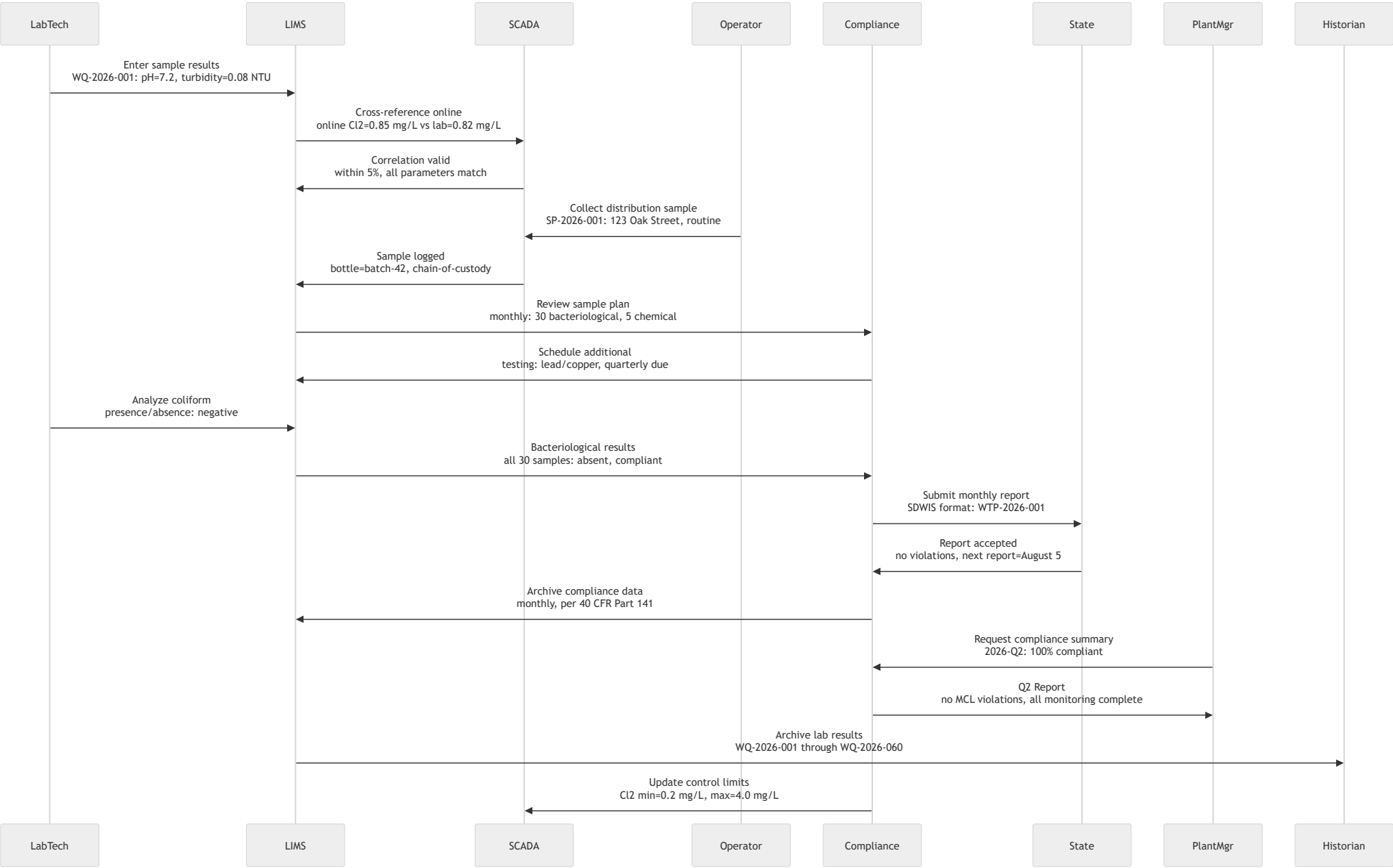
5.4 Water Quality Compliance & Reporting

Description: Laboratory team analyzes water samples throughout the treatment process and distribution system. Compliance reports are prepared for state and federal regulators covering SDWA (Safe Drinking Water Act) requirements.

Actors: Lab Manager, Compliance Officer, Plant Manager, State Regulator

Systems: LIMS (Lab Information System), SCADA (EcoStruxure), Compliance Reporting System, State Drinking Water Portal (SDWIS)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Water Quality	Sample	EnterLabResults	LIMS	SCADA (EcoStruxure)	LIMS API (REST)	SCADA API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Water Quality	Correlation	CrossReferenceOnline	SCADA (EcoStruxure)	LIMS	SCADA API (REST)	LIMS API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Water Quality	Sampling	CollectDistributionSample	Operator Mobile App	SCADA (EcoStruxure)	Mobile API (OAuth2)	SCADA API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Water Quality	Bacteriological	AnalyzeColiformSample	LIMS	LIMS	Internal	Internal	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Water Quality	Reporting	SubmitSDWISReport	Compliance System	State Drinking Water Portal (SDWIS)	Compliance API (REST)	SDWIS API (SOAP)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Water Quality	Archive	ArchiveComplianceData	Compliance System	LIMS	Compliance API (REST)	LIMS API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints
Water Quality	Limits	UpdateControlLimits	Compliance System	SCADA (EcoStruxure)	Compliance API (REST)	SCADA API (REST)	Event-driven	Simple	Simple event notification with fire-and-forget delivery semantics

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-WQ-01	Validate lab results within ±5% of online analyzer values for critical parameters	CrossReferenceOnline	Lab vs online correlation within 5% for all parameters	High
AC-WQ-02	Verify monthly bacteriological sample count meets regulatory minimum (30/month)	AnalyzeColiformSample	At least 30 bacteriological samples collected and analyzed monthly	High
AC-WQ-03	Confirm SDWIS report submission before regulatory deadline (10th of month)	SubmitSDWISReport	State acknowledges receipt before monthly deadline	High
AC-WQ-04>	Ensure all MCL violations are reported to state within 24 hours	SubmitSDWISReport	No unreported MCL violations	High

Water Distribution Network Management - UT-WDM UT-WQM

Management of the potable water distribution system including pressure zone management, reservoir tank level control, leak detection, flow monitoring, and hydrant/valve operations for the water utility network.

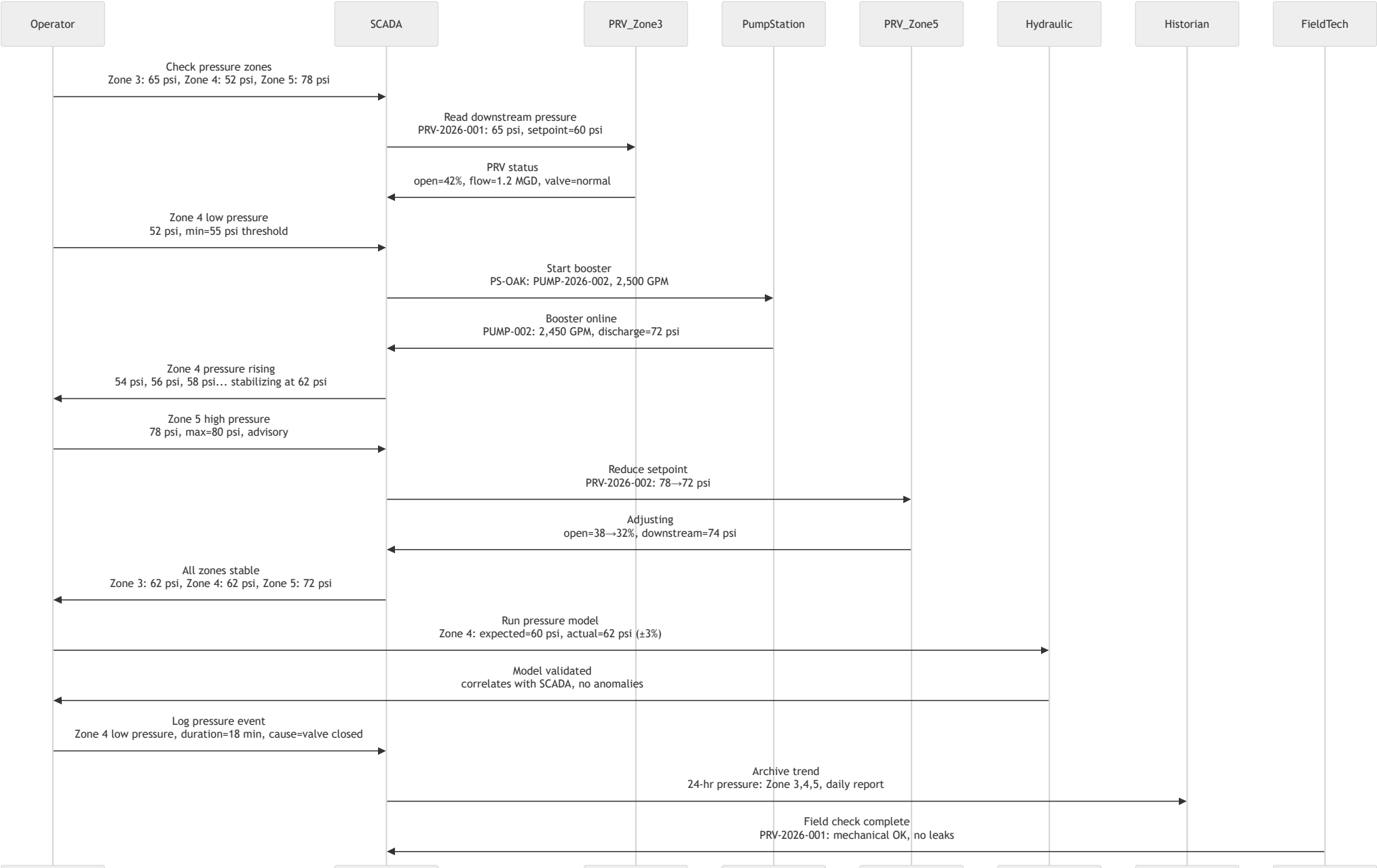
6.1 Pressure Zone Management & SCADA

Description: Operators monitor and control pressure zones across the distribution network. Pressure reducing valves (PRVs), booster pumps, and altitude valves are managed to maintain optimal pressure ranges throughout the system.

Actors: Distribution Operator, Field Technician, Hydraulic Engineer

Systems: SCADA (Schneider EcoStruxure), Pressure Monitoring System, PRV Controllers, Pump Station PLCs, Hydraulic Model

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Pressure	PRV	MonitorPRVDownstream	SCADA (EcoStruxure)	PRV Controller (Zone 3)	SCADA Modbus RTU	PRV Modbus RTU	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Pressure	Booster	StartBoosterPump	SCADA (EcoStruxure)	Pump Station PLC	SCADA Modbus TCP	PLC Modbus RTU	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Pressure	Monitoring	MonitorZonePressure	SCADA (EcoStruxure)	Operator HMI	SCADA API (REST)	HMI Display	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Pressure	PRV	AdjustPRVSetpoint	SCADA (EcoStruxure)	PRV Controller (Zone 5)	SCADA Modbus RTU	PRV Modbus RTU	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Pressure	Model	ValidateHydraulicModel	Hydraulic Model Software	SCADA (EcoStruxure)	Model API (REST)	SCADA API (REST)	Batch (Real-time)	Medium	Periodic near-real-time batch with data validation and transformation needs
Pressure	Log	LogPressureEvent	SCADA (EcoStruxure)	Plant Historian (OSIsoft PI)	SCADA API (REST)	PI API (REST)	Event-driven	Simple	Simple event notification with fire-and-forget delivery semantics
Pressure	Field	ConfirmFieldCheck	Field Tech Mobile App	SCADA (EcoStruxure)	Mobile API (OAuth2)	SCADA API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-PZ-01	Verify all pressure zones maintain pressure between 40-80 psi at critical nodes	MonitorZonePressure	All critical nodes within 40-80 psi range 99.5% of time	High
AC-PZ-02	Validate pressure recovery time under 30 minutes after low-pressure event	StartBoosterPump	Pressure returns to >55 psi within 30 min	Medium
AC-PZ-03	Confirm hydraulic model accuracy within ±5% of actual SCADA readings	ValidateHydraulicModel	Model vs actual pressure difference <5%	Medium
AC-PZ-04	Ensure PRV setpoint changes are logged with operator ID and reason	AdjustPRVSetpoint	Complete audit trail for every PRV adjustment	Medium

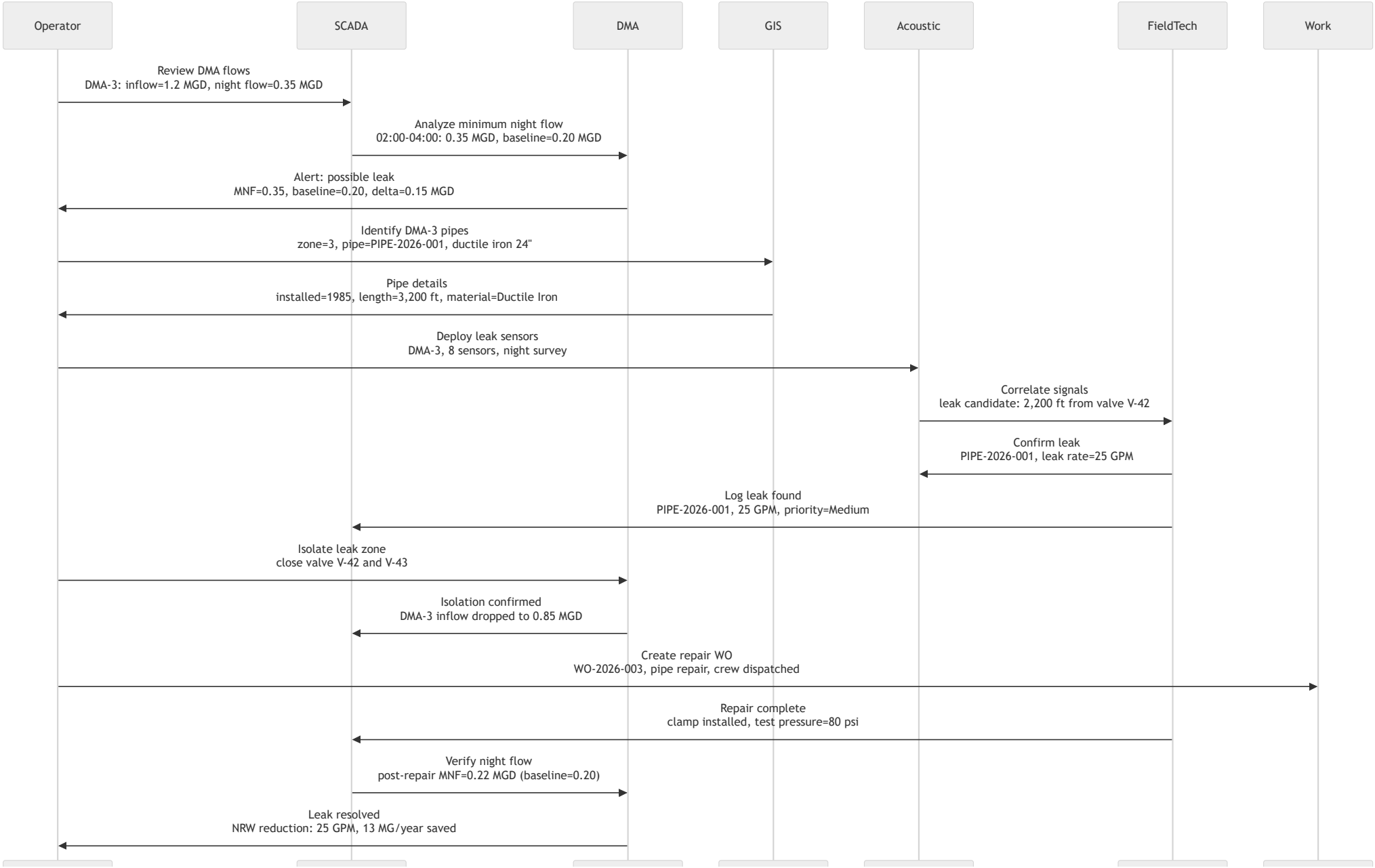
6.2 Leak Detection & Flow Monitoring

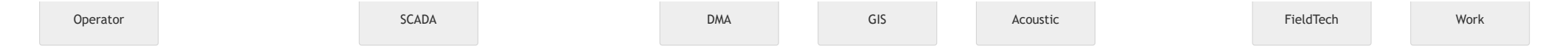
Description: The utility monitors distribution system flows to detect leaks, manage non-revenue water (NRW), and optimize system operations. District Metered Areas (DMAs) are analyzed for minimum night flow to identify potential leaks.

Actors: Water Loss Engineer, Distribution Operator, Field Technician

Systems: SCADA (EcoStruxure), DMA Flow Analyzer, Acoustic Leak Detection, GIS (ArcGIS), Hydraulic Model

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Leak	DMA	AnalyzeMinimumNightFlow	SCADA (EcoStruxure)	DMA Flow Analyzer	SCADA API (REST)	DMA API (REST)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Leak	Alert	FlagLeakCandidate	DMA Flow Analyzer	Operator Console	DMA API (REST)	SCADA HMI	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
Leak	GIS	IdentifyPipeAsset	GIS (ArcGIS)	Operator Console	GIS API (REST)	SCADA HMI	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Leak	Acoustic	DeployLeakSensors	Acoustic Leak Detection System	Field Tech Mobile App	Acoustic API (REST)	Mobile API (OAuth2)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Leak	Isolation	IsolateLeakZone	SCADA (EcoStruxure)	Valve Controllers	SCADA Modbus RTU	Valve PLC Modbus	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Leak	Repair	CreateRepairWorkOrder	Operator Console	CMMS (IBM Maximo)	SCADA API (REST)	Maximo API (REST)	Event-driven	Simple	Simple event notification with fire-and-forget delivery semantics
Leak	NRW	CalculateNRWReduction	DMA Flow Analyzer	Water Loss System	DMA API (REST)	Water Loss API	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-LK-01	Detect leaks > 10 GPM within 48 hours of occurrence	AnalyzeMinimumNightFlow	DMA alerts for leak candidates within 48 hrs	High
AC-LK-02	Validate minimum night flow is <25% of average daily flow for each DMA	FlagLeakCandidate	MNF/Average Daily Flow ratio <0.25	Medium
AC-LK-03	Confirm leak location accuracy within 10 ft of actual pipe	DeployLeakSensors	Correlation pinpoints leak within 10 ft	Medium
AC-LK-04	Ensure NRW is below 15% of system input volume	CalculateNRWReduction	Annual NRW below 15% of total production	High

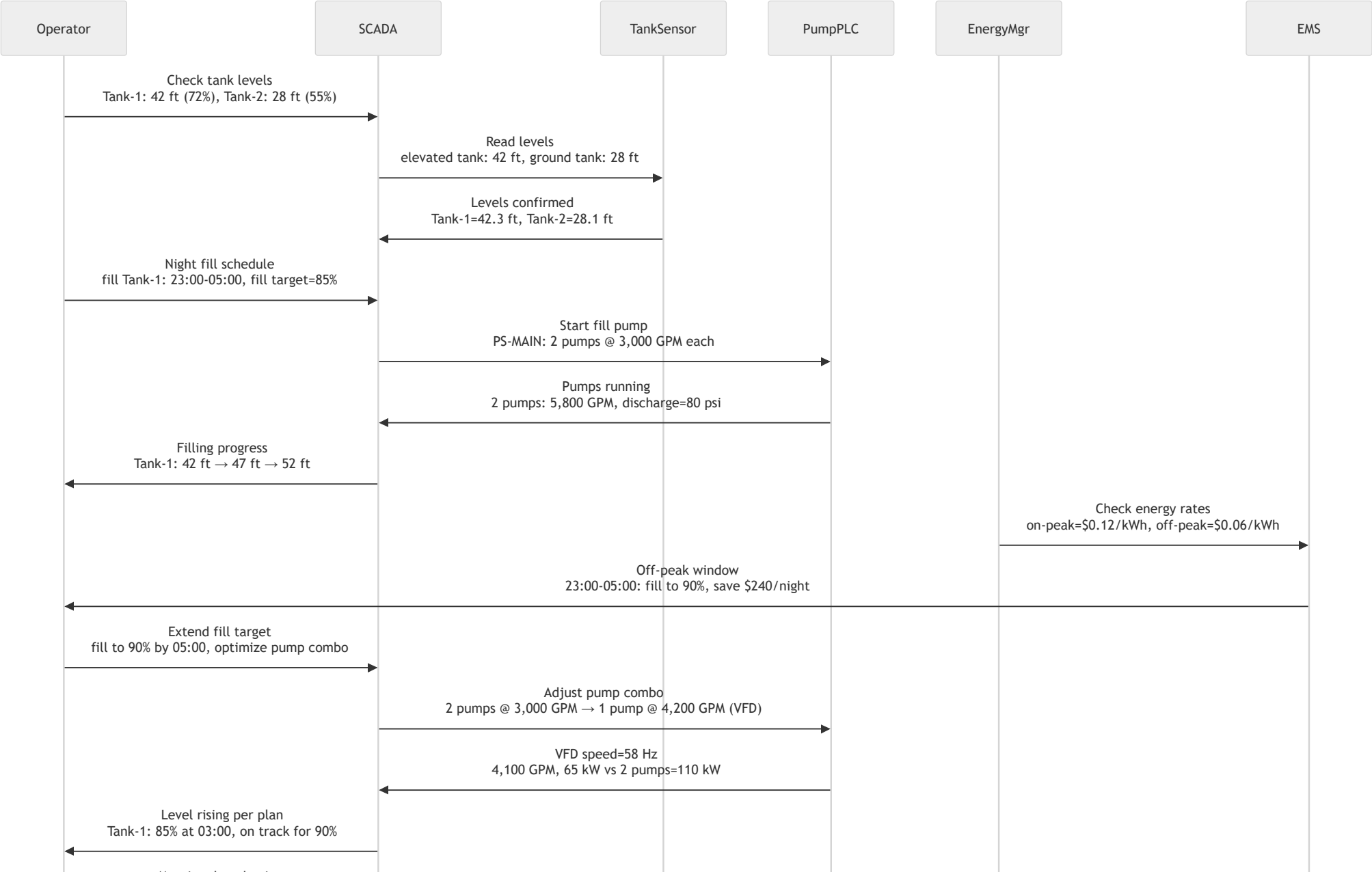
6.3 Reservoir & Tank Level Control

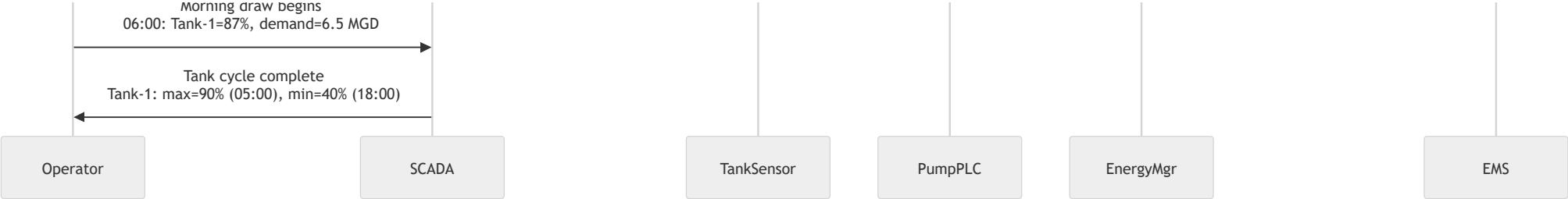
Description: Operators manage elevated storage tanks and ground reservoirs to maintain water levels, ensure fire flow capacity, and optimize pumping schedules for energy efficiency.

Actors: Distribution Operator, Energy Manager, Pump Station Technician

Systems: SCADA (EcoStruxure), Tank Level Sensors, Pump Station PLCs, Energy Management System (EMS), Hydraulic Model

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Tank	Level	ReadTankLevel	Tank Level Sensor	SCADA (EcoStruxure)	Sensor 4-20mA	SCADA Modbus RTU	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Tank	Fill	StartNightFillPump	SCADA (EcoStruxure)	Pump Station PLC	SCADA Modbus TCP	PLC Modbus RTU	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Tank	Energy	CheckOffPeakRates	Energy Management System	SCADA (EcoStruxure)	EMS API (REST)	SCADA API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Tank	Optimization	AdjustPumpCombo	SCADA (EcoStruxure)	Pump VFD	SCADA Modbus TCP	VFD Modbus RTU	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Tank	Schedule	MonitorFillProgress	SCADA (EcoStruxure)	Operator HMI	SCADA API (REST)	HMI Display	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Tank	Draw	MonitorMorningDraw	SCADA (EcoStruxure)	Plant Historian (OSIsoft PI)	SCADA API (REST)	PI API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Tank	Optimization	CalculateEnergySavings	Energy Management System	Plant Historian (OSIsoft PI)	EMS API (REST)	PI API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-TK-01	Verify tank levels return to target fill level by end of off-peak window (05:00)	StartNightFillPump	Tank reaches ≥90% by 05:00	Medium
AC-TK-02	Validate minimum storage capacity for fire flow (per NFPA 22)	ReadTankLevel	Tank never drops below fire storage reserve level	High
AC-TK-03	Confirm energy cost savings ≥15% vs day-fill schedule	CalculateEnergySavings	Night-fill reduces pumping energy cost by ≥15%	Medium
AC-TK-04	Ensure pump cycling frequency stays below 6 starts/hour	AdjustPumpCombo	Pump starts ≤6 per hour to prevent motor damage	Medium

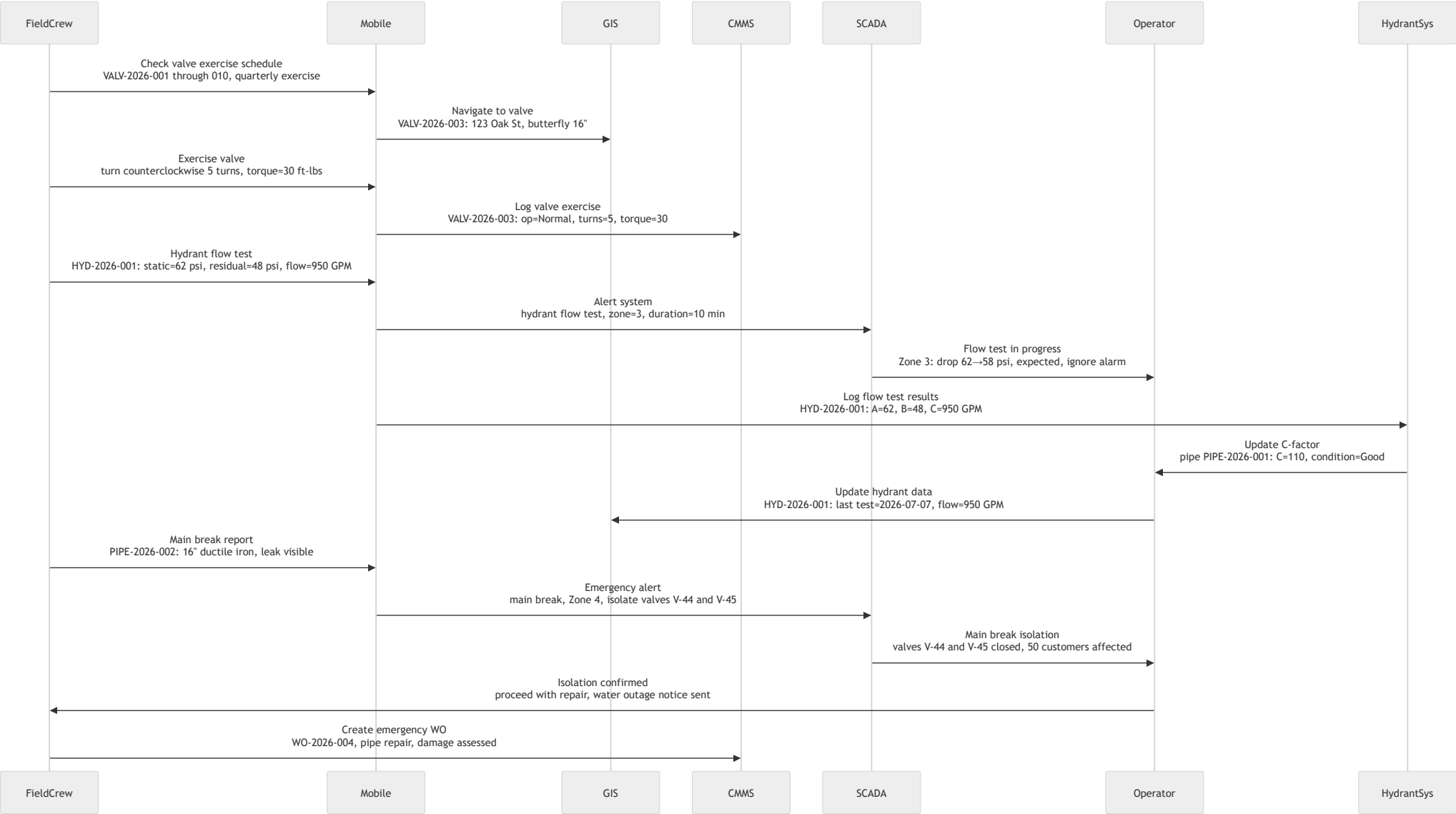
6.4 Hydrant & Valve Operations

Description: Field crews perform hydrant flushing and flow testing, valve exercising, and main breaks response. GIS tracks asset location and condition. SCADA monitors system impacts during operations.

Actors: Field Crew Lead, Distribution Operator, Hydrant Tester

Systems: GIS (ArcGIS), Mobile Field App, SCADA (EcoStruxure), CMMS (IBM Maximo), Hydrant Management System

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Hydrant	Valve	ExerciseValve	Field Crew Mobile App	GIS (ArcGIS)	Mobile API (OAuth2)	GIS API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Hydrant	Exercise	LogValveExercise	Field Crew Mobile App	CMMS (IBM Maximo)	Mobile API (OAuth2)	Maximo API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Hydrant	FlowTest	RecordHydrantFlowTest	Field Crew Mobile App	SCADA (EcoStruxure)	Mobile API (OAuth2)	SCADA API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Hydrant	CFactor	UpdatePipeCFactor	Hydrant Management System	GIS (ArcGIS)	Hydrant API (REST)	GIS API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Hydrant	MainBreak	AlertMainBreak	Field Crew Mobile App	SCADA (EcoStruxure)	Mobile API (OAuth2)	SCADA API (REST)	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
Hydrant	Isolation	IsolateMainBreak	SCADA (EcoStruxure)	Valve Controllers	SCADA Modbus RTU	Valve PLC Modbus	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Hydrant	WO	CreateEmergencyWorkOrder	Field Crew Mobile App	CMMS (IBM Maximo)	Mobile API (OAuth2)	Maximo API (REST)	Event-driven	Simple	Simple event notification with fire-and-forget delivery semantics

Acceptance Criteria

d>AC-HV-04

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-HV-01	Complete valve exercising within annual schedule (all valves every 12 months)	ExerciseValve	100% of valves exercised within 12-month cycle	Medium
AC-HV-02	Verify hydrant flow test data accuracy within ±5% (flow) and ±2 psi (pressure)	RecordHydrantFlowTest	Calibrated pitot gauge and pressure gauge used	Medium
AC-HV-03	Confirm main break isolation time under 45 minutes from notification	AlertMainBreak	Valves closed within 45 min of break notification	High
Ensure affected customers notified within 30 minutes of isolation	IsolateMainBreak	Customer notification sent ≤30 min after isolation	High	

Wastewater Collection & Treatment - UT-WWC UT-WQM

Management of wastewater collection system (lift stations, force mains, gravity sewers) and wastewater treatment processes (primary, secondary, tertiary treatment, and biosolids management) with regulatory effluent compliance.

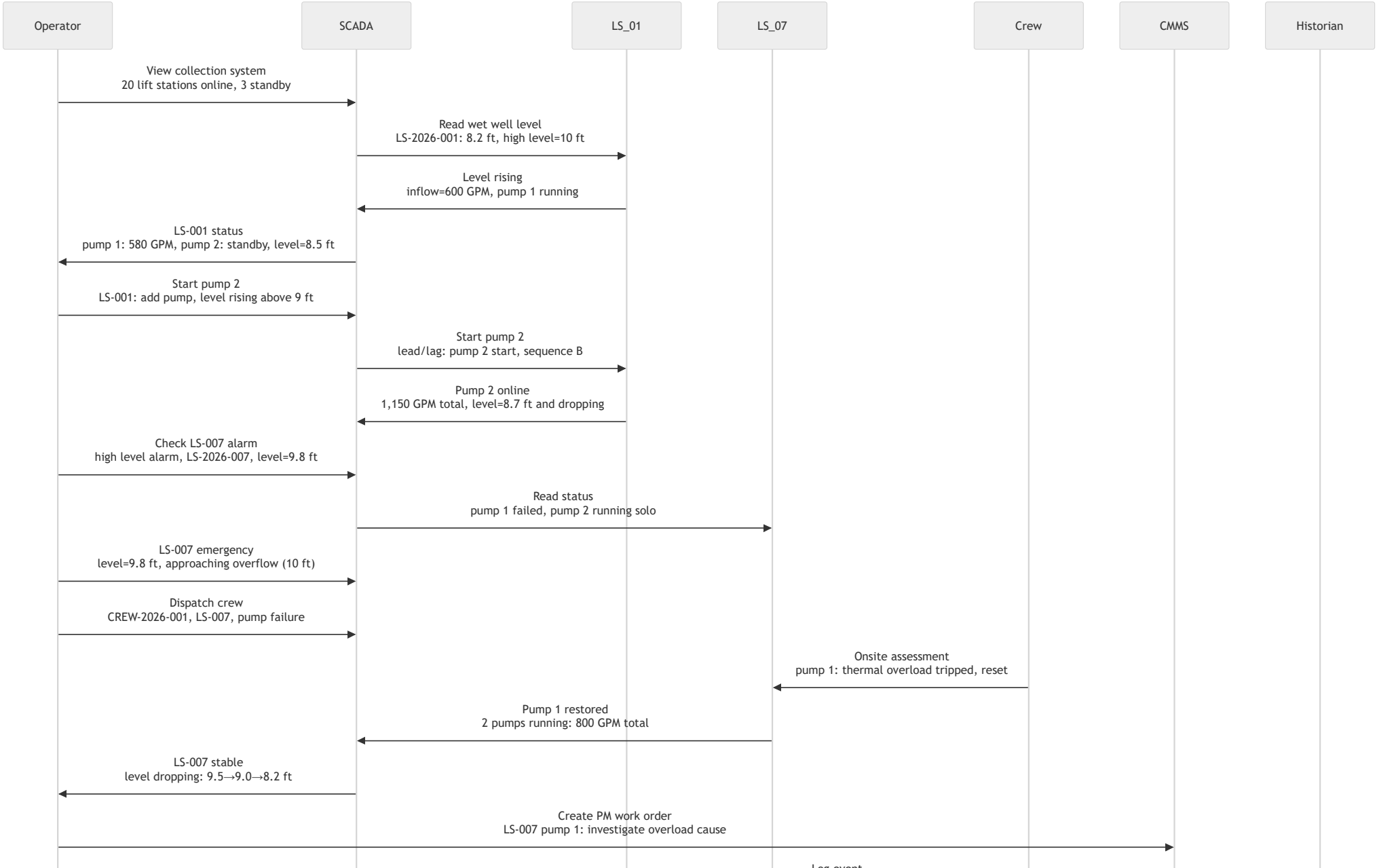
7.1 Collection System SCADA (Lift Stations)

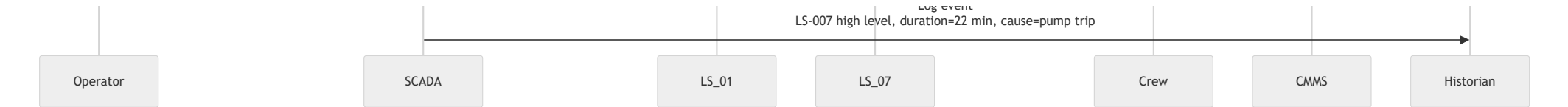
Description: Operators monitor and control wastewater lift stations across the collection network. SCADA system tracks wet well levels, pump status, flow rates, and alarms for overflow prevention.

Actors: Collection System Operator, Field Technician, Maintenance Supervisor

Systems: SCADA (Schneider EcoStruxure), Lift Station PLCs, Level Sensors, Flow Meters, CMMS (IBM Maximo)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
LiftStation	Level	ReadWetWellLevel	Level Sensor (LS-001)	Lift Station PLC (LS-001)	Sensor 4-20mA	PLC Analog Input	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
LiftStation	Pump	StartLiftStationPump	SCADA (EcoStruxure)	Lift Station PLC (LS-001)	SCADA Modbus TCP	PLC Modbus RTU	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
LiftStation	Alarm	MonitorHighLevelAlarm	Lift Station PLC (LS-007)	SCADA (EcoStruxure)	PLC Modbus RTU	SCADA Modbus TCP	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
LiftStation	Dispatch	DispatchCrew	SCADA (EcoStruxure)	CMMS (IBM Maximo)	SCADA API (REST)	Maximo API (REST)	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
LiftStation	Field	OnsiteAssessment	Field Tech Mobile App	CMMS (IBM Maximo)	Mobile API (OAuth2)	Maximo API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
LiftStation	PM	CreatePmWorkOrder	SCADA (EcoStruxure)	CMMS (IBM Maximo)	SCADA API (REST)	Maximo API (REST)	Event-driven	Simple	Simple event notification with fire-and-forget delivery semantics
LiftStation	Log	LogAlarmEvent	SCADA (EcoStruxure)	Plant Historian (OSIsoft PI)	SCADA API (REST)	PI API (REST)	Event-driven	Simple	Simple event notification with fire-and-forget delivery semantics

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-LS-01	Verify all lift stations maintain wet well level below 90% of high-level setpoint	ReadWetWellLevel	No lift station exceeds high-level alarm threshold	High
AC-LS-02	Validate pump run-time alternation to ensure equal wear across duty/standby pumps	StartLiftStationPump	Lead/lag alternation schedule maintained	Medium
AC-LS-03	Confirm overflow prevention: no sanitary sewer overflows (SSOs) from lift stations	MonitorHighLevelAlarm	Zero SSO events from monitored lift stations	High
AC-LS-04	Ensure crew dispatch and arrival on site within 60 minutes for high-priority alarms	DispatchCrew	Crew on-site within 60 min of alarm	Medium

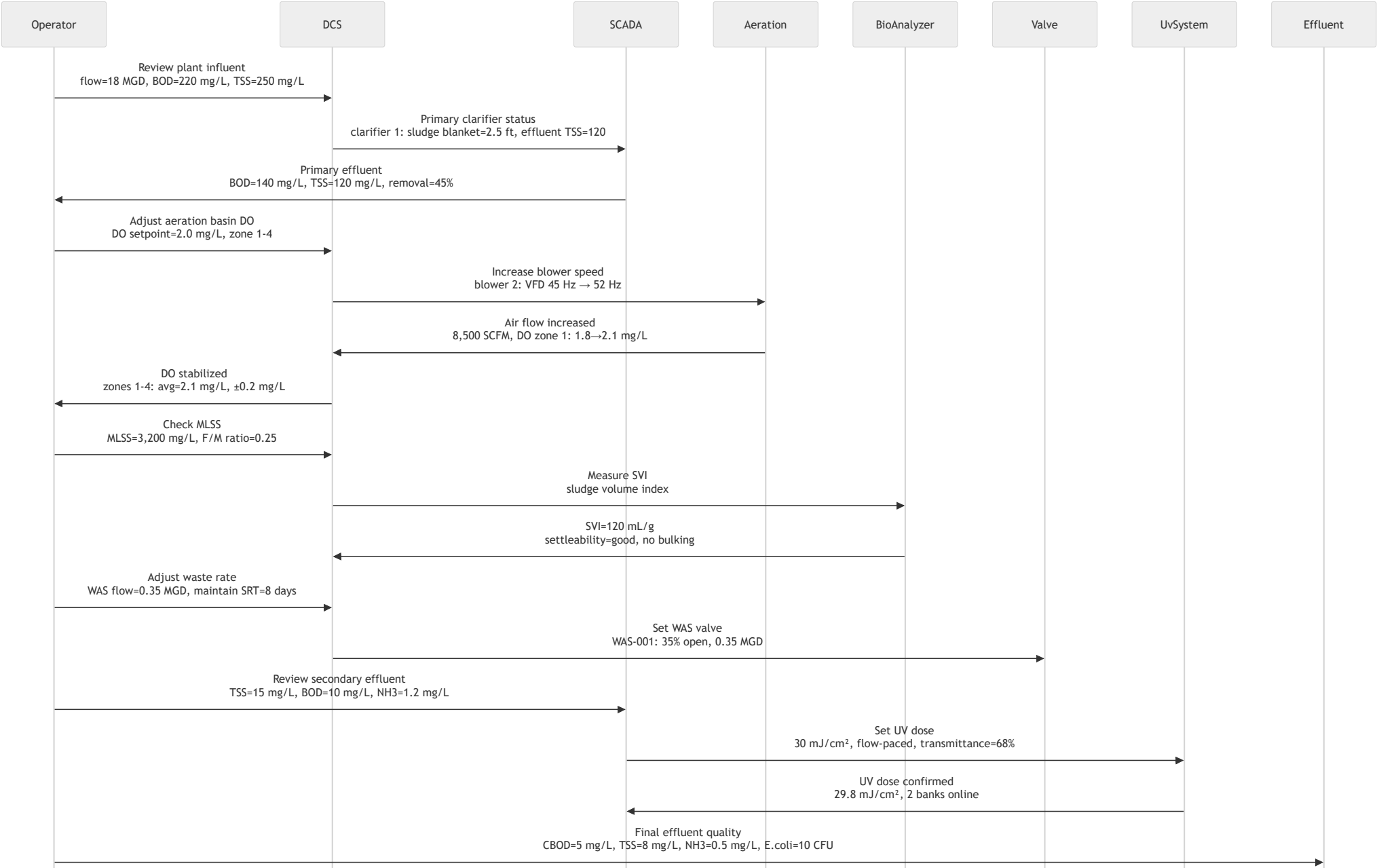
7.2 Wastewater Treatment Process Control

Description: Operators manage the wastewater treatment process including primary clarification, biological treatment (activated sludge), secondary clarification, and disinfection (UV/chlorine). Process control parameters are continuously monitored and adjusted.

Actors: WWTP Operator, Process Engineer, Lab Technician

Systems: DCS (Rockwell PlantPax), SCADA (EcoStruxure), Biological Process Analyzer, UV Disinfection System, Aeration Control System

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Treatment	Influent	MonitorPlantInfluent	SCADA (EcoStruxure)	DCS (Rockwell PlantPAx)	SCADA OPC-DA	DCS OPC-UA	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Treatment	DO	AdjustAerationDO	DCS (Rockwell PlantPAx)	Aeration Control System	DCS OPC-UA	Blower PLC Modbus	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Treatment	SVI	MeasureSludgeSettleability	Biological Process Analyzer	DCS (Rockwell PlantPAx)	Analyzer Modbus RTU	DCS OPC-UA	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Treatment	WAS	AdjustWasteActivatedSludge	DCS (Rockwell PlantPAx)	WAS Valve Actuator	DCS OPC-UA	Valve Modbus RTU	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Treatment	UV	SetUvDose	SCADA (EcoStruxure)	UV Disinfection System	SCADA Modbus TCP	UV System Modbus	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Treatment	Effluent	MonitorFinalEffluent	SCADA (EcoStruxure)	Effluent Analyzers	SCADA Modbus TCP	Analyzer 4-20mA	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Treatment	Compliance	LogEffluentQuality	DCS (Rockwell PlantPAx)	Plant Historian (OSIsoft PI)	DCS OPC-UA	PI API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-WW-01	Verify effluent BOD and TSS consistently below permit limits (30 mg/L each)	MonitorFinalEffluent	Daily average BOD <20 mg/L and TSS <20 mg/L	High
AC-WW-02	Validate DO control maintains setpoint within ±0.3 mg/L across all aeration zones	AdjustAerationDO	DO within ±0.3 mg/L of setpoint 95% of time	High
AC-WW-03	Confirm UV dose ≥30 mJ/cm² at peak flow for pathogen inactivation	SetUvDose	UV dose meets or exceeds 30 mJ/cm² at all flow conditions	High
AC-WW-04	Ensure SRT maintained within ±1 day of target for biological process stability	AdjustWasteActivatedSludge	SRT between 7-9 days for target of 8 days	Medium

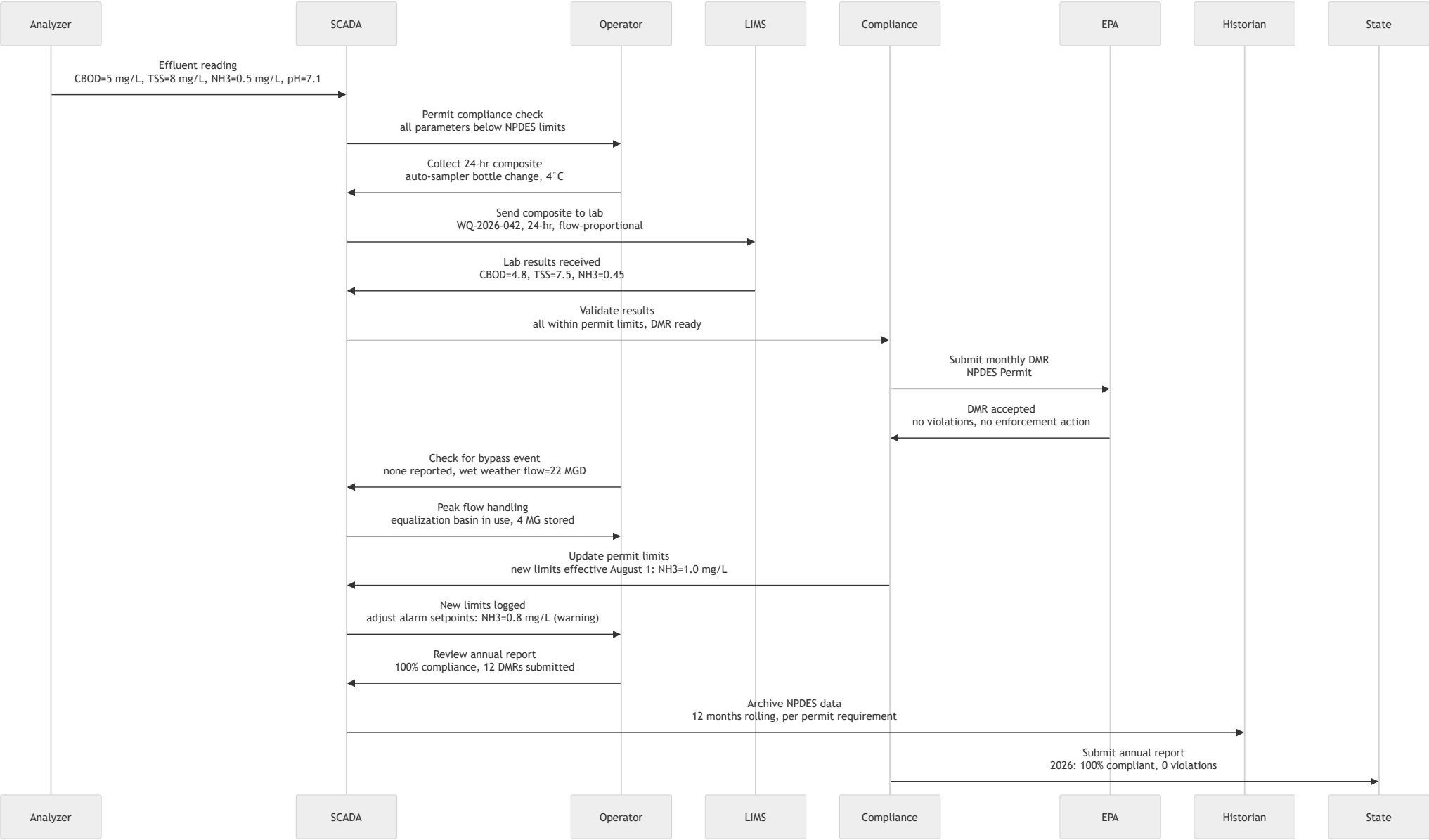
7.3 Effluent Quality Compliance

Description: Continuous monitoring of effluent quality parameters with real-time reporting to regulatory agencies. NPDES permit compliance is managed with automated sampling, data validation, and discharge reporting.

Actors: WWTP Operator, Compliance Manager, Lab Manager, State Regulator

Systems: SCADA (EcoStruxure), LIMS, Compliance Reporting System, EPA ICIS-NPDES Portal, Online Analyzers

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Effluent	Analyzer	ReadEffluentParameters	Online Analyzers	SCADA (EcoStruxure)	Analyzer Modbus RTU	SCADA Modbus TCP	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Effluent	Composite	CollectCompositeSample	Auto-Sampler	SCADA (EcoStruxure)	Sampler Modbus	SCADA Modbus TCP	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints
Effluent	Lab	AnalyzeLabComposite	LIMS	SCADA (EcoStruxure)	LIMS API (REST)	SCADA API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Effluent	DMR	SubmitMonthlyDMR	Compliance System	EPA ICIS-NPDES Portal	Compliance API (REST)	EPA CDX (SOAP)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Effluent	Bypass	MonitorWetWeatherFlow	SCADA (EcoStruxure)	Equalization Basin System	SCADA Modbus TCP	EQ Basin PLC Modbus	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Effluent	Permit	UpdatePermitLimits	Compliance System	SCADA (EcoStruxure)	Compliance API (REST)	SCADA API (REST)	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
Effluent	Archive	ArchiveNpdesData	SCADA (EcoStruxure)	Plant Historian (OSIsoft PI)	SCADA API (REST)	PI API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-EF-01	Validate DMR submission before regulatory deadline (10th of month)	SubmitMonthlyDMR	DMR submitted and acknowledged by EPA before deadline	High
AC-EF-02	Verify online analyzer calibration within ±10% of lab results for critical parameters	AnalyzeLabComposite	Lab vs online correlation within 10%	Medium
AC-EF-03	Confirm no un-reported bypass events or NPDES excursions	MonitorWetWeatherFlow	All bypass events logged and reported per permit	High
AC-EF-04	Ensure data retention of NPDES records meets minimum 3-year regulatory requirement	ArchiveNpdesData	All NPDES data archived for ≥3 years	Medium

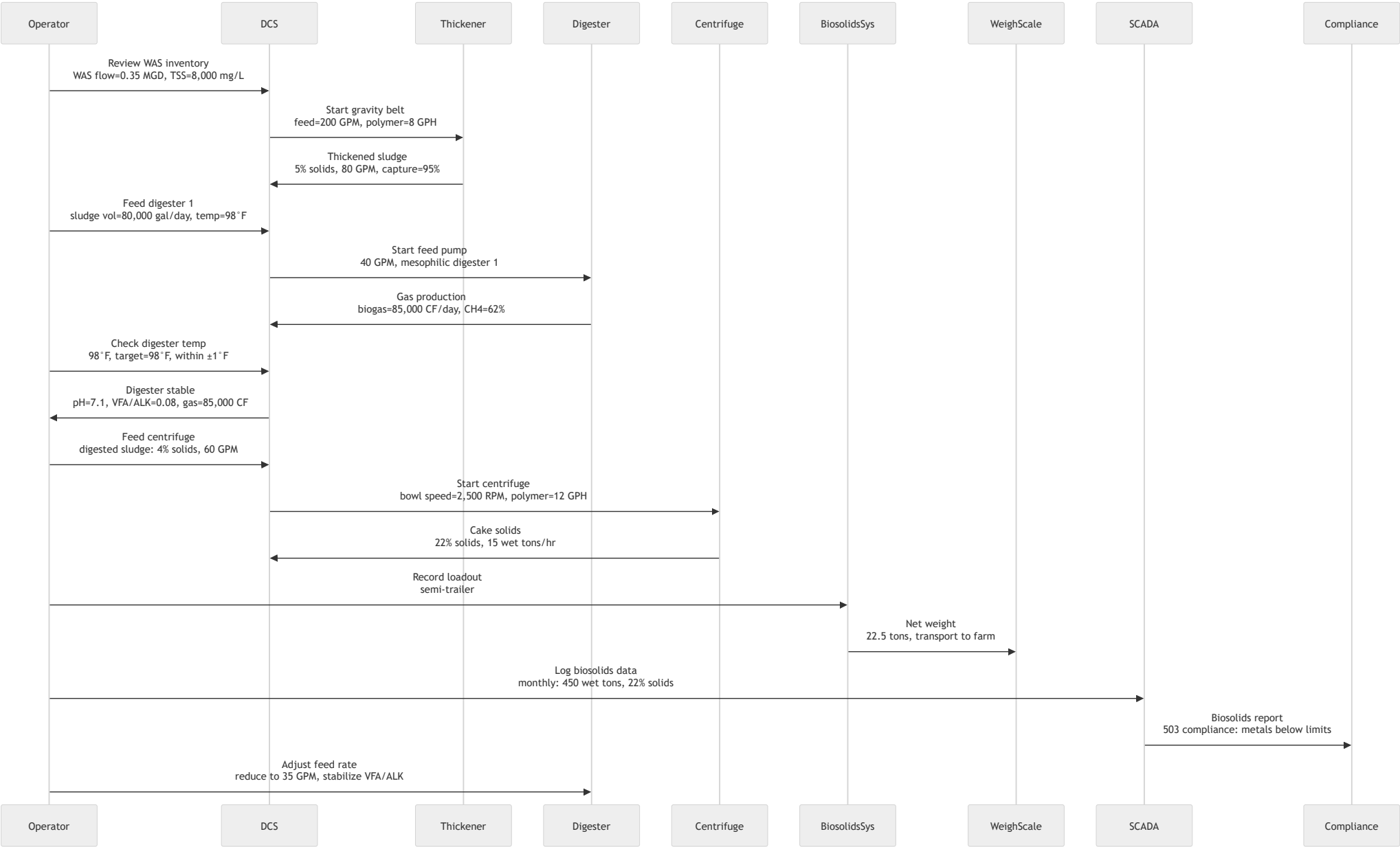
7.4 Biosolids Management

Description: Operators manage the biosolids treatment process including thickening, anaerobic digestion, dewatering (centrifuge/belt press), and beneficial reuse or disposal. Gas production from digesters is monitored for energy recovery.

Actors: WWTP Operator, Biosolids Coordinator, Hauler, Land Application Manager

Systems: DCS (Rockwell PlantPAX), SCADA (EcoStruxure), Digester Control System, Centrifuge PLC, Biosolids Tracking System, Weigh Scale

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Biosolids	Thickening	StartGravityBeltThickener	DCS (Rockwell PlantPAx)	Sludge Thickener PLC	DCS OPC-UA	Thickener PLC Modbus	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Biosolids	Digester	FeedAnaerobicDigester	DCS (Rockwell PlantPAx)	Digester Control System	DCS OPC-UA	Digester PLC Modbus	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Biosolids	Gas	MonitorBiogasProduction	Digester Control System	SCADA (EcoStruxure)	Digester Modbus	SCADA Modbus TCP	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Biosolids	Dewatering	StartCentrifuge	DCS (Rockwell PlantPAx)	Centrifuge PLC	DCS OPC-UA	Centrifuge Modbus RTU	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Biosolids	Loadout	RecordBiosolidsLoadout	Biosolids Tracking System	Weigh Scale System	Biosolids API (REST)	Scale API (Serial)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Biosolids	Compliance	ReportBiosolids503	SCADA (EcoStruxure)	Compliance System	SCADA API (REST)	Compliance API (REST)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Biosolids	Optimization	AdjustDigesterFeedRate	DCS (Rockwell PlantPAx)	Digester Control System	DCS OPC-UA	Digester PLC Modbus	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-BS-01	Verify digester temperature maintained at 98°F ±1°F for mesophilic operation	FeedAnaerobicDigester	Digester temp within 97-99°F range 99% of time	High
AC-BS-02	Validate biosolids pathogen reduction meets Class B requirements (Part 503)	ReportBiosolids503	Fecal coliform <2M MPN/g dry solids	High
AC-BS-03	Confirm centrifuge cake solids ≥20% for economical hauling	StartCentrifuge	Cake solids ≥20% averaged	Medium

AC ID	Description	Related Info Flow	Expected Outcome	Priority
			monthly	
AC-BS-04	Ensure trace metals in biosolids below 503 Table 3 ceiling concentrations	ReportBiosolids503	All metals below regulatory limits	High

Metering & Smart Grid - UT-AMI UT-MDM UT-SGO UT-GBD

Management of the advanced metering infrastructure (AMI) including head-end operations, meter data management (MDM), VEE processing, and Green Button data access for customer energy usage.

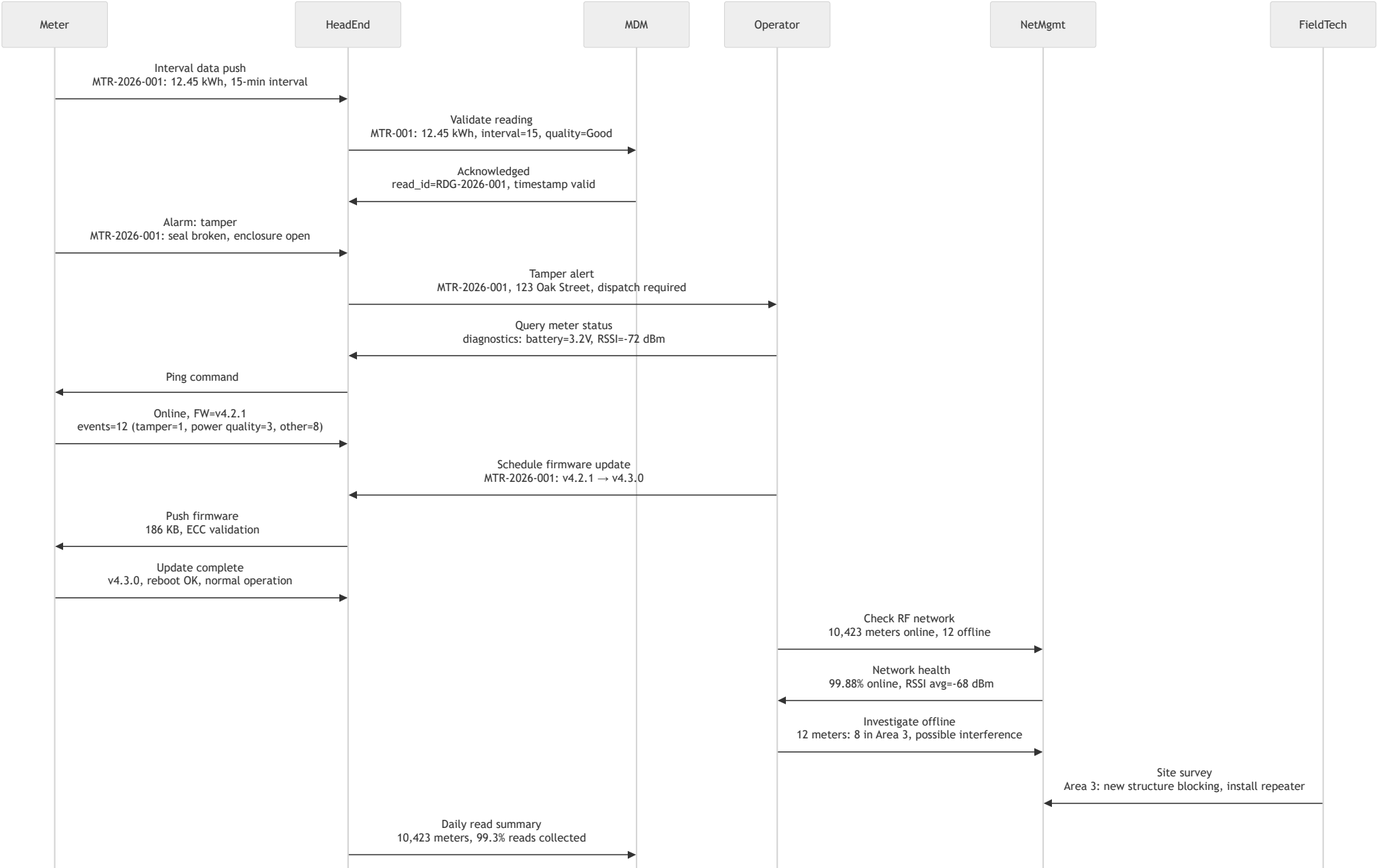
8.1 AMI Head-End Operations

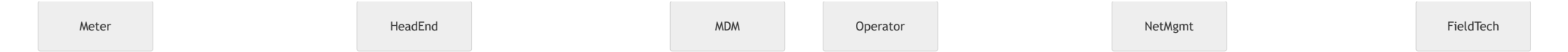
Description: AMI head-end system manages two-way communication with smart meters across the service territory. System collects interval reads, processes time-of-use data, manages firmware updates, and monitors network health.

Actors: Metering Specialist, Network Engineer, Field Technician

Systems: AMI Head-End (Itron OMS/IEE), Smart Meters (Itron OpenWay), Network Management System, MDM

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
AMI	Interval	CollectIntervalRead	Smart Meter (Itron OpenWay)	AMI Head-End (Itron OMS/IEE)	RF Mesh	Head-End WAN	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
AMI	VEE	ValidateMeterReading	AMI Head-End (Itron OMS/IEE)	MDM	Head-End API (SOAP)	MDM API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
AMI	Alarm	ProcessTamperAlarm	Smart Meter (Itron OpenWay)	AMI Head-End (Itron OMS/IEE)	RF Mesh	Head-End Event Bus	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
AMI	Diagnostics	QueryMeterDiagnostics	AMI Head-End (Itron OMS/IEE)	Smart Meter (Itron OpenWay)	Head-End API (SOAP)	RF Mesh	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
AMI	Firmware	PushFirmwareUpdate	AMI Head-End (Itron OMS/IEE)	Smart Meter (Itron OpenWay)	Head-End FW Server	RF Mesh (Over-the-Air)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
AMI	Network	MonitorNetworkHealth	Network Management System	Operator Console	NetMgmt API (SNMP)	SCADA HMI	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
AMI	Summary	GenerateDailyReadSummary	AMI Head-End (Itron OMS/IEE)	MDM	Head-End API (SOAP)	MDM API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-AM-01	Verify daily meter read success rate ≥99% of registered meters	GenerateDailyReadSummary	Daily read success rate at or above 99%	High
AC-AM-02	Validate tamper alerts are processed and dispatched within 2 hours	ProcessTamperAlarm	Tamper alert to dispatch ≤2 hours	Medium
AC-AM-03	Confirm firmware update completion rate ≥95% within scheduled window	PushFirmwareUpdate	95% of	Medium

AC ID	Description	Related Info Flow	Expected Outcome	Priority
			targeted meters updated successfully	
AC-AM-04	Ensure AMI network availability ≥99.5% (meters online/total registered)	MonitorNetworkHealth	Network availability meets 99.5% threshold	High

Customer Engagement & Demand Response - UT-CRM UT-DRP UT-OAD UT-DCE

Manages customer interactions, demand response programs, OpenADR communication, and DER customer enrollment. Includes customer portal, IVR, DR event management, and energy efficiency programs.

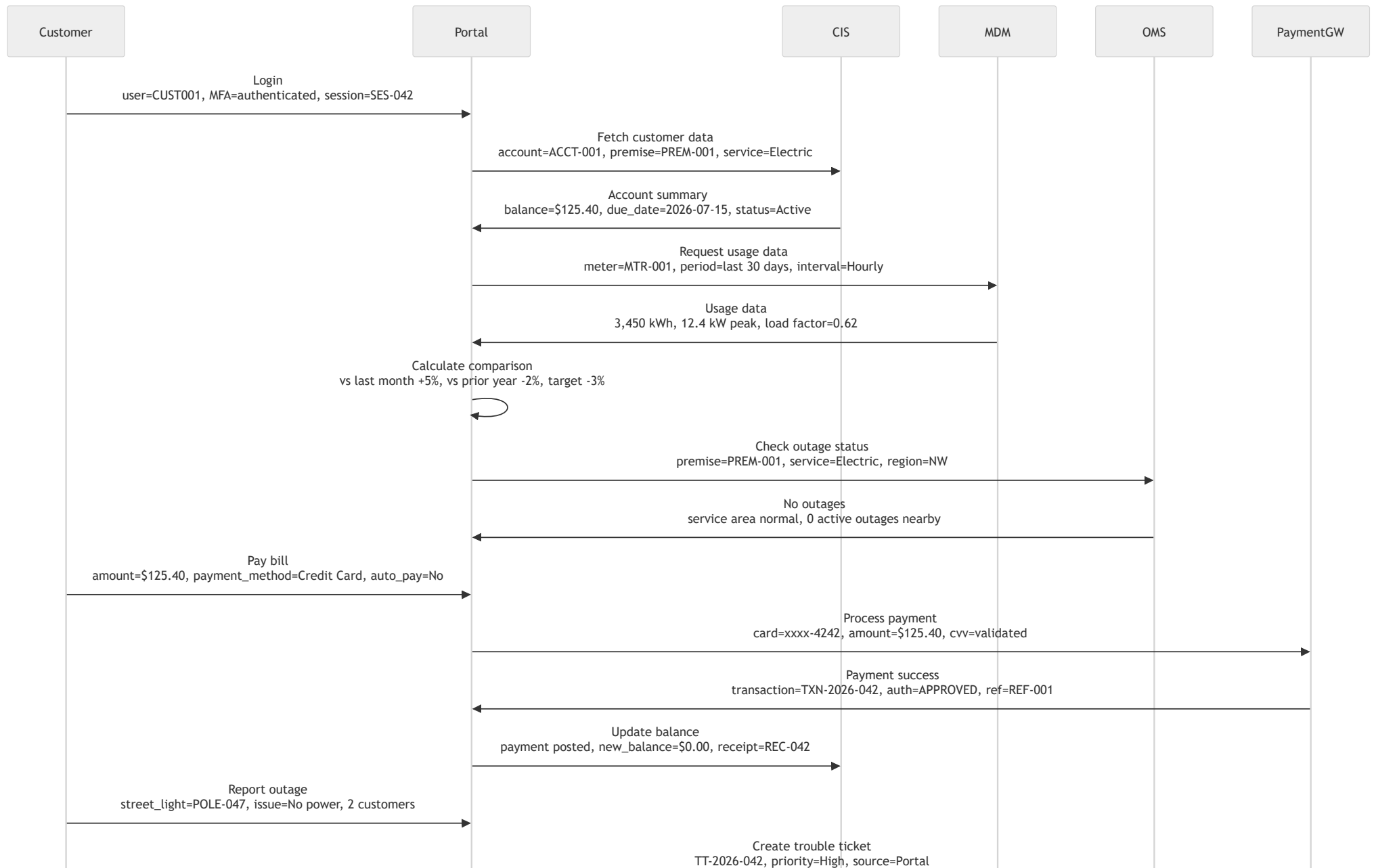
9.1 Customer Portal & Self-Service

Description: Customers access the web portal and mobile app to view usage data, pay bills, report outages, manage account preferences, and enroll in programs. Portal integrates with CIS, MDM, and OMS.

Actors: Customer, Customer Service Rep, Portal Admin

Systems: Customer Portal, CIS (Oracle Utilities), MDM, OMS, Payment Gateway (CyberSource)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Customer Portal	Login	AuthenticateCustomer	Customer Portal	CIS (Oracle Utilities)	Portal API (OAuth2)	CIS API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Customer Portal	Usage	FetchUsageData	Customer Portal	MDM	Portal API (OAuth2)	MDM API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Customer Portal	Outage	CheckOutageStatus	Customer Portal	OMS	Portal API (OAuth2)	OMS API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Customer Portal	Payment	ProcessBillPayment	Customer Portal	Payment Gateway (CyberSource)	Portal API (OAuth2)	Payment GW API (REST/SOAP)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Customer Portal	Balance	UpdateCustomerBalance	Customer Portal	CIS (Oracle Utilities)	Portal API (OAuth2)	CIS API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Customer Portal	Ticket	CreateTroubleTicket	Customer Portal	OMS	Portal API (OAuth2)	OMS API (REST)	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
Customer Portal	Survey	SubmitCSATSurvey	Customer Portal	Analytics Platform	Portal API (OAuth2)	Analytics API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-CP-01	Validate portal page load response time under 3 seconds for 95th percentile	AuthenticateCustomer	Portal dashboard loads within 3s for 95% of requests	High
AC-CP-02	Confirm payment processing completes with end-to-end encryption (PCI-DSS)	ProcessBillPayment	Payment data encrypted in transit and at rest, PCI compliant	High
AC-CP-03	Verify outage reporting accuracy within 100 meters of actual location	CreateTroubleTicket	Outage location mapped within 100m of customer premise	Medium
AC-CP-04	Ensure self-service adoption rate exceeds 60% of total customer interactions	SubmitCSATSurvey	Monthly self-service interactions >60% of all service contacts	Medium

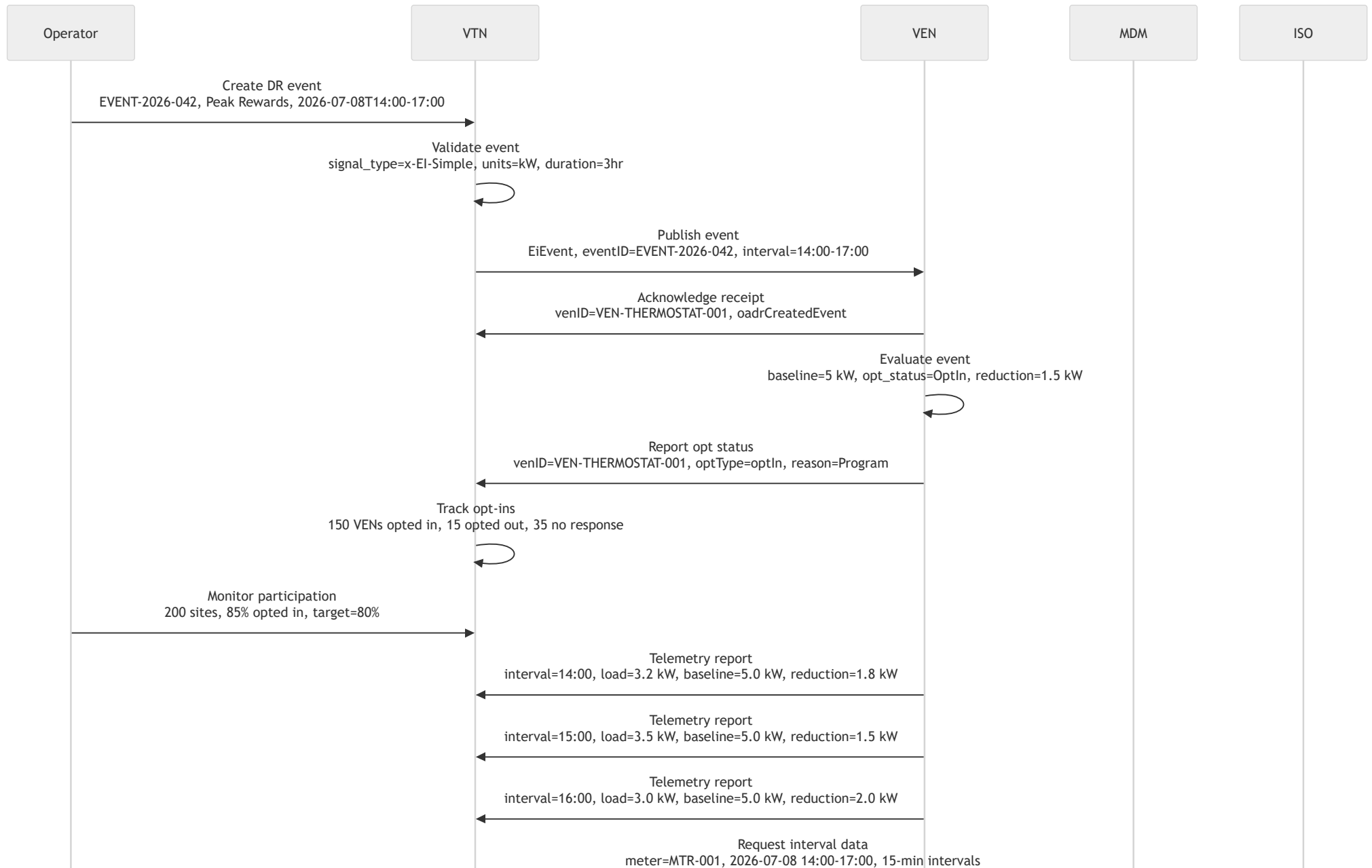
9.2 OpenADR VEN/VTN Operations

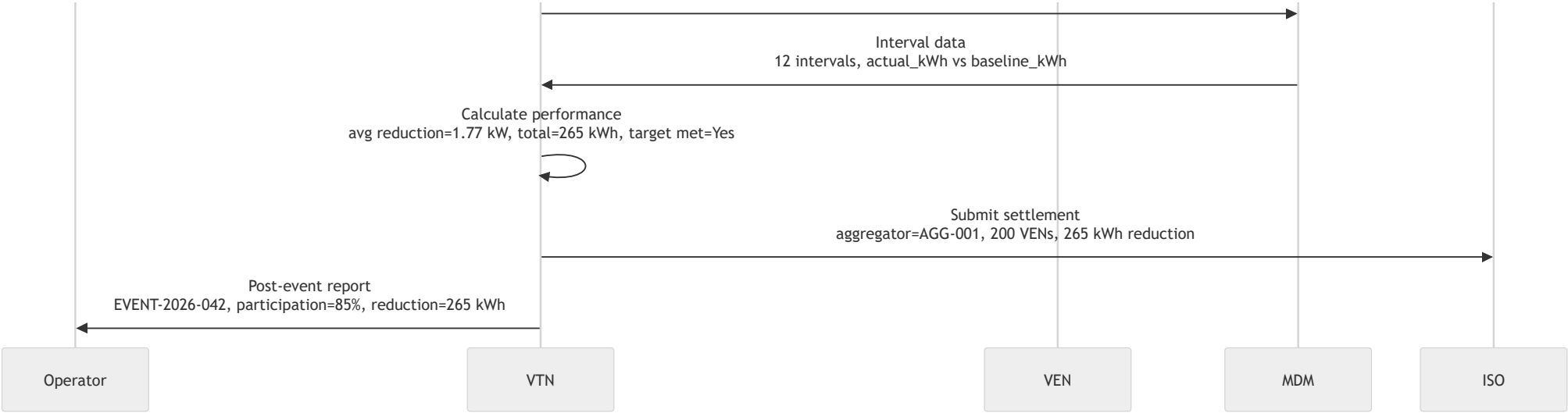
Description: Utility VTN communicates with customer VENs (smart thermostats, BMS, DER) using OpenADR 2.0b. DR events are created, published, and tracked with opt-in/opt-out and reporting.

Actors: DR Operator, Aggregator, Customer VEN

Systems: OpenADR VTN (EPRI OpenADR), DRMS, VEN (Customer), MDM, ISO Market System

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
OpenADR	Event	CreateDREvent	DR Operator HMI	OpenADR VTN (EPRI OpenADR)	Operator UI (REST)	VTN API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
OpenADR	Publish	PublishDREvent	OpenADR VTN (EPRI OpenADR)	VEN (Customer)	VTN OpenADR 2.0b	VEN OpenADR 2.0b	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
OpenADR	Opt	ProcessOptInOut	VEN (Customer)	OpenADR VTN (EPRI OpenADR)	VEN OpenADR 2.0b	VTN OpenADR 2.0b	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
OpenADR	Telemetry	ReportVENLoad	VEN (Customer)	OpenADR VTN (EPRI OpenADR)	VEN OpenADR 2.0b	VTN OpenADR 2.0b	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
OpenADR	Meter	FetchIntervalData	OpenADR VTN (EPRI OpenADR)	MDM	VTN API (REST)	MDM API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
OpenADR	Settlement	SubmitSettlementData	OpenADR VTN (EPRI OpenADR)	ISO Market System	VTN API (REST)	ISO API (REST)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
OpenADR	Report	GeneratePostEventReport	OpenADR VTN (EPRI OpenADR)	DRMS	VTN API (REST)	DRMS API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-OADR-01	Validate VTN-to- VEN event publication reaches 95% of VENs within 5 minutes	PublishDREvent	At least 95% of VENs acknowledge event within 5 min of publish	High
AC-OADR-02	Confirm opt-in/opt-out processing completes within 60 seconds of VEN response	ProcessOptInOut	VEN opt status reflected in VTN within 60s	Medium
AC-OADR-03	Verify telemetry report data accuracy within ±2% of meter interval data	ReportVENLoad	VEN-reported load matches MDM interval data within 2%	Medium
AC-OADR-04	Ensure settlement data submission completes within 24 hours of event end	SubmitSettlementData	Settlement data submitted to ISO within 24 hr window	High

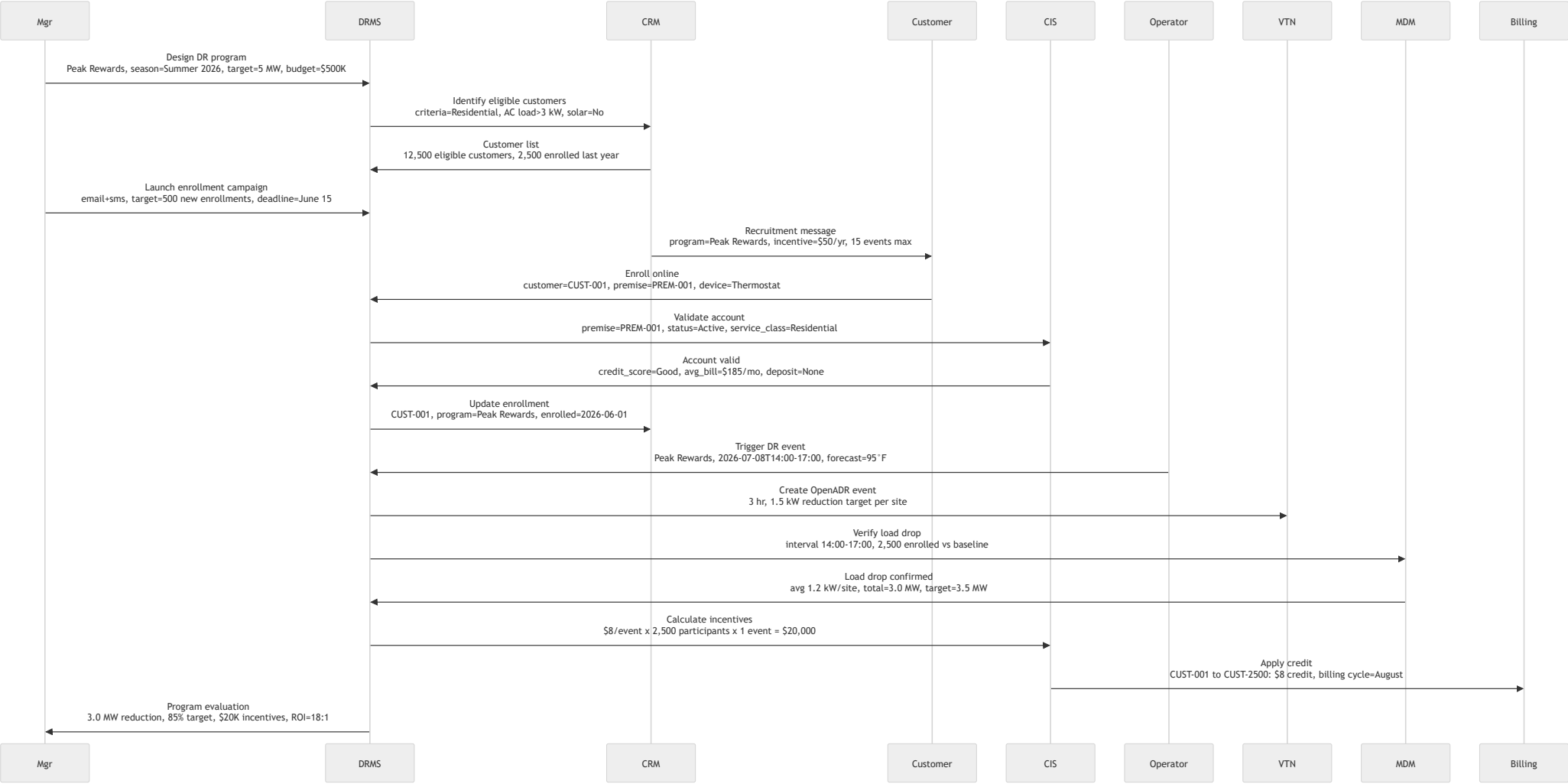
9.3 Demand Response Program Management

Description: Utility designs and manages DR programs (Peak Rewards, Critical Peak Pricing, Emergency). Customers are recruited, enrolled, and their performance tracked for settlement and incentive payments.

Actors: DR Program Manager, Customer, DRMS Administrator

Systems: DRMS, CRM (Salesforce), CIS (Oracle Utilities), MDM, Billing System

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
DRMS	Design	DesignDRProgram	DR Program Manager	DRMS	Manager UI (REST)	DRMS API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
DRMS	Eligibility	IdentifyEligibleCustomers	DRMS	CRM (Salesforce)	DRMS API (REST)	CRM API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
DRMS	Enroll	EnrollCustomerProgram	Customer Portal	DRMS	Portal API (OAuth2)	DRMS API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
DRMS	Validate	ValidateCustomerAccount	DRMS	CIS (Oracle Utilities)	DRMS API (REST)	CIS API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
DRMS	Event	TriggerDREvent	DR Program Manager	DRMS	Manager UI (REST)	DRMS API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
DRMS	Measurement	VerifyLoadReduction	DRMS	MDM	DRMS API (REST)	MDM API (REST)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
DRMS	Incentive	ApplyIncentiveCredit	DRMS	CIS (Oracle Utilities)	DRMS API (REST)	CIS API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-DRP-01	Validate enrollment completion within 2 business days of customer application	EnrollCustomerProgram	Customer confirmed enrolled within 48 hrs of application	High
AC-DRP-02	Confirm load drop measurement accuracy within ±5% of CBL methodology	VerifyLoadReduction	MDM baseline comparison matches CBL within 5%	High
AC-DRP-03	Verify incentive credits appear on customer bill within next billing cycle	ApplyIncentiveCredit	Incentive credit line item present on August bill	Medium
AC-DRP-04	Ensure program performance reports include participation, load drop, and incentive totals	DesignDRProgram	Monthly report contains all KPIs with trend analysis	Medium

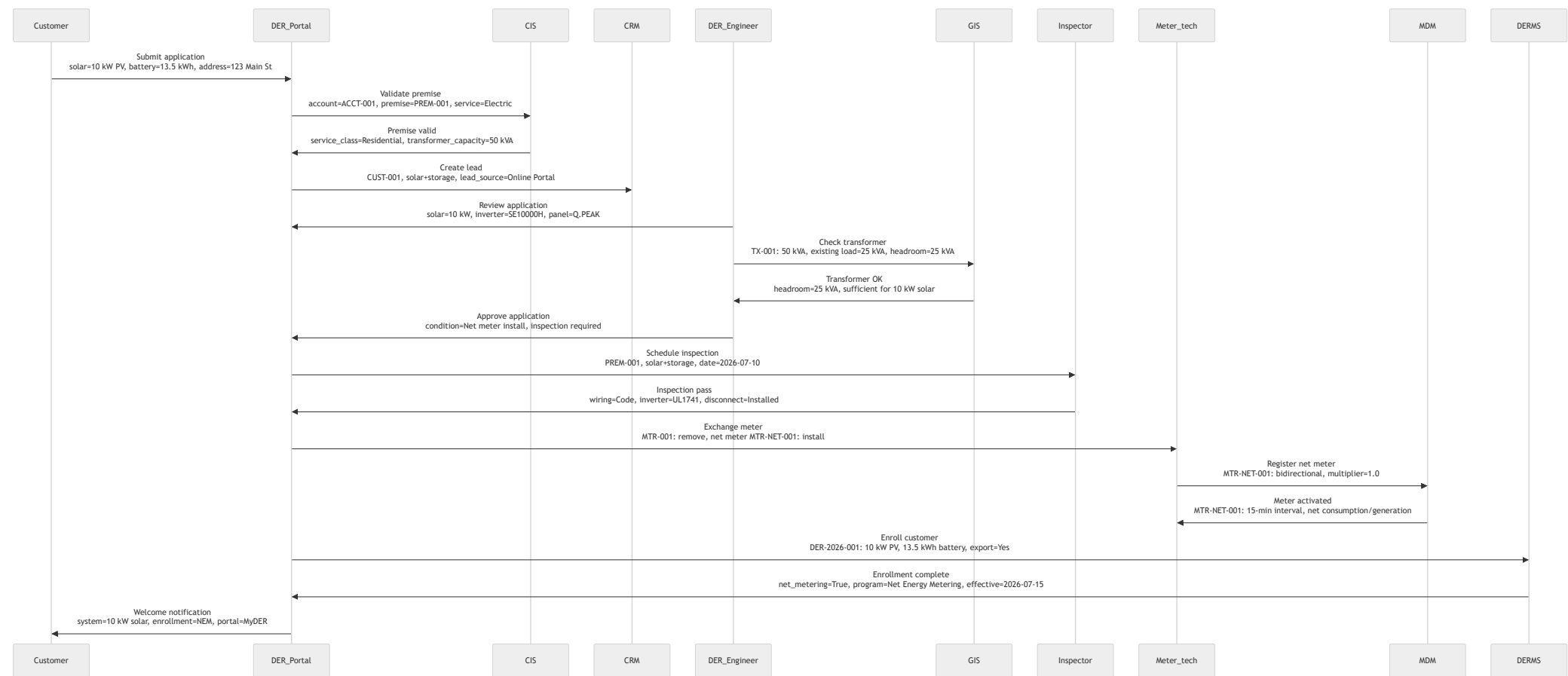
9.4 DER Customer Enrollment

Description: Customers with solar PV, battery storage, and EV chargers apply for interconnection. The utility processes the application, inspects installation, installs net meter, and enrolls in DER programs.

Actors: Customer, DER Engineer, Inspector, Metering Tech

Systems: DER Portal, DERMS, CIS, MDM, GIS, CRM

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
DER Enrollment	Apply	SubmitDERApplication	DER Portal	CIS (Oracle Utilities)	Portal API (OAuth2)	CIS API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
DER Enrollment	Lead	CreateCRMLead	DER Portal	CRM (Salesforce)	Portal API (OAuth2)	CRM API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
DER Enrollment	Review	CheckTransformerCapacity	GIS (ArcGIS)	DER Portal	GIS API (REST)	Portal API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
DER Enrollment	Inspect	ScheduleSiteInspection	DER Portal	Inspector Mobile App	Portal API (REST)	Field App API (OAuth2)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
DER Enrollment	Meter	ExchangeNetMeter	DER Portal	MDM	Portal API (REST)	MDM API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
DER Enrollment	DerMs	EnrollCustomerDERMS	DER Portal	DERMS	Portal API (REST)	DERMS API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
DER Enrollment	Notify	SendWelcomeNotification	DER Portal	Notification Service	Portal API (REST)	Notification API (Email/SMS)	Event-driven	Simple	Simple event notification with fire-and-forget delivery semantics

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-DER-EN-01	Validate application processing within 5 business days of submission	SubmitDERApplication	Engineering review complete within 5 business days	High
AC-DER-EN-02	Confirm inspection scheduled within 3 business days of application approval	ScheduleSiteInspection	Inspection date offered within 72 hrs of approval	Medium
AC-DER-EN-03	Verify net meter installation and activation within 5 business days of inspection pass	ExchangeNetMeter	Net meter activated in MDM within 5 days of inspection	High
AC-DER-EN-04	Ensure DERMS enrollment completed before net meter activation date	EnrollCustomerDERMS	DERMS enrollment effective date aligns with meter activation	Medium

Billing, CIS & Revenue Management - UT-CIMUT-RTMUT-BILUT-PAYUT-REV

Manages customer information, rate/tariff definitions, billing calculation and presentment, payment processing, and revenue assurance for multi-service (electric + water) utility billing.

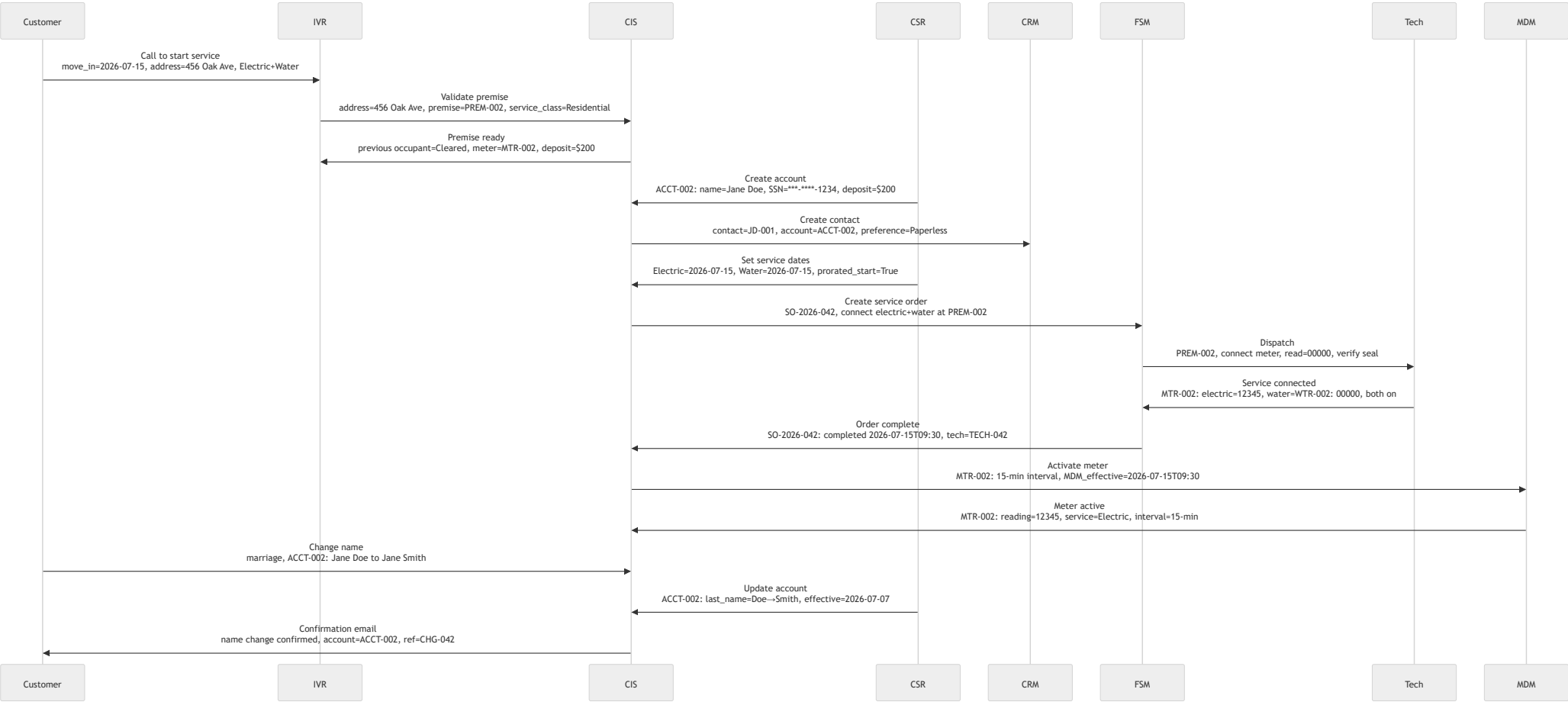
10.1 Customer Information Management

Description: Customer service representatives manage account creation, service orders, move-in/move-out, name changes, and account updates across electric and water services.

Actors: CSR, Customer, Field Service Tech

Systems: CIS (Oracle Utilities Customer Care & Billing), CRM (Salesforce), MDM, IVR, Field Service Management

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
CIM	Validation	ValidatePremiseAddress	IVR	CIS (Oracle Utilities)	IVR API (SOAP)	CIS API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
CIM	Account	CreateCustomerAccount	CSR HMI	CIS (Oracle Utilities)	CSR UI	CIS API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
CIM	Contact	CreateCRMContact	CIS (Oracle Utilities)	CRM (Salesforce)	CIS API (REST)	CRM API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
CIM	Service	CreateServiceOrder	CIS (Oracle Utilities)	Field Service Management	CIS API (REST)	FSM API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
CIM	Dispatch	DispatchFieldTech	Field Service Management	Tech Mobile App	FSM API (REST)	Mobile App API (OAuth2)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
CIM	Meter	ActivateMeterMDM	CIS (Oracle Utilities)	MDM	CIS API (REST)	MDM API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
CIM	Update	ProcessNameChange	CSR HMI	CIS (Oracle Utilities)	CSR UI	CIS API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-CIM-01	Validate account creation completes in under 10 minutes for walk-in/phone requests	CreateCustomerAccount	Account fully created and active within 10 min of request	High
AC-CIM-02	Confirm service order dispatched to field within 30 minutes of creation	CreateServiceOrder	Service order assigned and dispatched within 30 min	Medium
AC-CIM-03	Verify meter activation synchronizes between CIS and MDM within 5 minutes	ActivateMeterMDM	MDM reflects CIS effective date within 5 min tolerance	High
AC-CIM-04	Ensure customer data updates are audited with before/after values and user identity	ProcessNameChange	Audit trail shows previous and new values for all changes	Medium

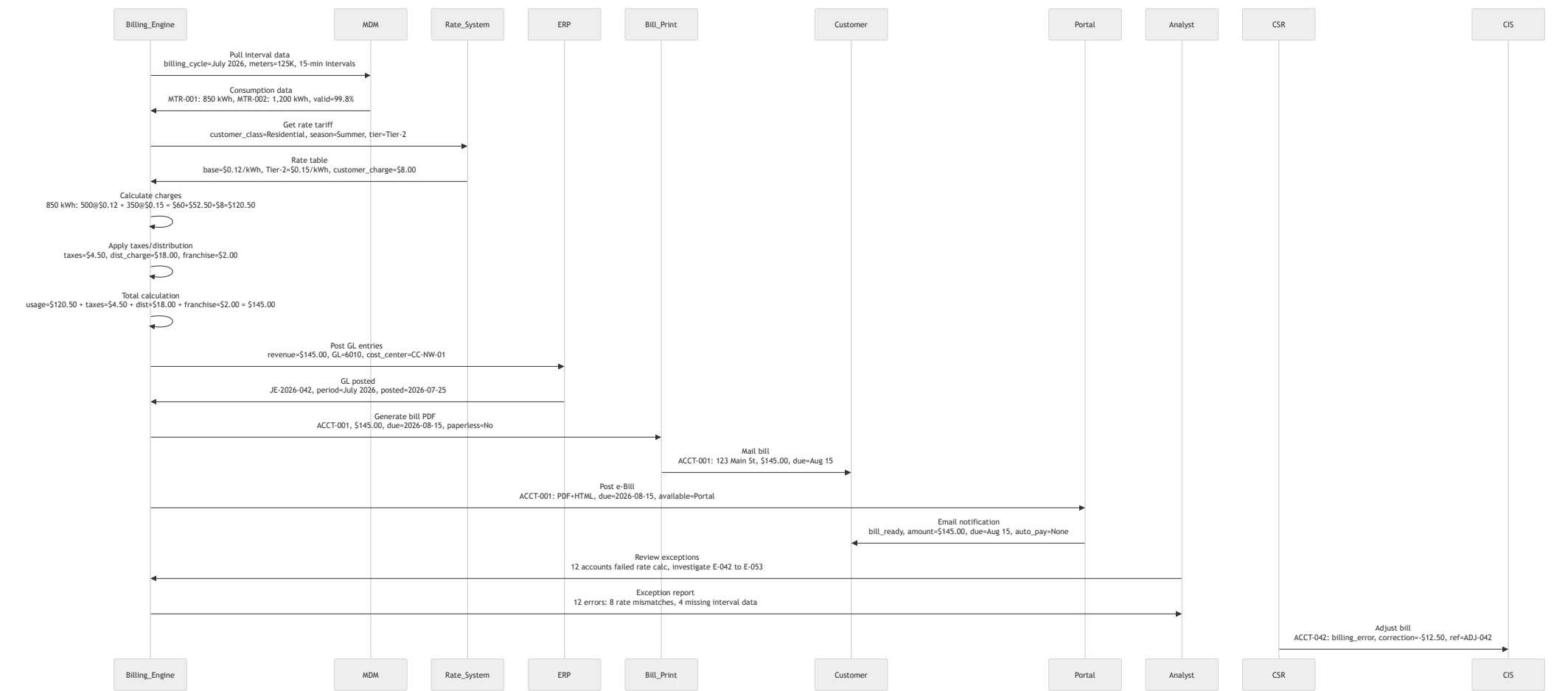
10.2 Billing Calculation & Presentment

Description: Billing engine calculates charges based on rate tariffs, meter data (MDM), and customer class. Bills are generated, rendered (paper/paperless), and delivered via Bill Print or e-Bill.

Actors: Billing Analyst, Customer, CSR

Systems: CIS Billing Engine, MDM, Rate/Tariff System, Bill Print Vendor, Customer Portal, ERP (SAP)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Billing	Data	PullIntervalData	CIS Billing Engine	MDM	Billing API (REST)	MDM API (REST)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Billing	Rate	FetchRateTariff	CIS Billing Engine	Rate/Tariff System	Billing API (REST)	Rate API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Billing	Calc	CalculateCharges	CIS Billing Engine	CIS Billing Engine	Internal	Internal	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Billing	GL	PostGLEntries	CIS Billing Engine	ERP (SAP)	CIS API (REST)	ERP API (SOAP/IDoc)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Billing	Present	GenerateBillPDF	CIS Billing Engine	Bill Print Vendor	CIS API (REST)	Print Vendor API (SFTP)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
Billing	Ebill	PostElectronicBill	CIS Billing Engine	Customer Portal	CIS API (REST)	Portal API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
Billing	Exception	ReviewBillingExceptions	CIS Billing Engine	Billing Analyst UI	CIS API (REST)	Analyst UI	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-BIL-01	Validate billing calculation accuracy within ±0.5% of independently computed values	CalculateCharges	Sample audit of 1% of bills shows accuracy within 0.5%	High
AC-BIL-02	Confirm bill generation completes within 4 hours for full customer base	GenerateBillPDF	All 125K customer bills generated and rendered within 4 hrs	High
AC-BIL-03	Verify GL posting matches billing totals with zero variance	PostGLEntries	GL control totals match billing system month-end totals	High
AC-BIL-04	Ensure billing exception rate remains under 0.5% of total accounts billed	ReviewBillingExceptions	Exception report shows <625 accounts (0.5% of 125K)	Medium

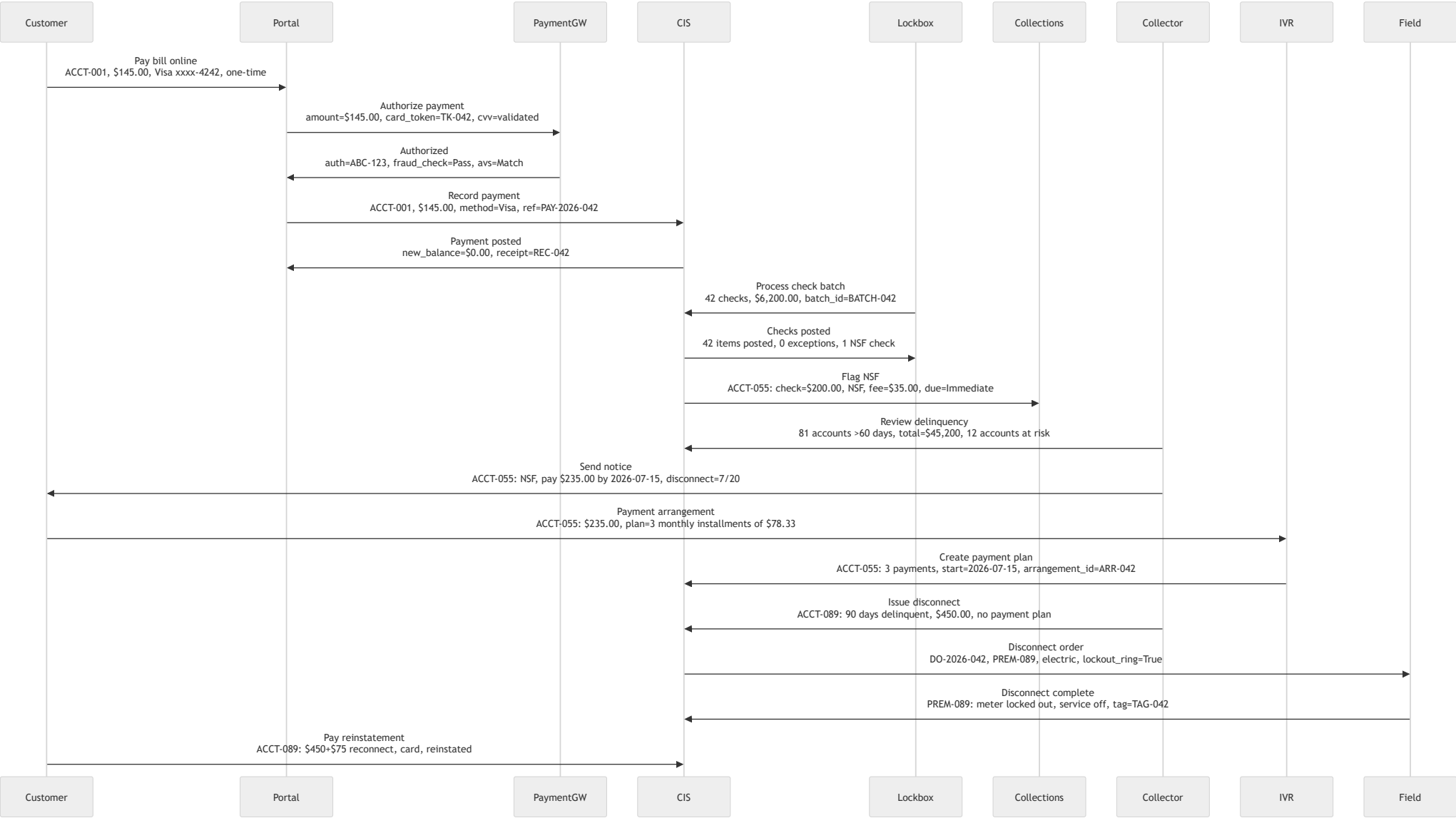
10.3 Payment & Collections

Description: Customers pay bills via multiple channels (web, auto-pay, walk-in, phone). Collections team manages delinquent accounts, payment plans, and disconnection/reconnection processes.

Actors: Customer, Cashier, Collections Specialist

Systems: CIS (Oracle Utilities), Payment Gateway (CyberSource), Lockbox, Collections System, IVR

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Payment	Online	ProcessOnlinePayment	Customer Portal	Payment Gateway (CyberSource)	Portal API (OAuth2)	Payment GW API (REST/SOAP)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Payment	Posting	RecordPaymentCIS	Customer Portal	CIS (Oracle Utilities)	Portal API (OAuth2)	CIS API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Payment	Lockbox	ProcessCheckBatch	Lockbox	CIS (Oracle Utilities)	Lockbox BAI2	CIS API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
Payment	NSF	ProcessNSFCheck	CIS (Oracle Utilities)	Collections System	CIS API (REST)	Collections API (REST)	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
Payment	Plan	CreatePaymentPlan	IVR	CIS (Oracle Utilities)	IVR API (SOAP)	CIS API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Payment	Disconnect	IssueDisconnectOrder	CIS (Oracle Utilities)	Field Service Management	CIS API (REST)	FSM API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Payment	Reconnect	ProcessReconnection	CIS (Oracle Utilities)	Field Service Management	CIS API (REST)	FSM API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-PAY-01	Validate payment processing completes within 5 seconds for online payments	ProcessOnlinePayment	Payment authorized and receipt generated within 5s	High
AC-PAY-02	Confirm lockbox check batch posts within 4 hours of batch receipt	ProcessCheckBatch	All checks posted within 4 hrs of batch file receipt	Medium
AC-PAY-03	Verify payment plan setup processes within 2 minutes of IVR confirmation	CreatePaymentPlan	Payment plan active in CIS within 2 min of IVR call end	Medium
AC-PAY-04	Ensure disconnect order execution within 24 hours for final notice expiry	IssueDisconnectOrder	Disconnect completed within 24 hrs of notice deadline	High

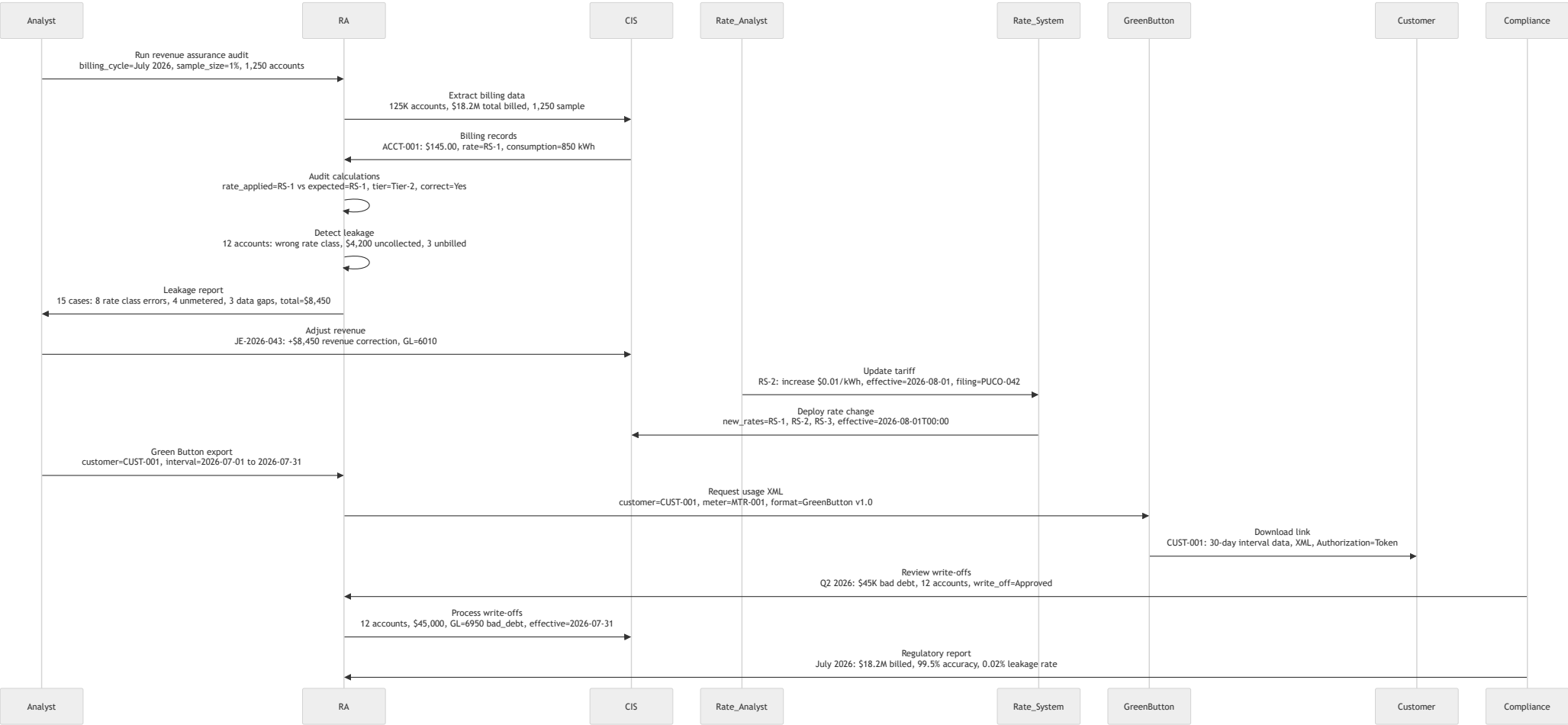
10.4 Rate & Revenue Assurance

Description: Revenue assurance team audits billing accuracy, detects leakage, manages write-offs, and ensures rates are correctly applied. Green Button data is provided for customer and third-party access.

Actors: Revenue Analyst, Rate Analyst, Compliance Officer

Systems: CIS, Rate/Tariff System, Revenue Assurance System, Green Button Platform, ERP (SAP)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Revenue	Audit	RunRevenueAssuranceAudit	Revenue Assurance Analyst	Revenue Assurance System	Analyst UI	RA API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
Revenue	Extract	ExtractBillingData	Revenue Assurance System	CIS (Oracle Utilities)	RA API (REST)	CIS API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
Revenue	Leakage	DetectRevenueLeakage	Revenue Assurance System	Revenue Assurance System	Internal	Internal	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Revenue	Tariff	DeployRateChange	Rate/Tariff System	CIS (Oracle Utilities)	Rate API (REST)	CIS API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Revenue	GreenButton	ExportGreenButtonXML	Revenue Assurance System	Green Button Platform	RA API (REST)	Green Button API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Revenue	WriteOff	ProcessWriteOffs	Revenue Assurance System	CIS (Oracle Utilities)	RA API (REST)	CIS API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
Revenue	Report	GenerateRegulatoryReport	Revenue Assurance System	Compliance Officer	RA API (REST)	Compliance UI	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-REV-01	Validate revenue assurance audit covers minimum 1% of accounts or 100% of high-risk	RunRevenueAssuranceAudit	Audit sample size meets minimum threshold per utility policy	High
AC-REV-02	Confirm revenue leakage detection identifies rate class errors, unmetered accounts, and data gaps	DetectRevenueLeakage	Leakage report includes all 3 categories with dollar impact	High

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-REV-03	Verify rate change deployment completes before effective date with zero billing interruption	DeployRateChange	All accounts billed using new rates effective 2026-08-01	Medium
AC-REV-04	Ensure Green Button data export includes all interval data with proper XML schema validation	ExportGreenButtonXML	Green Button XML passes NAESB schema validation	Medium

Outage & Workforce Management - UT-ODI UT-DAS UT-CDR UT-OAR

End-to-end outage management from detection (AMI, SCADA, IVR), damage assessment, crew dispatch, restoration tracking, and post-event analytics including SAIDI/SAIFI calculations.

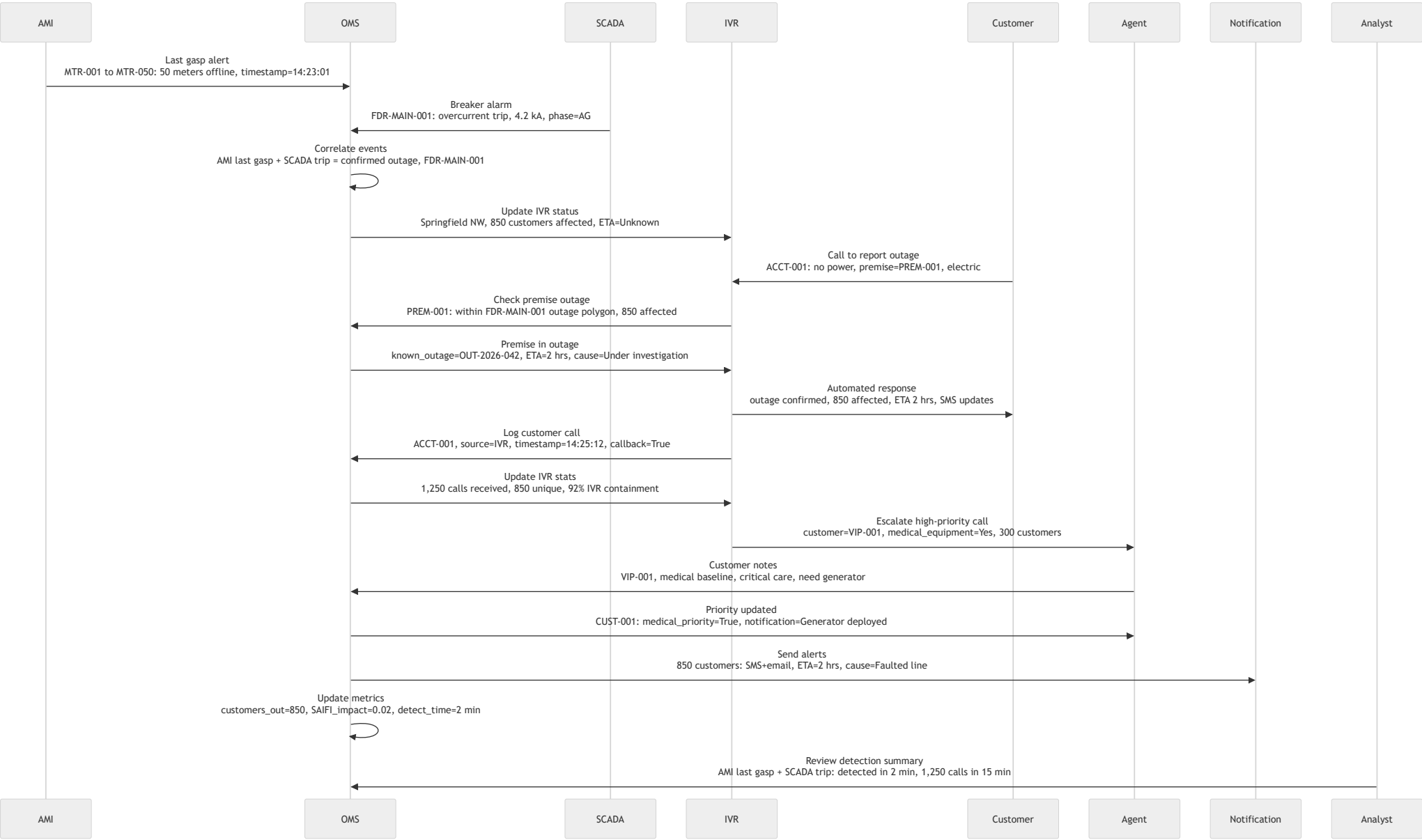
11.1 Outage Detection & IVR

Description: Outages are detected via AMI last gasp, SCADA breaker status, and customer calls. IVR system handles high call volumes, correlates with known outages, and updates OMS.

Actors: Call Center Agent, Customer, Outage Analyst

Systems: IVR, OMS (Oracle Utilities), AMI Head-End, SCADA, Outage Notification System

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
ODI	AMILastGasp	ProcessLastGaspAlert	AMI Head-End	OMS (Oracle Utilities)	AMI API (REST)	OMS API (REST)	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
ODI	SCADATrip	ProcessBreakerTrip	SCADA	OMS (Oracle Utilities)	SCADA API (ICCP)	OMS API (REST)	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
ODI	Correlation	CorrelateOutageEvents	OMS (Oracle Utilities)	OMS (Oracle Utilities)	Internal	Internal	Batch (Real-time)	High	High-frequency near-real-time batch processing with minimal latency tolerance
ODI	IVR	UpdateIVROutageStatus	OMS (OracleUtilities)	IVR	OMS API (REST)	IVR API (SOAP)	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
ODI	Customer	LogCustomerOutageCall	IVR	OMS (Oracle Utilities)	IVR API (SOAP)	OMS API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
ODI	Notification	SendOutageNotifications	OMS (Oracle Utilities)	Outage Notification System	OMS API (REST)	Notification API (Email/SMS)	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
ODI	Priority	UpdateMedicalPriority	Call Center Agent	OMS (Oracle Utilities)	Agent HMI	OMS API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-ODI-01	Validate outage detection time under 5 minutes from AMI last gasp or SCADA trip	ProcessLastGaspAlert	OMS outage event created within 5 min of first alert	High
AC-ODI-02	Confirm IVR containment rate exceeds 85% of total outage calls	UpdateIVROutageStatus	>85% of calls resolved without agent escalation	Medium
AC-ODI-03	Verify call-logging timestamp accuracy within ±1 second of customer call arrival	LogCustomerOutageCall	OMS call log timestamp matches ACD timestamp within 1s	Medium
AC-ODI-04	Ensure outage notifications reach 100% of affected customers within 15 minutes	SendOutageNotifications	SMS delivery receipt for all customers within 15 min	High

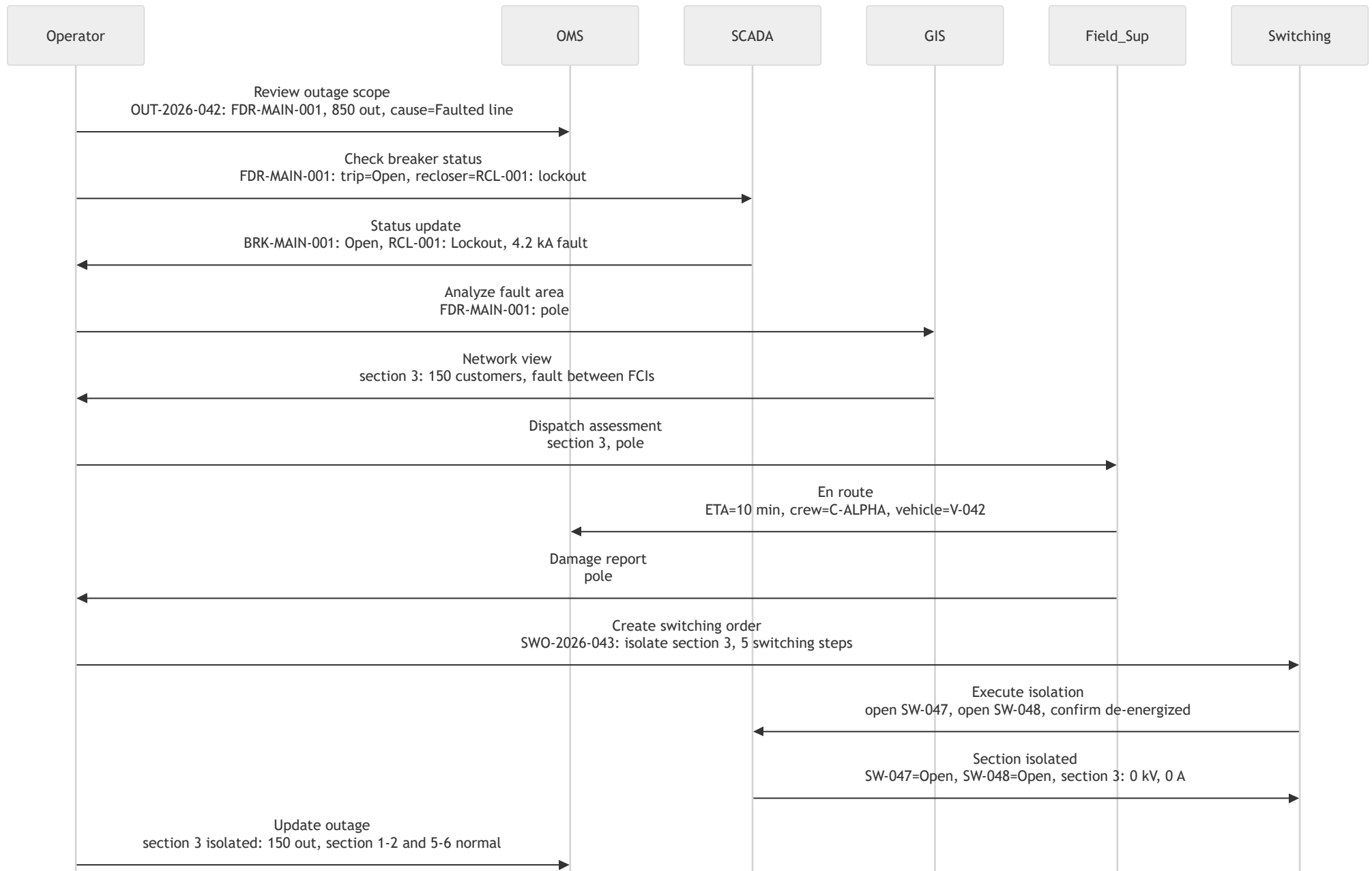
11.2 Damage Assessment & Switching

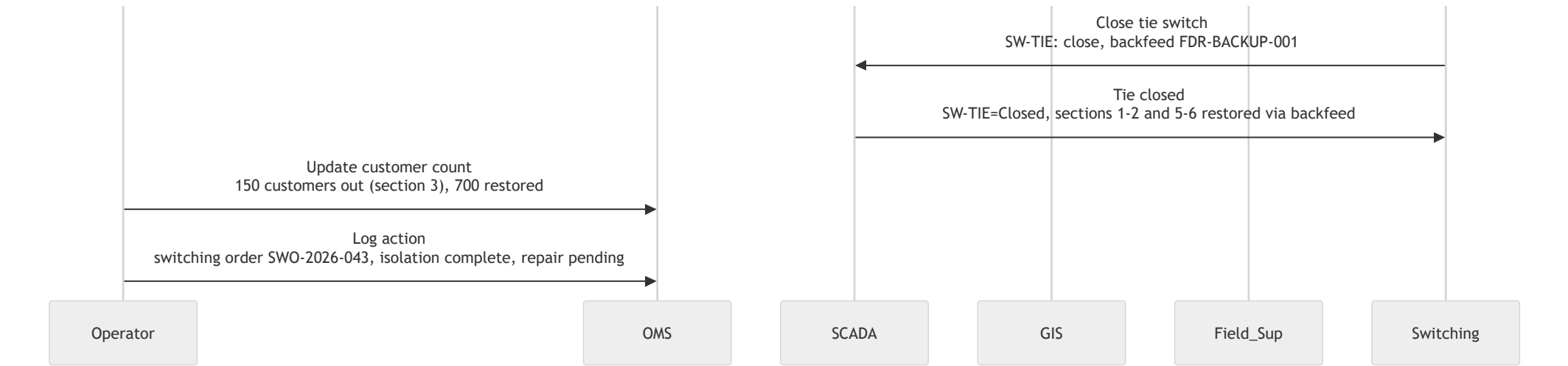
Description: After outage detection, system operator performs damage assessment via field reports, SCADA status, and GIS analysis. Switching orders isolate faulted sections and reconfigure the network.

Actors: System Operator, Field Supervisor, Damage Assessor

Systems: OMS, SCADA, GIS (ArcGIS), Switching Order Management, Mobile Crew App

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
DAS	Scope	ReviewOutageScope	System Operator HMI	OMS (Oracle Utilities)	Operator UI	OMS API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
DAS	Status	CheckBreakerStatus	System Operator HMI	SCADA	Operator UI	SCADA API (ICCP)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
DAS	GIS	AnalyzeFaultArea	System Operator HMI	GIS (ArcGIS)	Operator UI	GIS API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
DAS	Assessment	DispatchDamageAssessment	OMS (Oracle Utilities)	Mobile Crew App	OMS API (REST)	Mobile App API (OAuth2)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
DAS	Switching	CreateSwitchingOrder	System Operator HMI	Switching Order Management	Operator UI	Switching API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
DAS	Isolate	ExecuteSwitchingIsolation	Switching Order Management	SCADA	Switching API (REST)	SCADA DNP3	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
DAS	Restore	CloseTieSwitchRestore	Switching Order Management	SCADA	Switching API (REST)	SCADA DNP3	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-DAS-01	Validate damage assessment dispatched within 15 minutes of outage confirmation	DispatchDamageAssessment	Field crew dispatched within 15 min of outage creation	High
AC-DAS-02	Confirm switching order creation includes engineer-approved isolation sequence	CreateSwitchingOrder	Switching order reviewed and approved by system operator	High
AC-DAS-03	Verify isolation switching completes within 30 minutes of switching order approval	ExecuteSwitchingIsolation	All switches operated and confirmed within 30 min	High
AC-DAS-04	Ensure backfeed restoration does not exceed tie switch ampacity rating	CloseTieSwitchRestore	Post-restoration loading <90% of tie switch rating	Medium

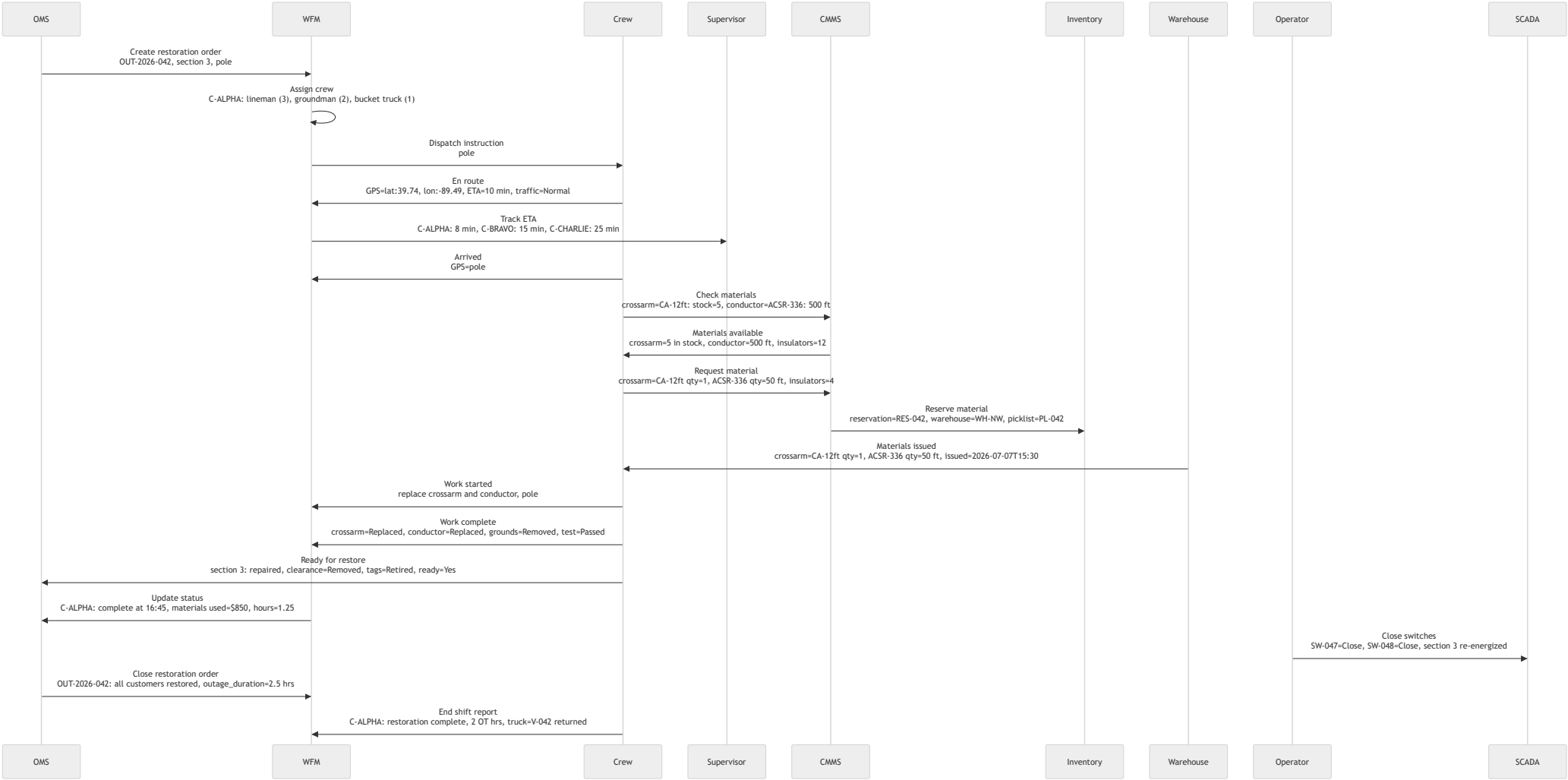
11.3 Crew Dispatch & Restoration

Description: Crews are dispatched based on location, skillset, and priority. Mobile workforce management tracks ETA, work progress, and materials used. Supervisors coordinate restoration activities.

Actors: Dispatch Supervisor, Field Crew, Warehouse Clerk

Systems: WFM (Mobile Workforce), OMS, GIS, CMMS (Maximo), Inventory System, AVL/GPS

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
CDR	Order	CreateRestorationOrder	OMS (Oracle Utilities)	WFM (Mobile Workforce)	OMS API (REST)	WFM API (REST)	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
CDR	Dispatch	AssignCrewToJob	WFM (Mobile Workforce)	Mobile Crew App	WFM API (REST)	Mobile App API (OAuth2)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
CDR	Tracking	TrackCrewETA	Mobile Crew App (GPS)	WFM (Mobile Workforce)	Mobile GPS (AVL)	WFM API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
CDR	Materials	CheckMaterialAvailability	Mobile Crew App	CMMS (Maximo)	Mobile App API (OAuth2)	CMMS API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
CDR	Inventory	ReserveAndIssueMaterial	CMMS (Maximo)	Inventory System	CMMS API (REST)	Inventory API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
CDR	Restore	NotifyRestorationReady	Mobile Crew App	OMS (Oracle Utilities)	Mobile App API (OAuth2)	OMS API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
CDR	Report	CloseRestorationOrder	OMS (OracleUtilities)	WFM (Mobile Workforce)	OMS API (REST)	WFM API (REST)	Event-driven	Simple	Simple event notification with fire-and-forget delivery semantics

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-CDR-01	Validate crew dispatch within 5 minutes of restoration order creation	AssignCrewToJob	Crew assigned and notified within 5 min of order creation	High
AC-CDR-02	Confirm GPS location accuracy within 10 meters for crew ETA calculation	TrackCrewETA	AVL GPS coordinates accurate within 10m of actual location	Medium
AC-CDR-03	Verify material availability check completes within 60 seconds of mobile request	CheckMaterialAvailability	CMMS responds with stock status within 60s	Medium
AC-CDR-04	Ensure restoration order is closed within 30 minutes of all customers restored	CloseRestorationOrder	OMS and WFM synchronized within 30 min of full restoration	High

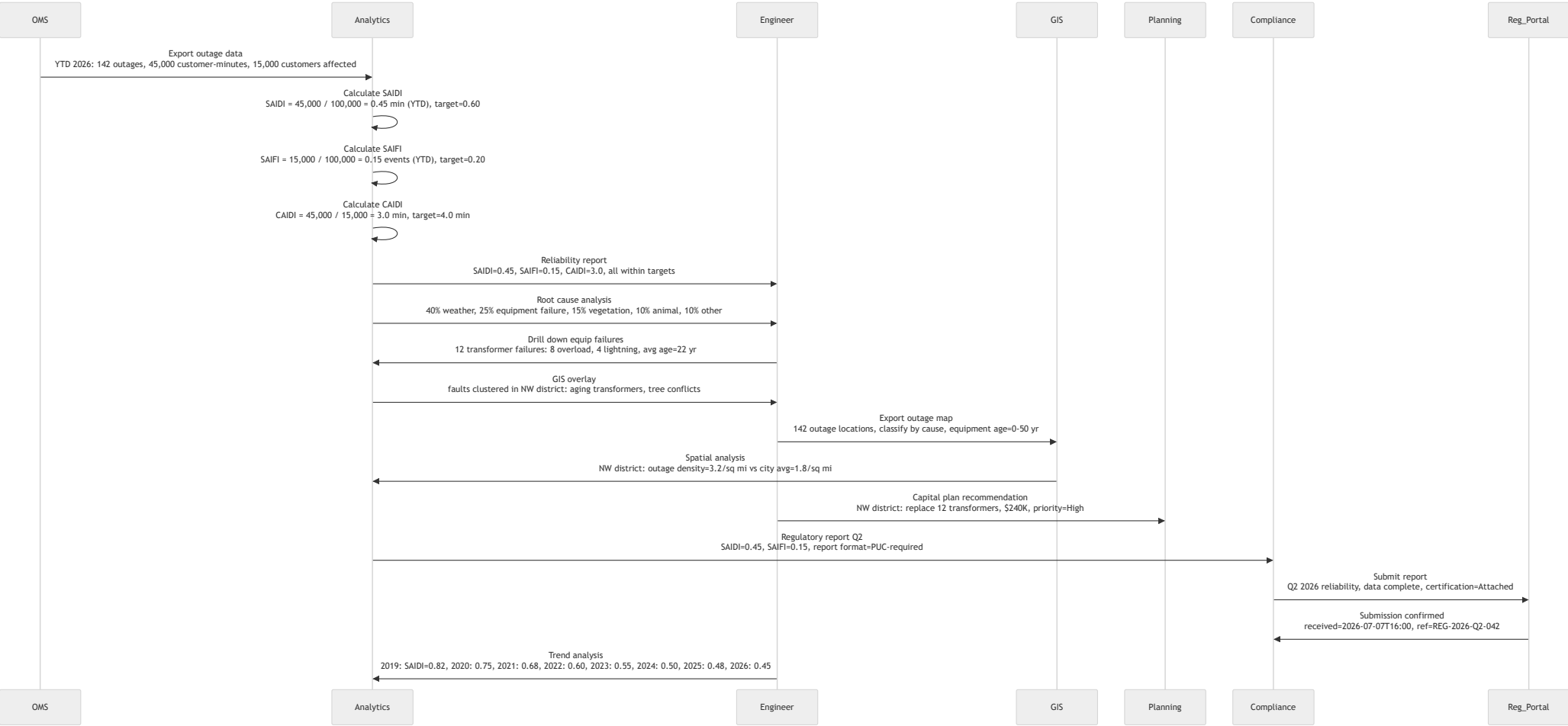
11.4 Outage Analytics & Reporting

Description: Post-event analysis calculates reliability metrics (SAIDI, SAIFI, CAIDI), identifies root causes, and generates regulatory reports. Trends are analyzed for system improvement planning.

Actors: Reliability Engineer, Compliance Manager, Planning Engineer

Systems: OMS, Analytics Platform, Compliance Reporting, GIS, Regulatory Portal

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
OAR	Data	ExportOutageData	OMS (Oracle Utilities)	Analytics Platform	OMS API (REST)	Analytics API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
OAR	Metrics	CalculateReliabilityMetrics	Analytics Platform	Analytics Platform	Internal	Internal	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
OAR	RootCause	AnalyzeRootCause	Analytics Platform	Reliability Engineer	Analytics API (REST)	Engineer UI	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
OAR	Spatial	ExportOutageGISMap	GIS (ArcGIS)	Analytics Platform	GIS API (REST)	Analytics API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints
OAR	Plan	UpdateCapitalPlan	Reliability Engineer	Planning System	Engineer UI	Planning API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
OAR	Compliance	GenerateRegulatoryReport	Analytics Platform	Compliance Reporting	Analytics API (REST)	Compliance API (REST)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
OAR	Submit	SubmitToRegulatoryPortal	Compliance Manager	Regulatory Portal	Compliance UI	Reg Portal API (REST/SOAP)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-OAR-01	Validate SAIDI/SAIFI/CAIDI calculations match IEEE 1366 standard methodology	CalculateReliabilityMetrics	Metrics recalculated independently, match within 0.1%	High
AC-OAR-02	Confirm root cause analysis categorizes 100% of outages with plausible cause	AnalyzeRootCause	All outages assigned cause code with <1% unknown category	Medium
AC-OAR-03	Verify regulatory report submitted withi	SubmitToRegulatoryPortal	Regulatory portal confirms receipt before filing deadline	High

AC ID	Description	Related Info Flow	Expected Outcome	Priority
	n filing deadline (30 days post-quarter)			
AC-OAR-04	Ensure trend analysis includes minimum 5-year historical comparison	ExportOutageData	Trend report shows 5+ years of annual SAIDI/SAIFI values	Medium

Asset Management & GIS - UT-ARH UT-GIS UT-CMI UT-PMC

Comprehensive asset management covering asset registry, GIS network model, IoT condition monitoring, and predictive maintenance for electrical and water infrastructure assets.

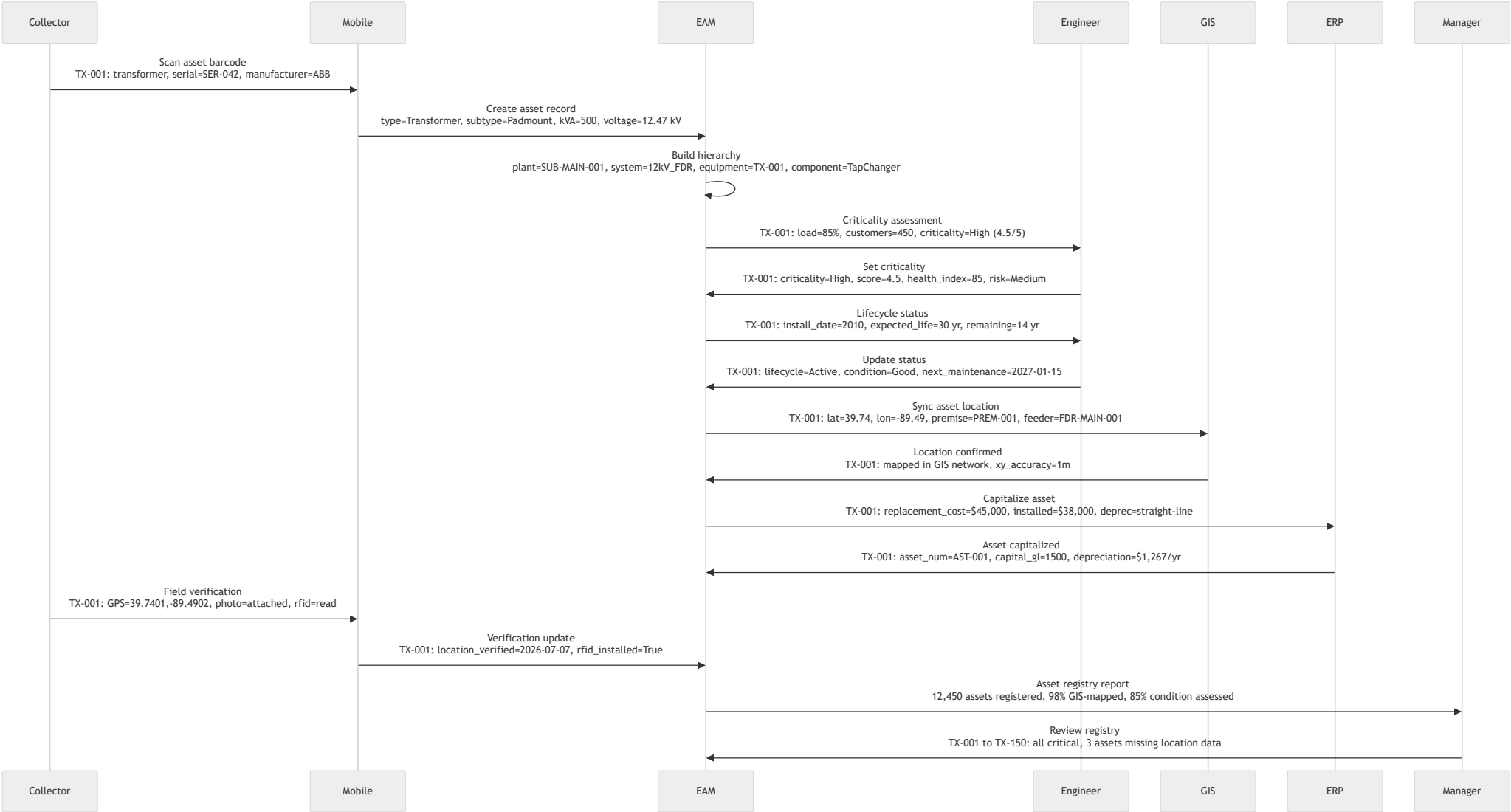
12.1 Asset Registry & Hierarchy

Description: Asset management team maintains the asset register with criticality ratings, lifecycle status, and hierarchical relationships (plant→system→equipment→component).

Actors: Asset Manager, Engineer, Field Data Collector

Systems: EAM (IBM Maximo), GIS (ArcGIS), Mobile Field App, ERP (SAP)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
ARH	Create	CreateAssetRecord	Mobile Field App	EAM (IBM Maximo)	Mobile App API (OAuth2)	EAM API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
ARH	Hierarchy	BuildAssetHierarchy	EAM (IBM Maximo)	EAM (IBM Maximo)	Internal	Internal	Batch (Real-time)	Medium	Periodic near-real-time batch with data validation and transformation needs
ARH	Criticality	AssessAssetCriticality	Engineer HMI	EAM (IBM Maximo)	Engineer UI	EAM API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
ARH	GIS	SyncAssetToGIS	EAM (IBM Maximo)	GIS (ArcGIS)	EAM API (REST)	GIS API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
ARH	Financial	CapitalizeAssetERP	EAM (IBM Maximo)	ERP (SAP)	EAM API (REST)	ERP API (SOAP/IDoc)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
ARH	Field	VerifyAssetLocation	Mobile Field App	EAM (IBM Maximo)	Mobile App API (OAuth2)	EAM API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
ARH	Report	GenerateAssetRegistryReport	EAM (IBM Maximo)	Asset Manager UI	EAM API (REST)	Manager UI	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-ARH-01	Validate asset hierarchy completeness (100% of physical assets in registry)	BuildAssetHierarchy	All field-verified assets have hierarchy parent-child relationships	High
AC-ARH-02	Confirm GIS synchronization accuracy within 1 meter for all mapped assets	SyncAssetToGIS	Asset GPS coordinates match GIS location within 1m	Medium
AC-ARH-03	Verify asset capitalization in ERP within 30 days of installation	CapitalizeAssetERP	ERP depreciation starts within 30 days of in-service date	High
AC-ARH-04	Ensure field verification cycle covers 100% of critical assets annually	VerifyAssetLocation	100% of critical assets verified within calendar year	Medium

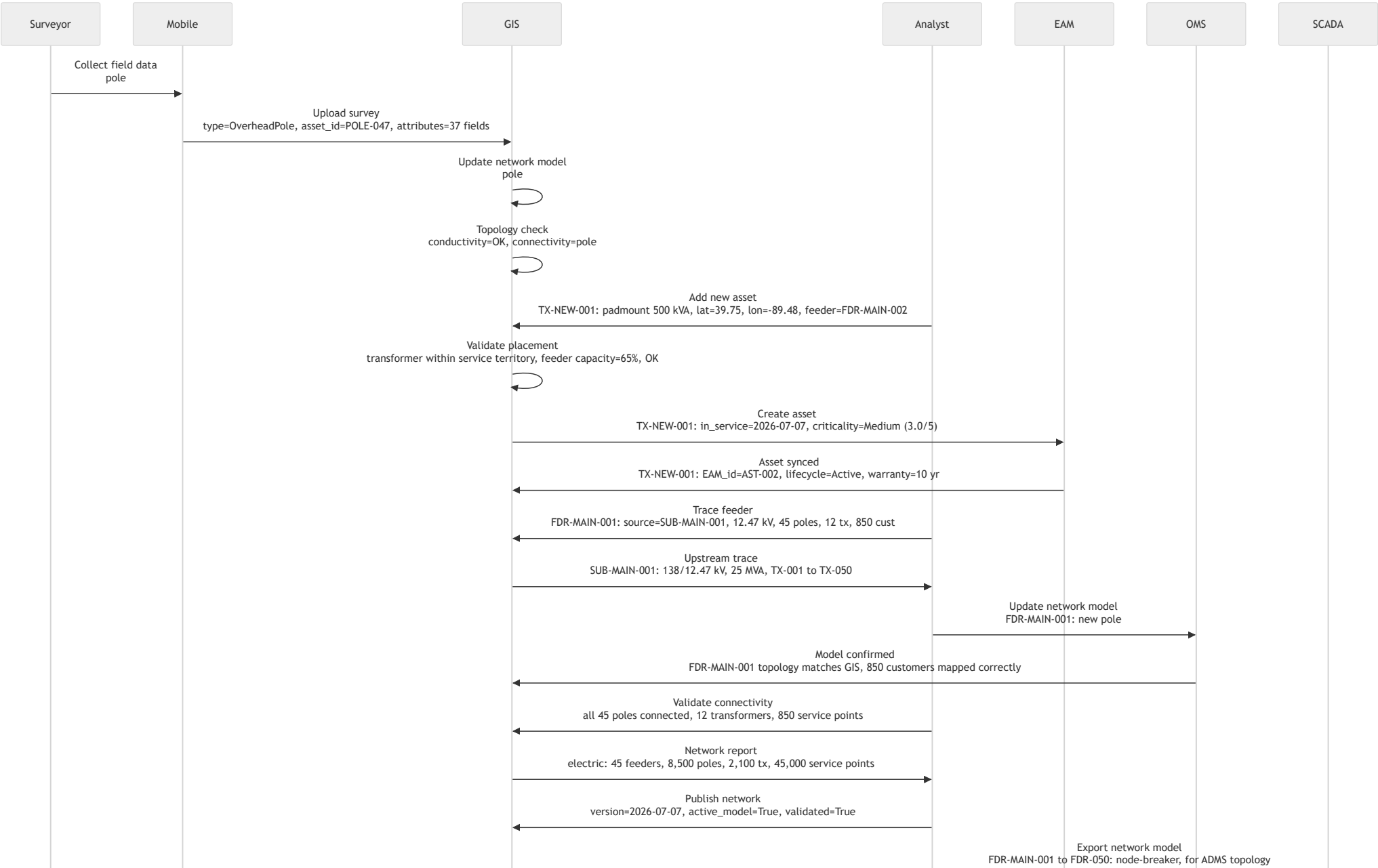
12.2 GIS Network Model

Description: GIS team manages the network model for electric (primary/secondary, substations, feeders, transformers) and water (mains, hydrants, valves, meters) distribution systems.

Actors: GIS Analyst, Network Modeler, Field Surveyor

Systems: GIS (ArcGIS Utility Network), EAM (Maximo), SCADA, OMS, Mobile Data Collector

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
GIS	Survey	UploadFieldSurvey	Mobile Data Collector	GIS (ArcGIS Utility Network)	Mobile App API (OAuth2)	GIS API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
GIS	Topology	ValidateNetworkTopology	GIS (ArcGIS Utility Network)	GIS (ArcGIS Utility Network)	Internal	Internal	Batch (Real-time)	High	High-frequency near-real-time batch processing with minimal latency tolerance
GIS	Asset	CreateGISAsset	GIS (ArcGIS Utility Network)	EAM (IBM Maximo)	GIS API (REST)	EAM API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
GIS	Trace	TraceFeederSource	GIS Analyst HMI	GIS (ArcGIS Utility Network)	Analyst UI	GIS API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
GIS	OMS	ExportNetworkToOMS	GIS (ArcGIS Utility Network)	OMS	GIS API (REST)	OMS API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
GIS	SCADA	ExportNetworkToSCADA	GIS (ArcGIS Utility Network)	SCADA	GIS API (REST)	SCADA API (ICCP)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
GIS	Report	GenerateNetworkReport	GIS (ArcGIS Utility Network)	GIS Analyst HMI	GIS API (REST)	Analyst UI	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-GIS-01	Validate GIS network connectivity (100% of assets have valid parent-child topology)	ValidateNetworkTopology	Topology check passes for all network features with 0 errors	High
AC-GIS-02	Confirm field survey data integrated into GIS within 24 hours of collection	UploadFieldSurvey	Survey data available in GIS within 24 hrs of field upload	Medium
AC-GIS-03	Verify feeder trace returns complete upstream path from any service point to source	TraceFeederSource	Trace returns all intermediate nodes from service point to substation	Medium
AC-GIS-04	Ensure OMS network model reflects GIS changes within 1 hour of GIS publish	ExportNetworkToOMS	OMS topology matches GIS within 1 hr of publication	High

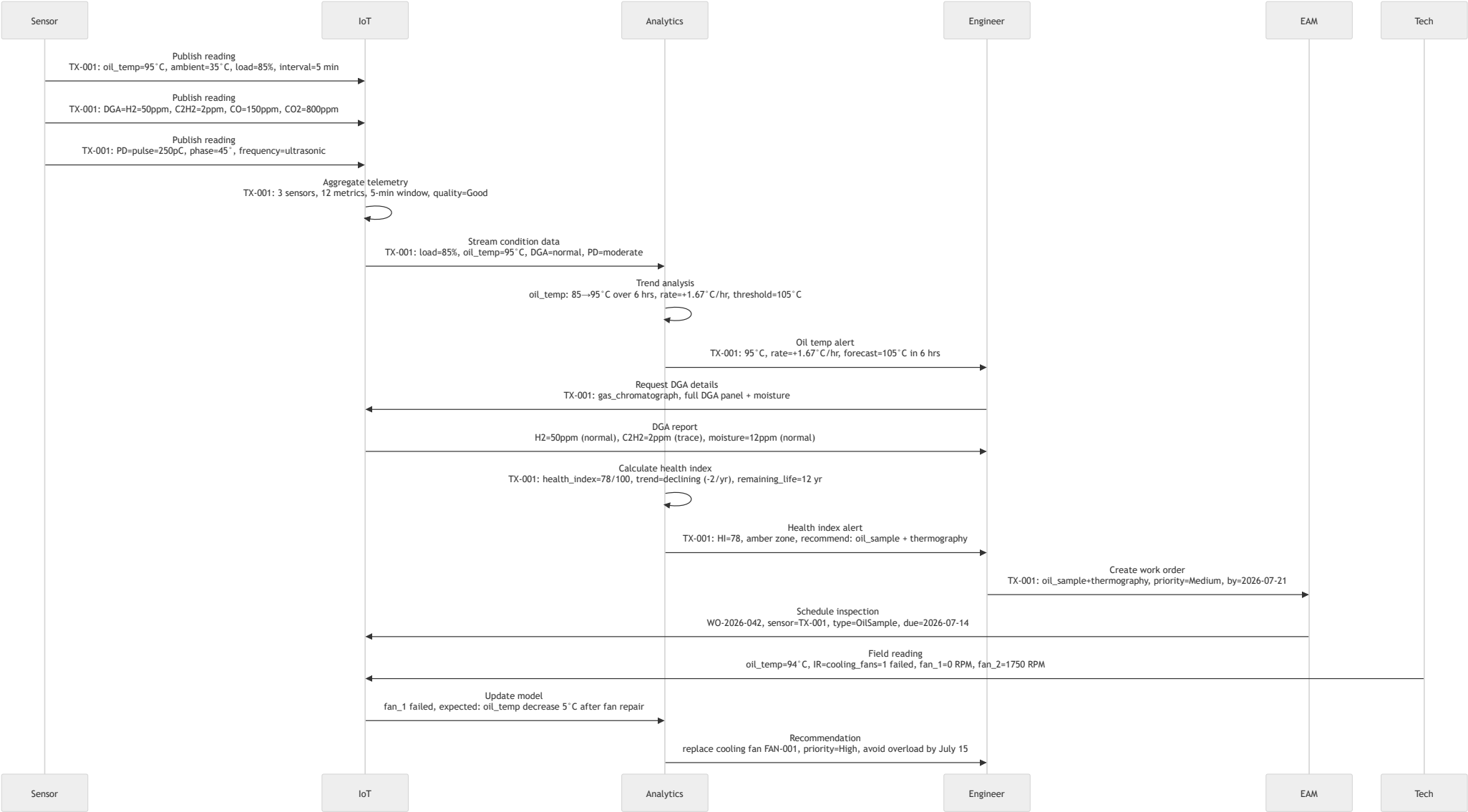
12.3 IoT Condition Monitoring

Description: IoT sensors monitor asset condition in real-time: transformer temperature, dissolved gas, partial discharge, pole tilt, pipe corrosion, and pump vibration.

Actors: Condition Monitoring Engineer, Asset Manager, Field Technician

Systems: IoT Platform (C3 AI / AWS IoT), EAM (Maximo), SCADA, Sensor Network, Analytics Platform

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
CMI	Sensor	PublishSensorReading	Sensor (IoT)	IoT Platform (C3 AI / AWS IoT)	MQTT (JSON)	IoT API (MQTT/HTTP)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
CMI	DGA	PublishDGADData	DGA Monitor	IoT Platform (C3 AI / AWS IoT)	DGA Serial (Modbus)	IoT API (MQTT/HTTP)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
CMI	PD	PublishPartialDischarge	PD Sensor (HVPD)	IoT Platform (C3 AI / AWS IoT)	PD Sensor (IEC 61850)	IoT API (MQTT/HTTP)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
CMI	Analytics	StreamConditionData	IoT Platform (C3 AI / AWS IoT)	Analytics Platform	IoT API (REST)	Analytics API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
CMI	Alert	GenerateConditionAlert	Analytics Platform	Condition Monitoring Engineer	Analytics API (REST)	Engineer HMI	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
CMI	WorkOrder	CreateInspectionWorkOrder	Condition Monitoring Engineer	EAM (IBM Maximo)	Engineer UI	EAM API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
CMI	Field	LogFieldInspectionData	Field Technician (Mobile)	IoT Platform (C3 AI / AWS IoT)	Mobile App API (OAuth2)	IoT API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-CMI-01	Validate sensor data latency under 60 seconds from sensor to IoT platform	PublishSensorReading	IoT platform timestamp within 60s of sensor reading timestamp	High
AC-CMI-02	Confirm DGA analysis identifies fault gases (H2, C2H2, CO, CO2, C2H4, CH4) within 24 hours	PublishDGADData	Full DGA panel reported with IEEE C57.104 interpretation	High
AC-CMI-03	Verify health index calculation uses at least 3 condition parameters	CalculateHealthIndex	Health index weighted: DGA (30%), PD (25%), oil_temp (25%), load (20%)	Medium
AC-CMI-04	Ensure alert-to-work-order creation completes within 1 hour of condition threshold breach	GenerateConditionAlert	Work order created in EAM within 1 hr of analytics alert	Medium

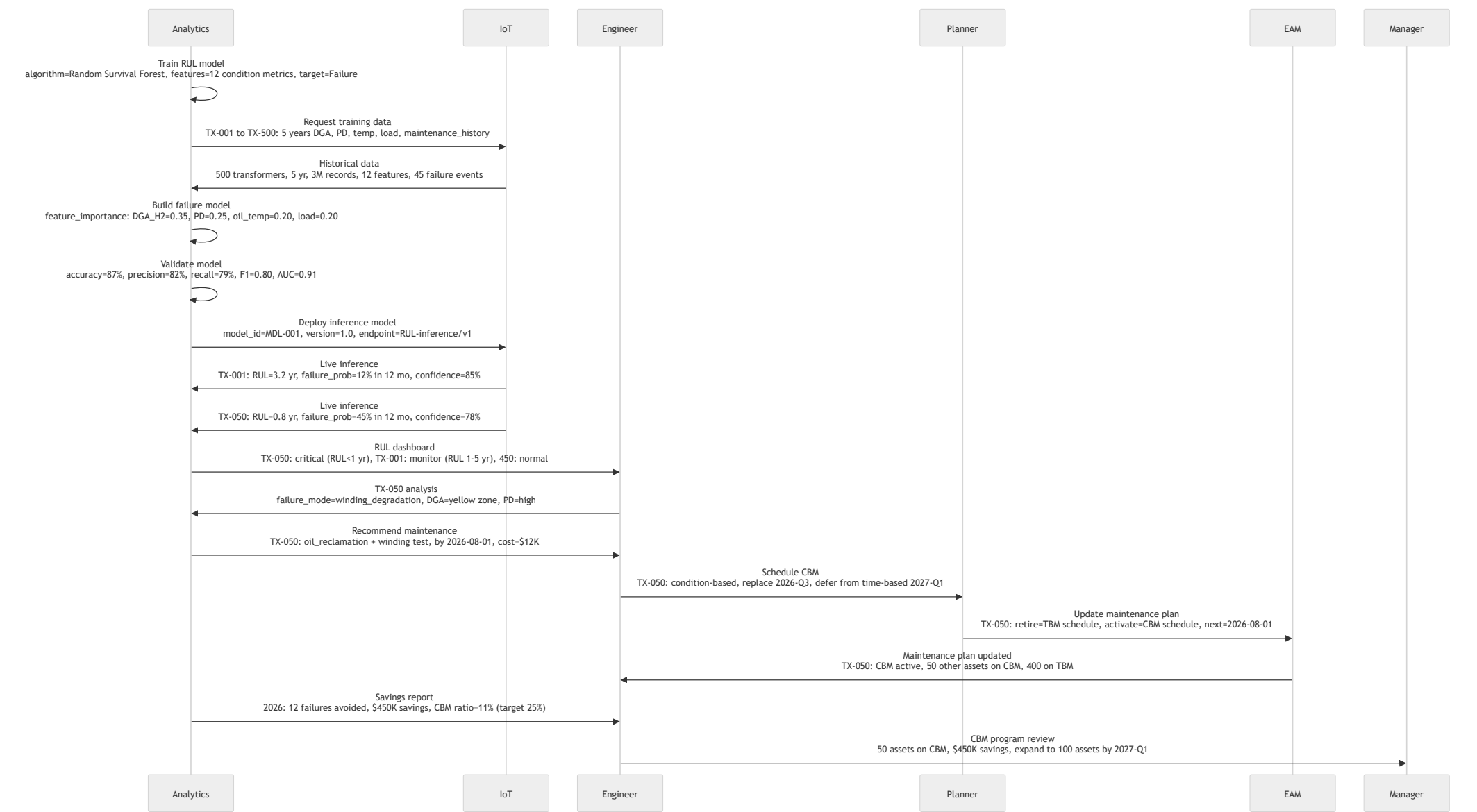
12.4 Predictive Maintenance & CBM

Description: Using condition data and AI models, the utility predicts asset failures, calculates remaining useful life (RUL), and optimizes maintenance schedules to transition from time-based to condition-based maintenance.

Actors: Reliability Engineer, Asset Manager, Maintenance Planner

Systems: EAM (Maximo), Analytics Platform (SAS / Azure ML), IoT Platform, SCADA Historian, CMMS

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
PMC	Model	TrainRULModel	Analytics Platform (SAS / Azure ML)	Analytics Platform (SAS / Azure ML)	Internal	Internal	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
PMC	Data	RequestTrainingData	Analytics Platform (SAS / Azure ML)	IoT Platform (C3 AI / AWS IoT)	Analytics API (REST)	IoT API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
PMC	Deploy	DeployInferenceModel	Analytics Platform (SAS / Azure ML)	IoT Platform (C3 AI / AWS IoT)	Analytics API (REST)	IoT API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
PMC	Inference	RunLiveInference	IoT Platform (C3 AI / AWS IoT)	Analytics Platform (SAS / Azure ML)	IoT API (REST)	Analytics API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
PMC	Alert	AlertRULCritical	Analytics Platform (SAS / Azure ML)	Reliability Engineer	Analytics API (REST)	Engineer HMI	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
PMC	Schedule	UpdateCBMSchedule	Maintenance Planner	EAM (IBM Maximo)	Planner UI	EAM API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
PMC	Report	GenerateCBMSavingsReport	Analytics Platform (SAS / Azure ML)	Asset Manager	Analytics API (REST)	Manager UI	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-PMC-01	Validate RUL prediction accuracy within ±15% of actual failure date (historical backtest)	TrainRULModel	Backtested RUL predictions within 15% of actual failure dates	High
AC-PMC-02	Confirm inference model processes 100% of monitored assets within each 24-hour cycle	RunLiveInference	All 500 transformers scored with RUL within daily batch window	High
AC-PMC-03	Verify maintenance plan transition from TBM to CBM is trac	UpdateCBMSchedule	EAM audit trail shows TBM termination and CBM activation dates	Medium

AC ID	Description	Related Info Flow	Expected Outcome	Priority
	ked with before/after metrics			
AC-PMC-04	Ensure CBM savings report includes failures avoided, maintenance cost reduction, and ROI	GenerateCBMSavingsReport	Report shows quantitative savings with auditable failure avoidance evidence	Medium

Enterprise Resource Planning - UT-FIN UT-HRP UT-PSC UT-INV

Manages enterprise financial operations, human resources, payroll, procurement, supply chain, inventory, and warehousing for the multi-utility organization (electric + water).

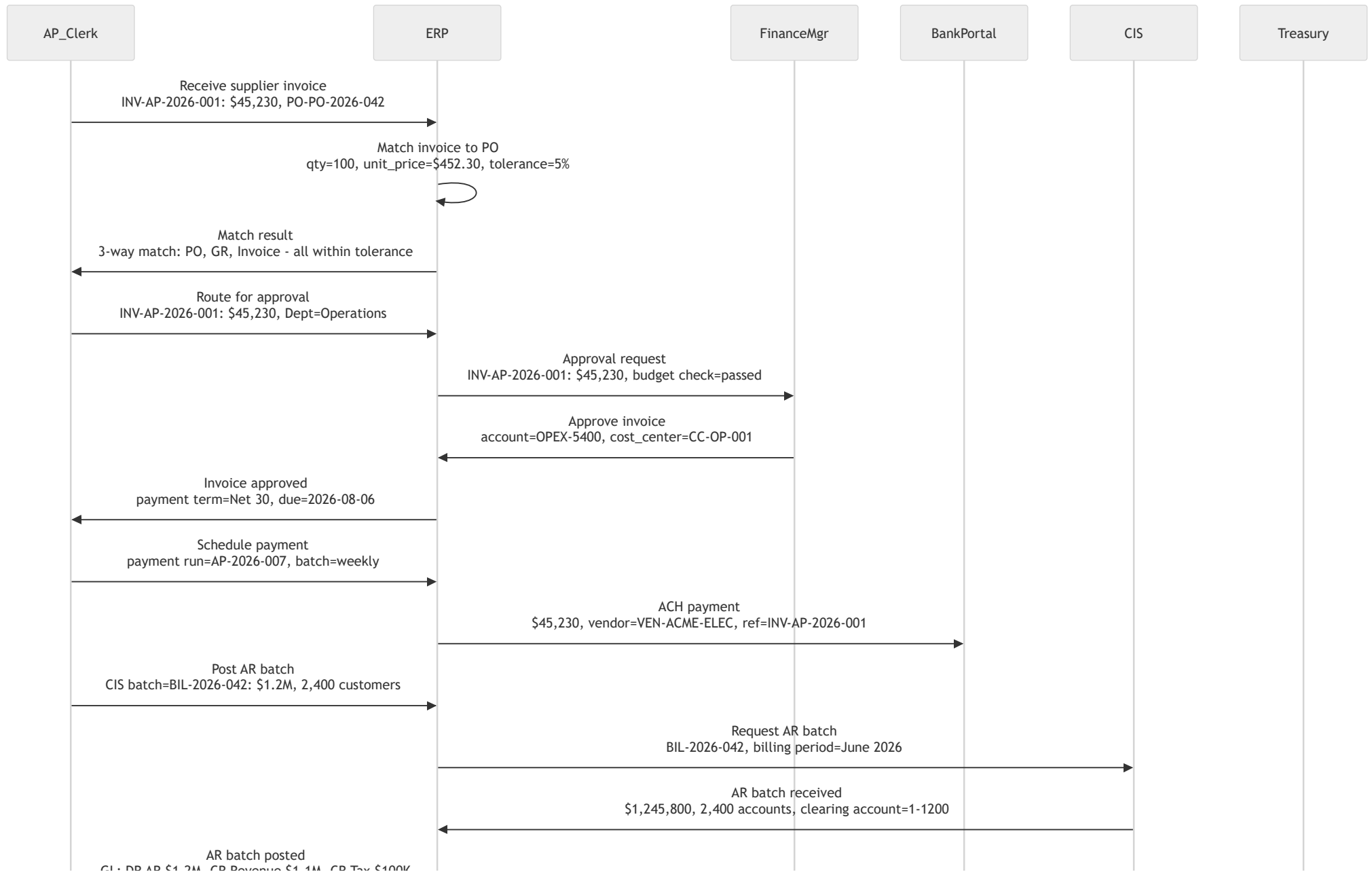
13.1 Financial Management & Budgeting

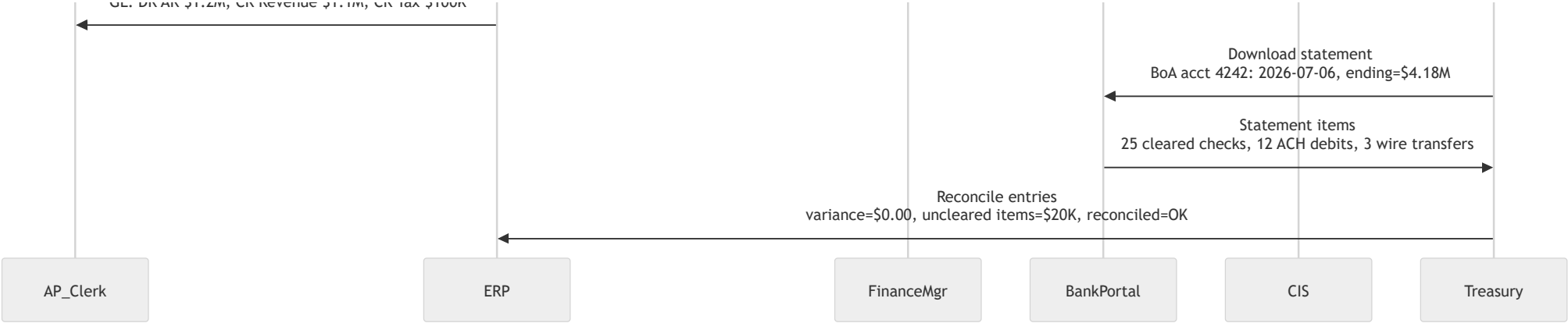
Description: AP invoice processing workflow from receipt through matching, approval, and payment. AR batch from CIS billing, GL journal entries, monthly close, and budget planning cycle where departments submit requests, finance consolidates, and leadership approves. Cash management reconciliations between bank portal and ERP.

Actors: AP Clerk, Finance Manager, Department Budget Owner, Treasury Analyst

Systems: ERP (SAP S/4HANA), CIS (Oracle Utilities), Bank Portal (BoA), Budget System (SAP BPC)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Financial	AP	MatchInvoiceToPO	ERP (SAP S/4HANA)	CIS (Oracle Utilities)	SAP SOAP	CIS SOAP	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Financial	AR	ReceiveARBatch	CIS (Oracle Utilities)	ERP (SAP S/4HANA)	CIS REST API	SAP REST API	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Financial	GL	PostJournalEntry	ERP (SAP S/4HANA)	ERP (SAP S/4HANA)	Internal	Internal	Batch (Real-time)	Medium	Periodic near-real-time batch with data validation and transformation needs
Financial	Budget	SubmitBudgetRequest	Department Budget Owner	Budget System (SAP BPC)	Budget Web UI	SAP BPC API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Financial	Cash	ReconcileCash	ERP (SAP S/4HANA)	Bank Portal (BoA)	SAP SFTP	Bank Portal SFTP	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
Financial	Payment	ExecutePaymentRun	ERP (SAP S/4HANA)	Bank Portal (BoA)	SAP EDI	Bank Portal EDI	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Financial	Close	RunMonthlyClose	ERP (SAP S/4HANA)	Reporting Database	SAP JDBC	Reporting SQL JDBC	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-FIN-01	Validate 3-way PO match tolerance within configured thresholds (±5% price, ±10% qty)	MatchInvoiceToPO	All invoice matches completed within tolerance before approval	High
AC-FIN-02	Verify AR batch posting balances to GL within ±\$0.01	ReceiveARBatch	Batch totals match GL posting with zero variance	High
AC-FIN-03	Confirm bank reconciliation completes daily with variance resolution	ReconcileCash	Daily reconciliation completed with all variances investigated	Medium
AC-FIN-04	Ensure budget submission workflow enforces department manager approval before consolidation	SubmitBudgetRequest	All budget submissions approved by department manager before finance consolidation	Medium

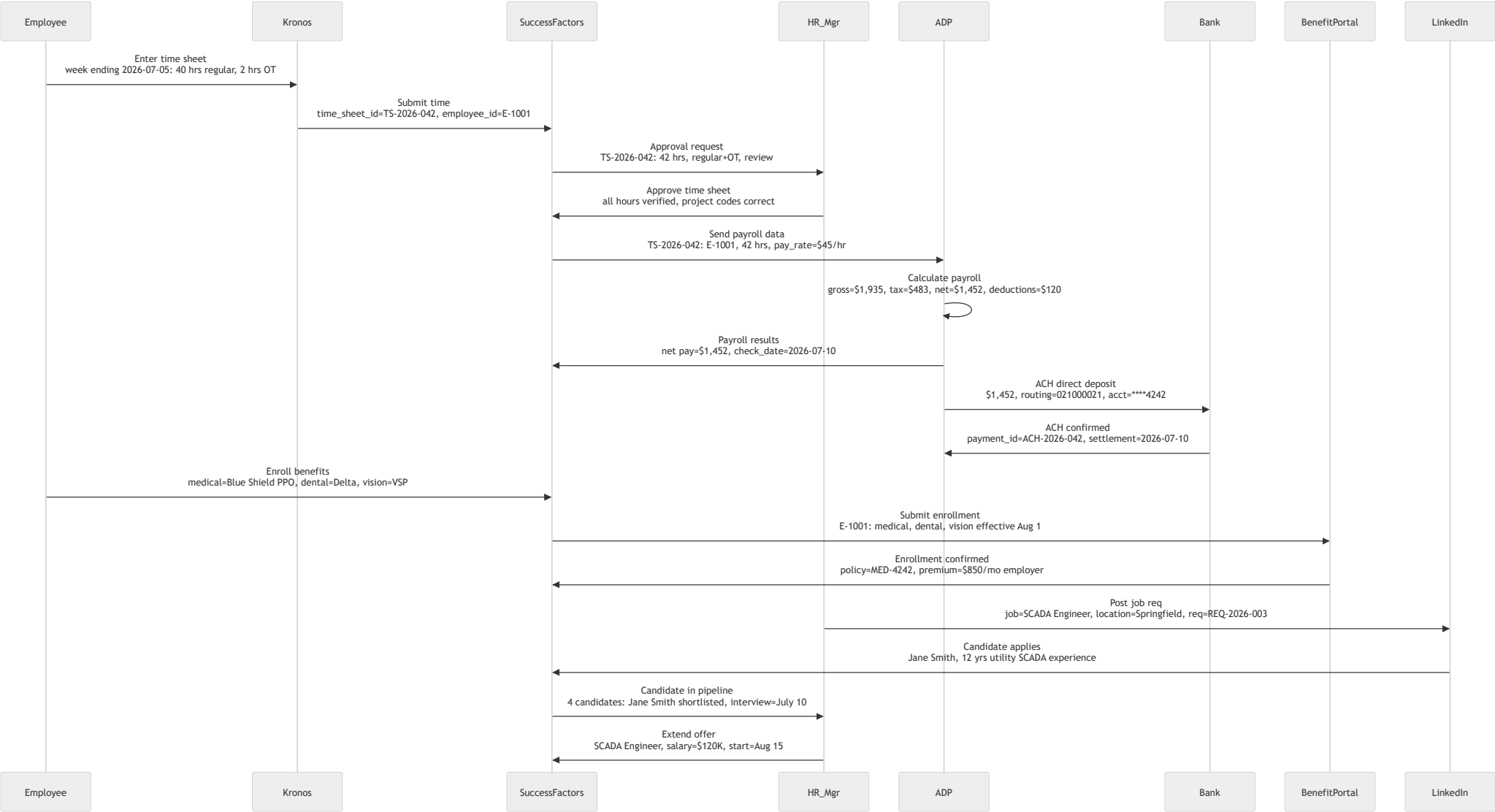
13.2 HR & Payroll

Description: Employee time entry through HR portal with supervisor approval and SAP processing. Payroll calculation and disbursement from HR to SAP Payroll to ADP to bank. Benefits enrollment, recruiting lifecycle, training enrollment, and performance review cycle management.

Actors: Employee, HR Manager, Payroll Specialist, Benefits Admin

Systems: ERP (SAP SuccessFactors), Payroll (ADP), Benefit Carrier Portal, Bank, Time System (Kronos)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
HR	Time	ImportTimeSheet	Time System (Kronos)	ERP (SAP SuccessFactors)	Kronos OData	SuccessFactors OData	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
HR	Payroll	CalculatePayroll	ERP (SAP SuccessFactors)	Payroll (ADP)	SuccessFactors SFTP	ADP SFTP	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
HR	Disbursement	DisburseNetPay	Payroll (ADP)	Bank	ADP EDI	Bank EDI	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
HR	Benefits	EnrollBenefits	ERP (SAP SuccessFactors)	Benefit Carrier Portal	SuccessFactors API (REST)	Carrier API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
HR	Recruiting	PostJobRequisition	LinkedIn	ERP (SAP SuccessFactors)	LinkedIn API (REST)	SuccessFactors API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
HR	Training	EnrollInTraining	ERP (SAP SuccessFactors)	LMS (Cornerstone)	SuccessFactors SCORM	LMS SCORM	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
HR	OrgChart	MaintainOrgChart	ERP (SAP SuccessFactors)	ERP (SAP SuccessFactors)	Internal	Internal	Batch (Real-time)	Simple	Minimal near-real-time batch with basic data pass-through

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-HRP-01	Validate time sheet approval completes within 48 hours of employee submission	ImportTimeSheet	All time sheets approved within 48 hrs of period end	Medium
AC-HRP-02	Verify payroll calculation accuracy within ±\$0.01 of manual calculation for test cases	CalculatePayroll	Payroll register matches calculated gross/net/deductions exactly	High
AC-HRP-03	Confirm ACH direct deposit settles on scheduled pay date	DisburseNetPay	Bank confirms ACH settlement on scheduled pay date	High
AC-HRP-04	Ensure benefits enrollment changes are reflected in carrier systems within 24 hours	EnrollBenefits	Carrier portal confirms enrollment within 24 hrs of submission	Medium

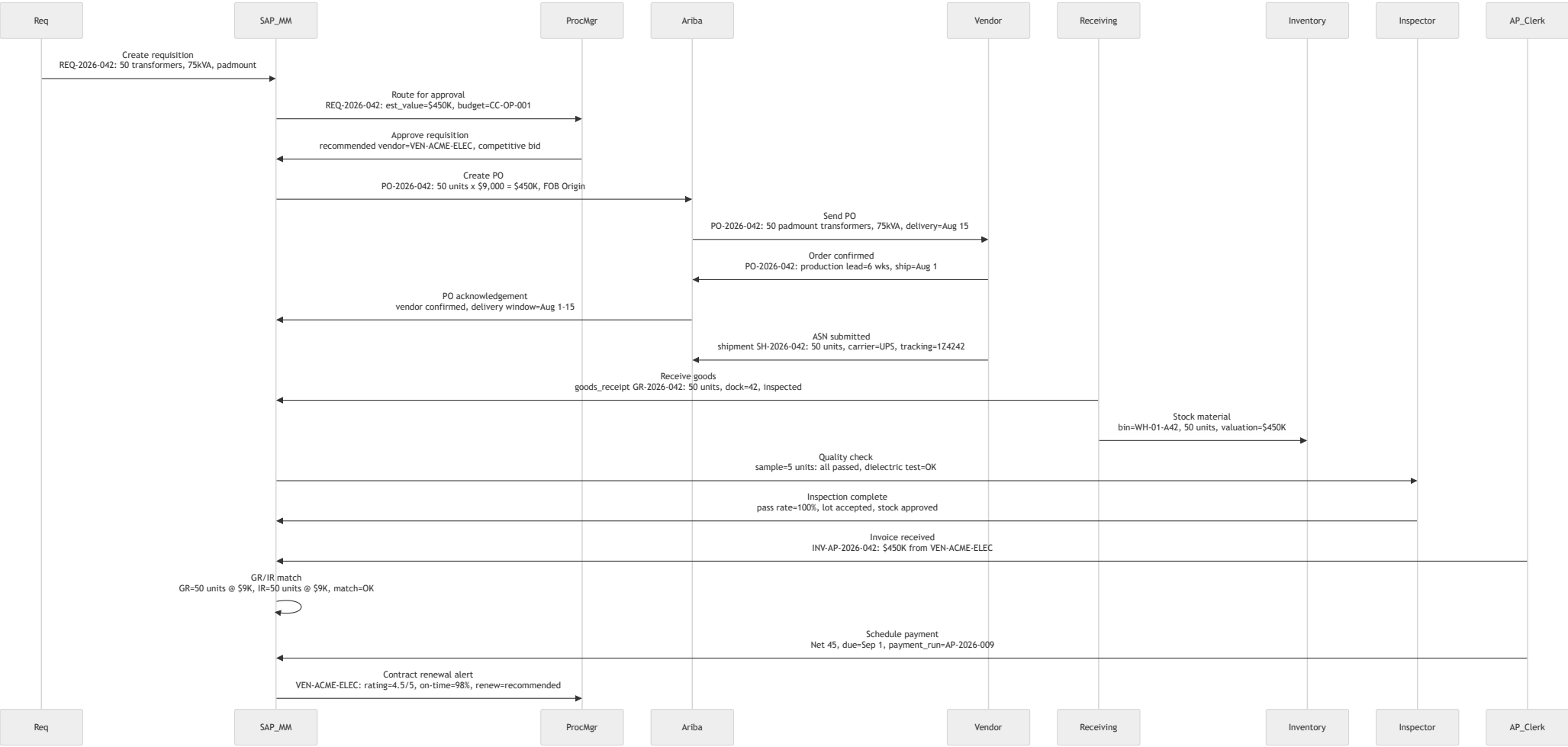
13.3 Procurement & Supply Chain

Description: End-to-end procurement process: requisition through approval, purchase order creation, vendor confirmation, shipment, receipt, inspection, stock, invoice matching, and payment. Includes emergency procurement for outage materials, vendor evaluation scoring, and contract management lifecycle.

Actors: Requisitioner, Procurement Manager, Vendor, Receiving Clerk, AP Clerk

Systems: ERP (SAP MM/SRM), Vendor Portal (Ariba), EAM (Maximo), Inventory System, GRIR Reconciliation

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Procurement	Requisition	CreateRequisition	Requisitioner	ERP (SAP MM/SRM)	SAP Web UI	SAP Internal	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Procurement	PO	SendPurchaseOrder	ERP (SAP MM/SRM)	Vendor Portal (Ariba)	SAP cXML	Ariba cXML	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Procurement	ASN	ReceiveAdvancedShipNotice	Vendor Portal (Ariba)	ERP (SAP MM/SRM)	Ariba EDI 856	SAP EDI	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
Procurement	Receipt	ReceiveGoods	Receiving Clerk (RF Scan)	ERP (SAP MM/SRM)	RF Scanner API	SAP REST API	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Procurement	Invoice	MatchInvoiceGRIR	ERP (SAP MM/SRM)	ERP (SAP MM/SRM)	Internal	Internal	Batch (Real-time)	High	High-frequency near-real-time batch processing with minimal latency tolerance
Procurement	Contract	CreateContract	ERP (SAP MM/SRM)	ERP (SAP MM/SRM)	Internal	Internal	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Procurement	Eval	EvaluateVendor	ERP (SAP MM/SRM)	Supplier Portal	SAP REST API	Supplier Portal API (REST)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-PSC-01	Validate PO-to-GR-to-IR 3-way match within tolerance before payment processing	MatchInvoiceGRIR	No payment processed without complete GR/IR match	High
AC-PSC-02	Verify emergency procurement requests are processed within 4 hours for outage materials	CreateRequisition	Emergency PO issued within 4 hrs of requisition approval	High
AC-PSC-03	Confirm ASN receipt triggers automatic dock scheduling 24+ hours before delivery	ReceiveAdvancedShipNotice	ASN processed and dock slot assigned before shipment arrival	Medium
AC-PSC-04	Ensure vendor evaluations are updated quarterly with on-time delivery and quality scores	EvaluateVendor	All active vendors scored within 45 days of quarter end	Medium

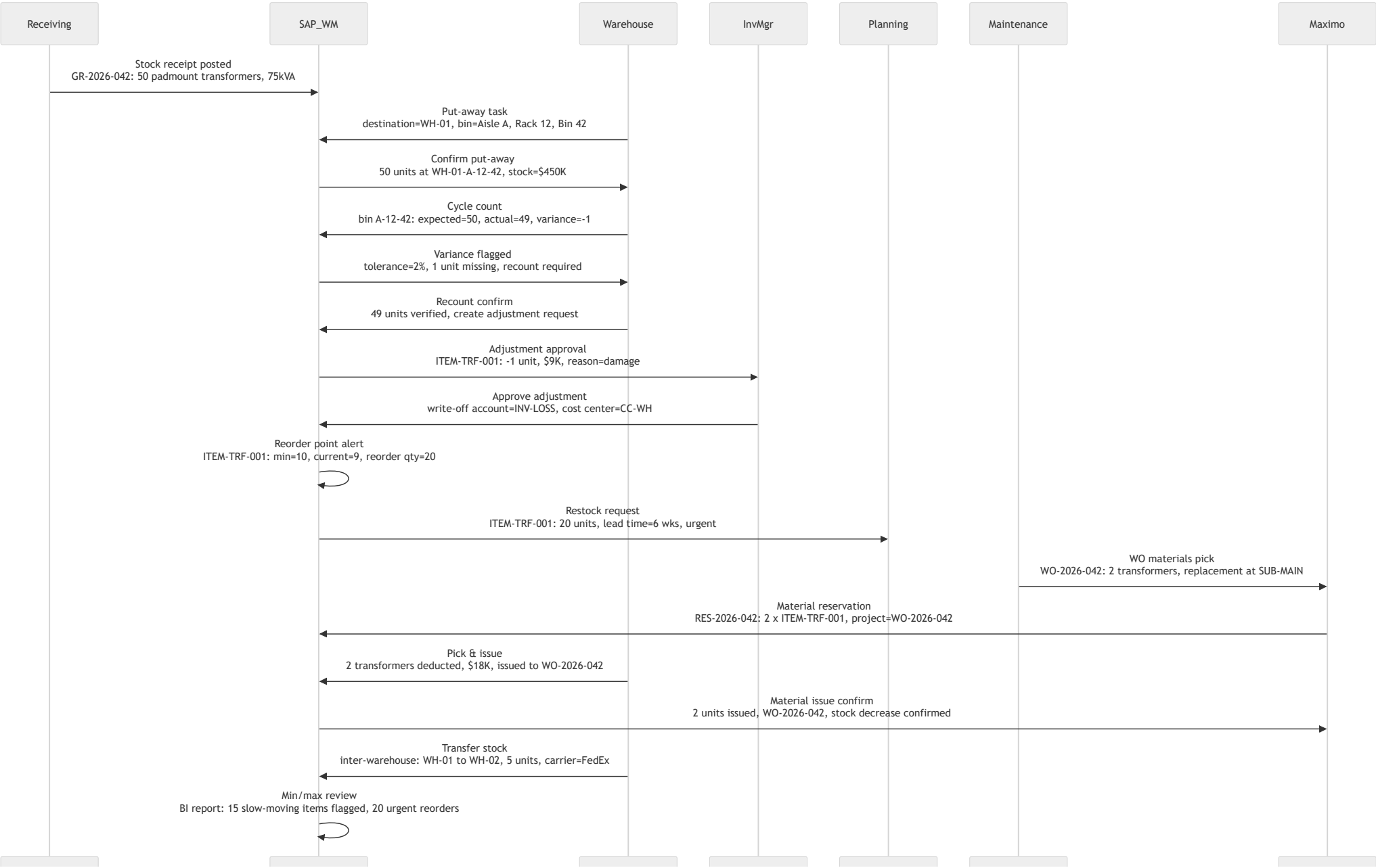
13.4 Inventory & Warehousing

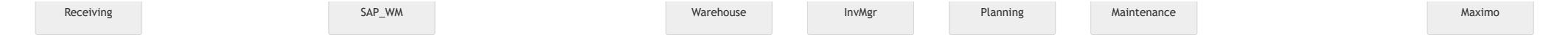
Description: Stock receipt through put-away, cycle counting, reorder point monitoring, picking for work order issue, inter-warehouse transfer, and inventory adjustments. Critical spares management for transmission transformers, substation breakers, and SCADA components with min/max stock replenishment.

Actors: Warehouse Clerk, Inventory Manager, Maintenance Planner

Systems: ERP (SAP WM), EAM (Maximo), Barcode/RF Scanner, Inventory Planning, Logistics (UPS/FedEx)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Inventory	Stock	ExecuteStockTransfer	ERP (SAP WM)	ERP (SAP WM)	SAP Internal	SAP Internal	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
Inventory	Count	PerformCycleCount	Barcode/RF Scanner	ERP (SAP WM)	Scanner API (REST)	SAP REST API	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Inventory	Reorder	TriggerReorderPoint	ERP (SAP WM)	Inventory Planning	SAP Internal	Planning API (REST)	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
Inventory	Issue	IssueMaterialToWO	ERP (SAP WM)	EAM (Maximo)	SAP REST API	Maximo REST API	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Inventory	Review	RunMinMaxReview	ERP (SAP WM)	BI Reporting	SAP JDBC	BI SQL JDBC	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints
Inventory	PhysInv	ConfirmPhysicalInventory	Barcode/RF Scanner	ERP (SAP WM)	Scanner RF API	SAP REST API	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
Inventory	Reserve	CreateStockReservation	EAM (Maximo)	ERP (SAP WM)	Maximo REST API	SAP REST API	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation

Acceptance Criteria

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AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-INV-01	Validate cycle count accuracy within ±1% of system inventory for A-class items	PerformCycleCount	A-class items counted monthly with <1% variance	High
AC-INV-02	Verify reorder point triggers purchase requisition within 1 hour of threshold breach	TriggerReorderPoint	Requisition created within 1 hr of stock below reorder point	Medium
AC-INV-03	Confirm stock-out rate for critical spares remains below 2% annually	RunMinMaxReview	Critical spares available 98% of time when requested	High
AC-INV-04	Ensure material issue to work order updates both SAP inventory and Maximo within 5 minutes	IssueMaterialToWO	Inventory balance and WO materials synchronized within 5 min	Medium

Cybersecurity & Compliance - UT-IAM UT-SSM UT-VPM UT-NCC

Manages cybersecurity across OT and IT domains: identity and access management, security monitoring (SIEM), vulnerability and patch management, and NERC CIP compliance for critical infrastructure protection.

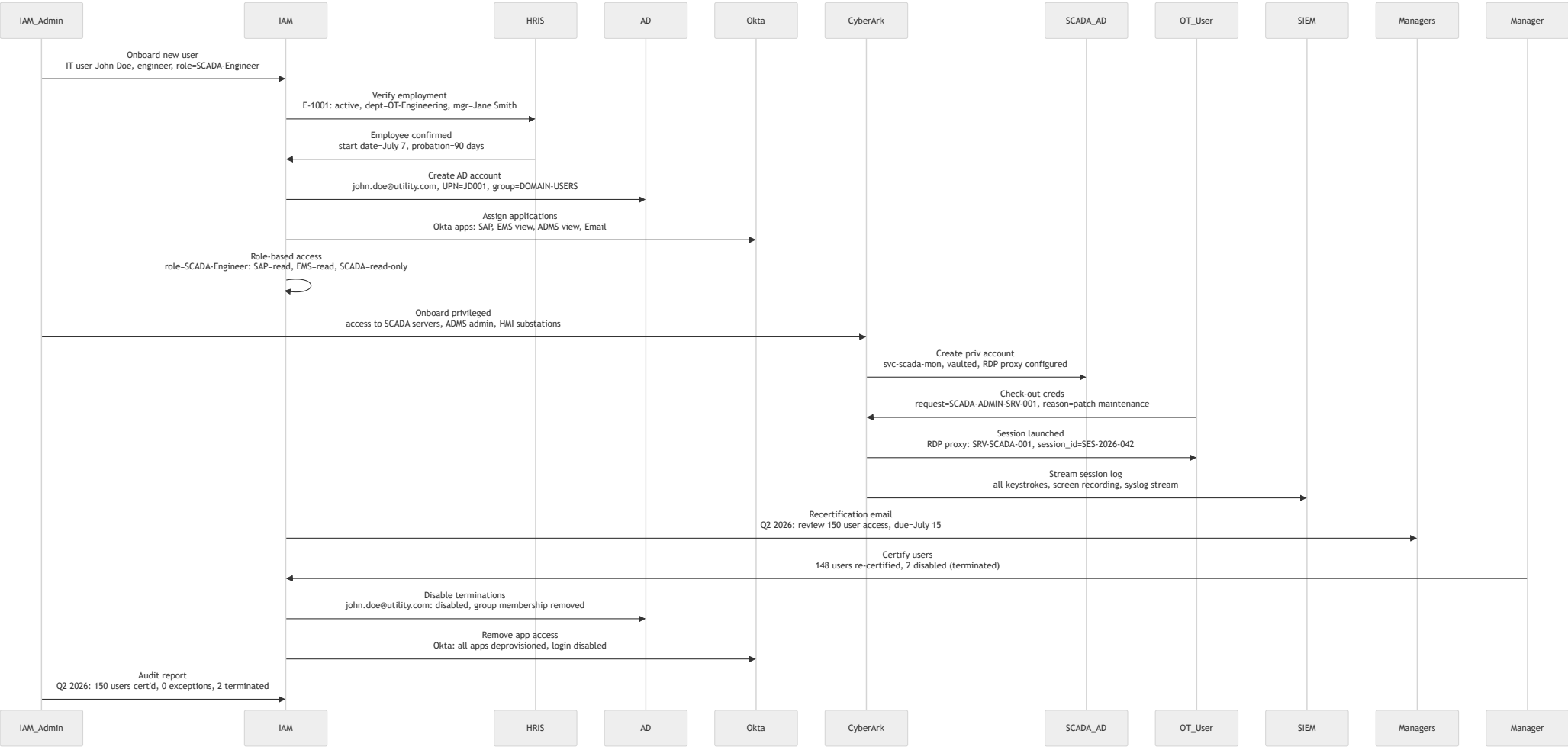
14.1 Identity & Access Management (IAM/PAM)

Description: User onboarding through role assignment, access provisioning for IT applications and OT systems, privileged access vaulting via PAM, session monitoring, periodic access recertification, and deprovisioning on termination. Privileged access to substation HMIs, SCADA servers, and ADMS consoles managed with CyberArk.

Actors: IAM Admin, OT User, IT User, Security Auditor, PAM Administrator

Systems: IAM (Okta/SailPoint), PAM (CyberArk), AD/Azure AD, SCADA AD, HMI Workstations, HRIS (SuccessFactors)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
IAM	Provision	ProvisionUserFromHR	HRIS (SuccessFactors)	IAM (Okta/SailPoint)	SuccessFactors SCIM	IAM SCIM	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
IAM	Access	AssignApplicationAccess	IAM (Okta/SailPoint)	Okta	IAM REST API	Okta REST API	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
IAM	PAM	CheckoutPrivilegedAccount	PAM (CyberArk)	SCADA AD	CyberArk RDP Proxy	SCADA RDP	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
IAM	Recert	RunAccessRecertification	IAM (Okta/SailPoint)	Manager Email	IAM REST API	Email Webhook (SMTP)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
IAM	Deprovision	DeprovisionOnTermination	IAM (Okta/SailPoint)	AD/Azure AD	IAM REST API	AD LDAP	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
IAM	Session	RecordPrivilegedSession	PAM (CyberArk)	SIEM (Splunk)	CyberArk Syslog	SIEM Syslog	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
IAM	RoleMining	AnalyzeRoleAssignments	IAM (Okta/SailPoint)	AD/Azure AD	IAM LDAP	AD LDAP	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-IAM-01	Verify user provisioning completes within 4 hours of HR onboarding event	ProvisionUserFromHR	Active Directory account created and applications assigned within 4 hrs	High
AC-IAM-02	Confirm privileged access session recording captures all keystrokes and screen activity	RecordPrivilegedSession	SIEM contains complete session recording for every PAM session	High
AC-IAM-03	Validate quarterly access recertification completed with 100% manager attestation	RunAccessRecertification	All active users recertified within 45 days of quarter end	Medium
AC-IAM-04	Ensure terminated user access is revoked within 2 hours of HR notification	DeprovisionOnTermination	AD disabled, Okta deprovisioned within 2 hrs of termination event	High

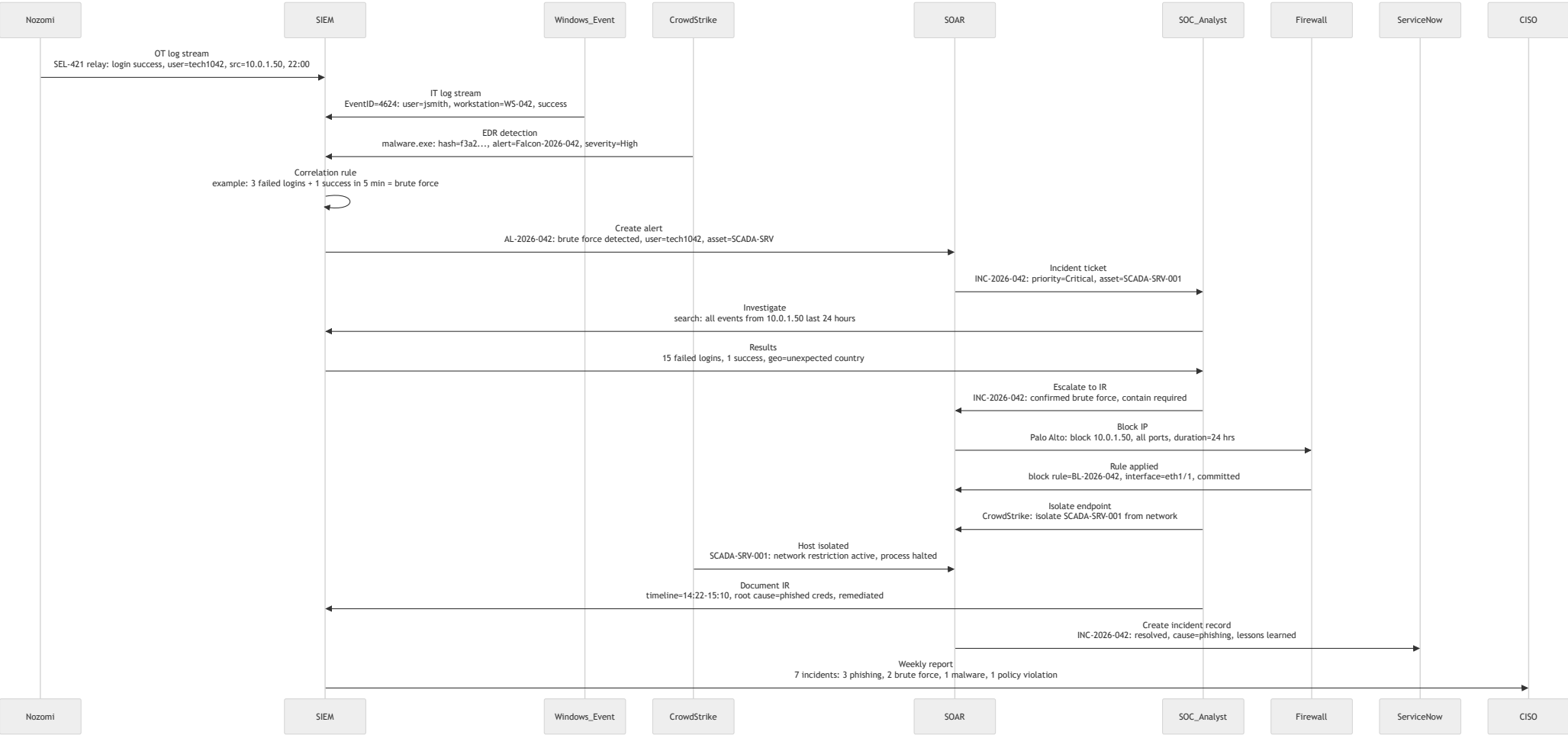
14.2 Security Monitoring & SIEM

Description: Log collection from OT environments (firewalls, IEDs, RTUs) and IT (servers, applications, endpoints) into SIEM for correlation, threat detection, alert triage, investigation, incident response, remediation, and documentation. Phishing incident response walkthrough included.

Actors: SOC Analyst, Threat Hunter, OT Security Engineer, CISO

Systems: SIEM (Splunk/ArcSight), OT Monitoring (Nozomi/Claroty), EDR (CrowdStrike), Firewall (Palo Alto), SOAR

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
SIEM	OT	IngestOTLogs	OT Monitoring (Nozomi/Claroty)	SIEM (Splunk/ArcSight)	Nozomi Syslog	SIEM Syslog	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
SIEM	IT	IngestITLogs	Windows Event Collector	SIEM (Splunk/ArcSight)	WinCollect	SIEM Syslog	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
SIEM	ThreatIntel	ReceiveThreatIntelligence	Threat Feed (AlienVault)	SIEM (Splunk/ArcSight)	Threat Feed API (STIX)	SIEM API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
SIEM	Alert	CreateSOARAlert	SIEM (Splunk/ArcSight)	SOAR	SIEM REST API	SOAR REST API	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
SIEM	Ticket	CreateIncidentTicket	SOAR	ServiceNow	SOAR REST API	ServiceNow REST API	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
SIEM	Block	BlockIndicators	SOAR	Firewall (Palo Alto)	SOAR REST API	PAN-OS REST API	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
SIEM	Report	GenerateComplianceReport	SIEM (Splunk/ArcSight)	Compliance Email	SIEM REST API	SMTP Email	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-SSM-01	Validate OT log ingestion under 5 seconds from Nozomi to SIEM for critical assets	IngestOTLogs	SIEM receives OT events within 5s of Nozomi detection	High
AC-SSM-02	Confirm SOAR playbook executes containment actions within 5 minutes of alert	BlockIndicators	Firewall block rule applied within 5 min of alert creation	High
AC-SSM-03	Verify correlation rules cover all MITRE ATT&CK techniques relevant to utility OT	CreateSOARAlert	100% of defined ICS ATT&CK techniques have active correlation rules	Medium
AC-SSM-04	Ensure incident tickets include full timeline, root cause, and remediation steps	CreateIncidentTicket	All closed incidents contain required IR fields	Medium

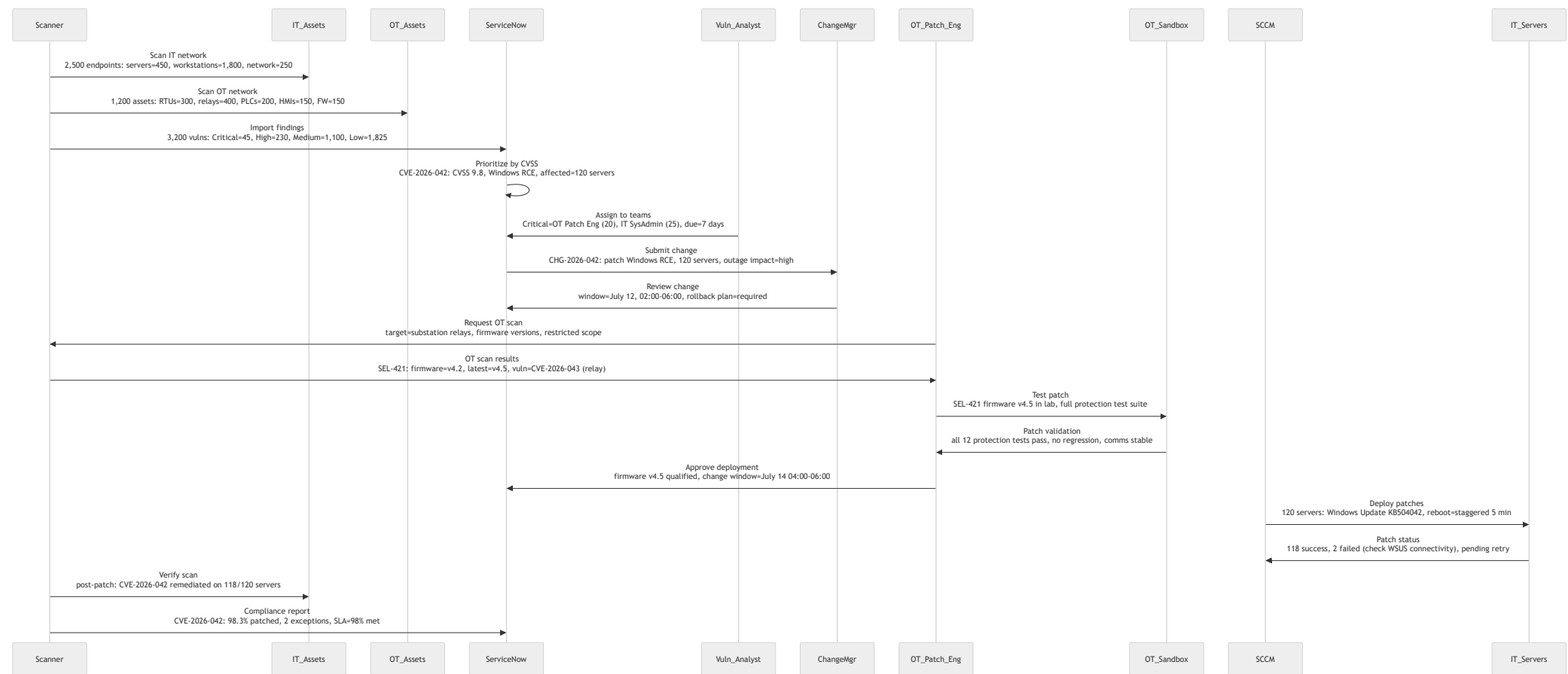
14.3 Vulnerability & Patch Management

Description: Vulnerability scanning across OT and IT networks, reporting, prioritization by CVSS, approval workflow, patch testing in sandbox, deployment to production, and verification scanning. OT patching considerations for substation relays, RTU firmware, and PLC logic with change management windows.

Actors: Vuln Mgmt Analyst, OT Patch Engineer, Change Manager, IT SysAdmin

Systems: Vulnerability Scanner (Tenable/Qualys), Patch Mgmt (WSUS/SCCM), OT Asset Inventory, ServiceNow Change Mgmt, EAM (Maximo)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
VulnMgmt	Scan	ScheduleVulnerabilityScan	Vulnerability Scanner (Tenable/Qualys)	IT/OT Assets	Scanner API (REST)	Asset Agent	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
VulnMgmt	Report	ImportVulnerabilityReport	Vulnerability Scanner (Tenable/Qualys)	ServiceNow Change Mgmt	Scanner REST API	ServiceNow REST API	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
VulnMgmt	Patch	DeployPatchSCCM	Patch Mgmt (WSUS/SCCM)	Windows Servers	SCCM GPO	Windows Update Agent	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
VulnMgmt	OTTest	TestOTPatchSandbox	Vulnerability Scanner (Tenable/Qualys)	OT Sandbox Environment	Scanner REST API (restricted)	OT Sandbox Agent	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
VulnMgmt	Change	CreateChangeTicket	ServiceNow Change Mgmt	ServiceNow Change Mgmt	Internal	Internal	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
VulnMgmt	Approval	ApprovePatchDeployment	ServiceNow Change Mgmt	Manager Email	ServiceNow Email	SMTP Email	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
VulnMgmt	Compliance	GenerateComplianceReport	Vulnerability Scanner (Tenable/Qualys)	GRC System	Scanner JDBC	GRC SQL JDBC	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-VPM-01	Validate critical vulnerability patches deployed within 7 days of discovery	DeployPatchSCCM	100% of critical vulns patched or exception-approved within 7 days	High
AC-VPM-02	Confirm OT patch testing in sandbox environment completes before production deployment	TestOTPatchSandbox	All protection tests pass and no regression observed in sandbox	High
AC-VPM-03	Verify patch compliance SLA of 98% coverage for critical and high severity vulnerabilities	GenerateComplianceReport	Patch coverage >=98% for critical and high vulns	Medium
AC-VPM-04	Ensure change management approval obtained for all patching with outage windows	CreateChangeTicket	All patch deployments have approved change ticket with rollback plan	High

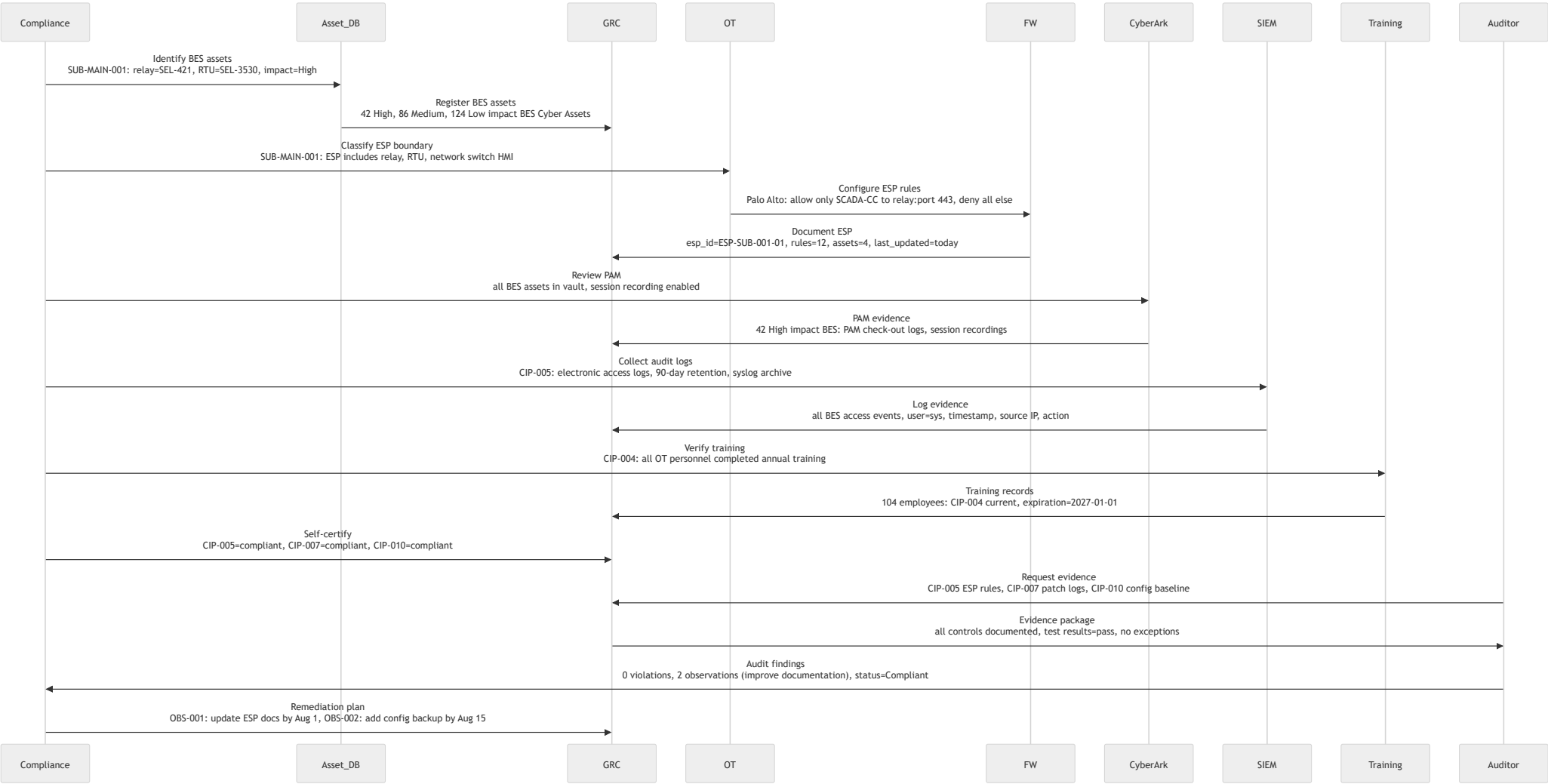
14.4 NERC CIP Compliance

Description: Identify BES Cyber Assets, classify by impact rating (High/Medium/Low), document Electronic Security Perimeter (ESP), implement controls for electronic access, physical security, and personnel training. Self-certification, audit preparation, remediation tracking, and evidence reporting for CIP-005 (ESP), CIP-007 (patch mgmt), and CIP-010 (configuration management).

Actors: CIP Compliance Lead, System Administrator, Auditor (Regional Entity), IT/OT Engineer

Systems: GRC (Archer/ServiceNow IRM), OT Asset Inventory, PAM (CyberArk), SIEM (Splunk), Training System, NERC Portal

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
CIP	Inventory	RegisterBESAssets	OT Asset Inventory	GRC (Archer/ServiceNow IRM)	Asset DB REST API	GRC REST API	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
CIP	ESP	ManageESPBoundary	OT Network	Firewall (Palo Alto)	OT REST API	PAN-OS API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
CIP	Evidence	CollectAuditEvidence	ARC (Asset/Log Source)	GRC (Archer/ServiceNow IRM)	ARC REST API	GRC REST API	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
CIP	Violation	TrackViolations	GRC (Archer/ServiceNow IRM)	NERC Portal	GRC REST API	NERC Web Portal	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
CIP	Training	ReportTrainingRecords	Training System (LMS)	GRC (Archer/ServiceNow IRM)	LMS SCORM	GRC REST API	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints
CIP	Remediation	TrackRemediation	ServiceNow	GRC (Archer/ServiceNow IRM)	ServiceNow REST API	GRC REST API	Event-driven	Medium	Event-driven integration with intermediate complexity in event routing and state management
CIP	Report	GenerateComplianceReport	GRC (Archer/ServiceNow IRM)	Management Email	GRC REST API	SMTP Email	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-NCC-01	Validate all High Impact BES Cyber Assets have documented ESP with firewall rules enforced	ManageESPBoundary	100% of High impact BES assets inside defined ESP with access rules	High
AC-NCC-02	Confirm audit evidence collection covers CIP-005, CIP-007, and CIP-010 with 90-day retention	CollectAuditEvidence	Evidence package contains logs, configs, and access records within retention period	High<

AC ID	Description	Related Info Flow	Expected Outcome	Priority
				/td>
AC-NCC-03	Verify CIP-004 personnel training completion rate at 100% for OT personnel	ReportTrainingRecords	All OT personnel with access to BES assets have current training	High
AC-NCC-04	Ensure audit findings and observations are tracked to closure with documented remediation	TrackRemediation	All audit findings closed or on track with approved remediation plan	Medium

Data, Analytics & AI - UT-ODL UT-GAF UT-CAN UT-AHP

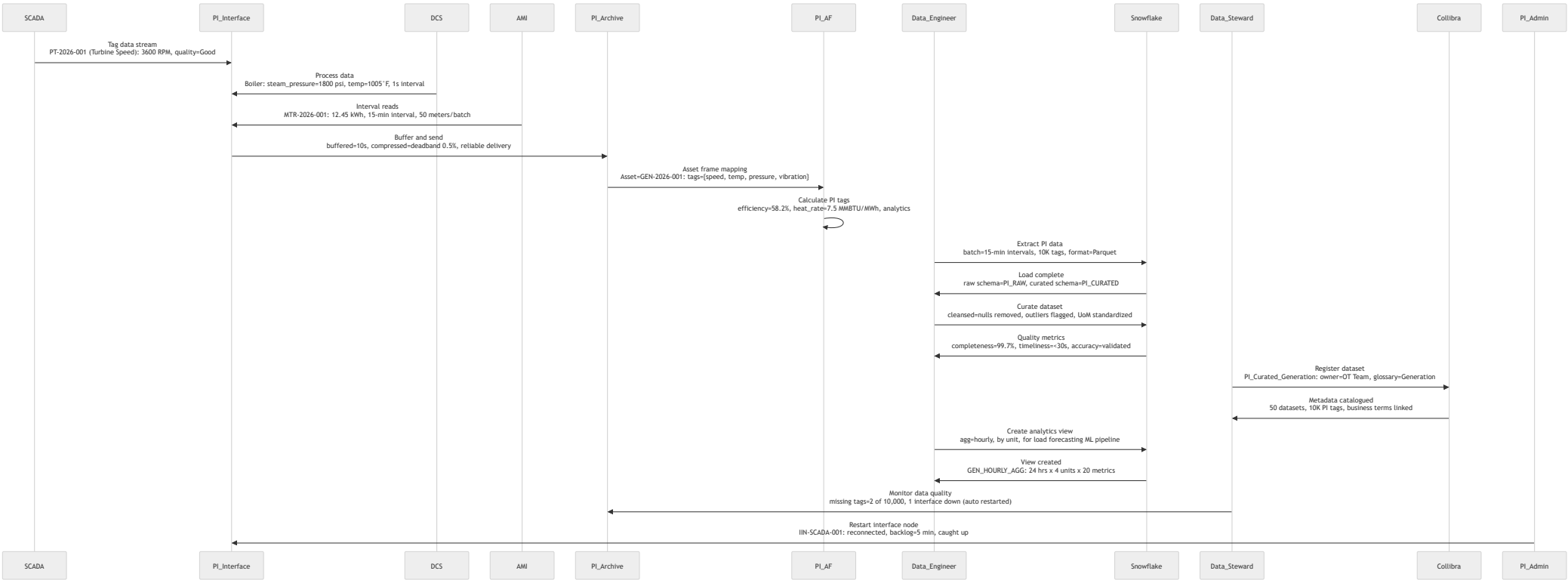
Enterprise data and analytics covering operational data lake (PI System), grid analytics (load forecasting, DER integration), customer analytics (segmentation, EE targeting), and asset health predictive analytics.

15.1 Operational Data Lake (PI System)

Description: Data ingestion from SCADA, DCS, AMI, GIS, and CMMS through PI Interface nodes into PI Asset Framework and PI Data Archive, then to Enterprise Data Lake (Azure/Snowflake) for curation and analytics datasets. Data quality monitoring dashboards and metadata management via data catalog.

Actors: Data Engineer, OT Data Steward, PI Administrator
Systems: OSIsoft PI System (Data Archive / AF), SCADA, DCS (ABB), AMI Head-End, Data Lake (Snowflake), Data Catalog (Collibra)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
DataLake	SCADA	IngestSCADADData	SCADA (Siemens Spectrum Power)	PI Interface Node	SCADA API (ICCP)	PI Interface (OPC-DA)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
DataLake	DCS	IngestDCSProcessData	DCS (ABB 800xA)	PI Interface Node	DCS OPC-UA	PI Interface (OPC-UA)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
DataLake	AF	MapAssetFramework	PI Interface Node	PI Asset Framework	PI API (REST)	PI AF SDK	Batch (Real-time)	Medium	Periodic near-real-time batch with data validation and transformation needs
DataLake	Lake	ExtractToDataLake	PI Data Archive	Data Lake (Snowflake)	PI ODBC/JDBC	Snowflake SQL	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
DataLake	Quality	MonitorDataQuality	Data Lake (Snowflake)	Data Catalog (Collibra)	Snowflake SQL	Collibra REST API	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
DataLake	Catalog	RegisterDataset	Data Lake (Snowflake)	Data Catalog (Collibra)	Snowflake SQL	Collibra REST API	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
DataLake	PI_Admin	ManagePIConnections	PI Administrator	PI Interface Node	PI Admin Console	PI API (REST)	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-ODL-01	Validate PI tag data latency from source to Data Archive under 30 seconds for critical tags	IngestSCADADData	Critical tag timestamps within 30s of source scan time	High
AC-ODL-02	Confirm Snowflake data completeness above 99% for all PI tags in curated layer	ExtractToDataLake	Curated datasets have < 1% missing values after PI extraction	Medium
AC-ODL-03	Verify PI Interface deadband compression preserves data within 0.5% accuracy	IngestDCSProcessData	Compressed data reconstructs within 0.5% of original values	Medium
AC-ODL-04	Ensure data catalog coverage is 100% for all production analytics datasets	RegisterDataset	All curated datasets have business terms and owners in Collibra	Medium

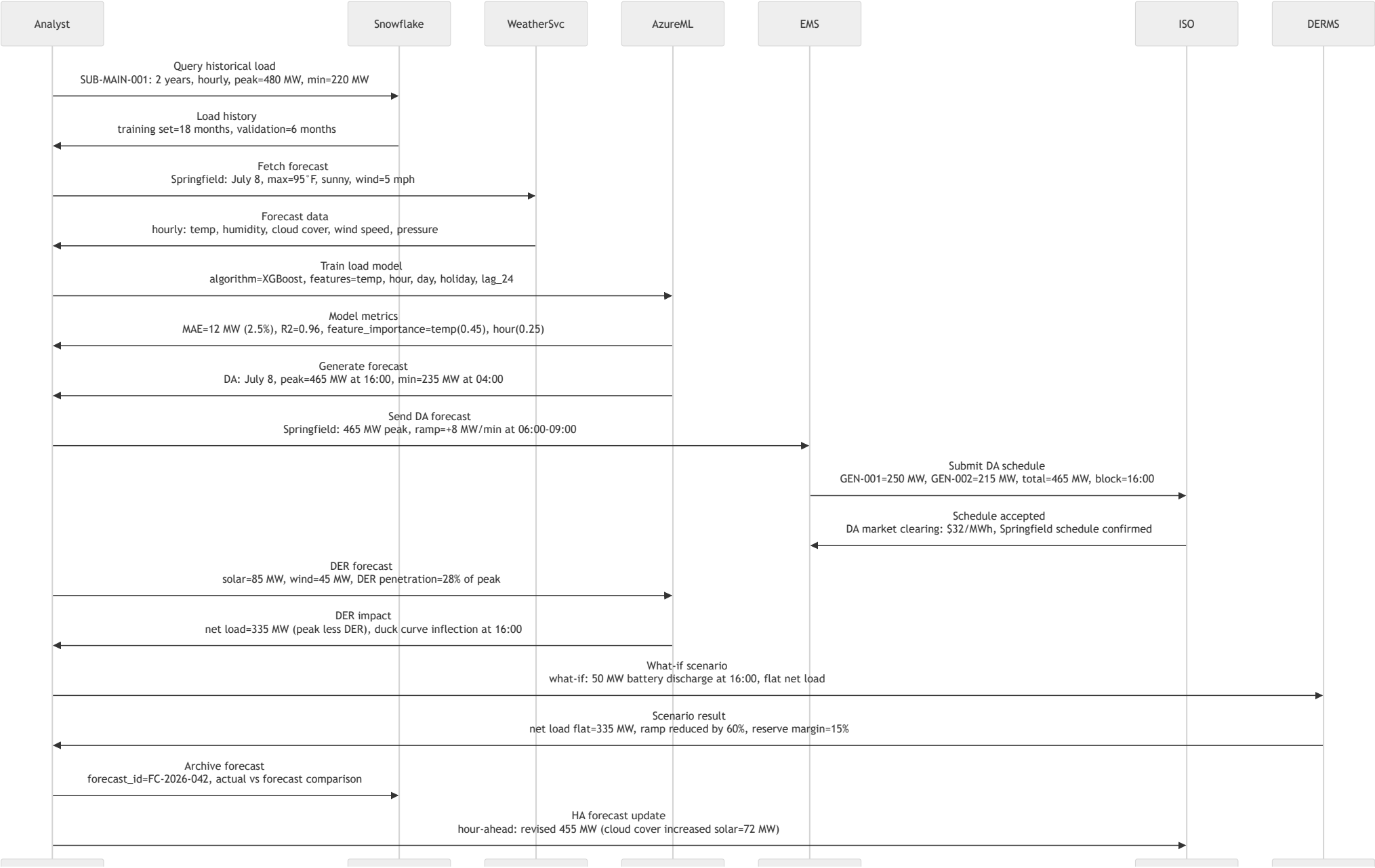
15.2 Grid Analytics & Forecasting

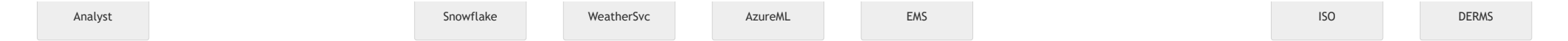
Description: Historical load data combined with weather forecasts to run ML-based load forecaster producing day-ahead and hour-ahead forecasts for ISO submission and grid operator decision support. DER penetration forecasting, renewable integration analysis, and what-if scenarios for capacity planning.

Actors: Grid Analyst, Load Forecaster, ISO Scheduler

Systems: Analytics Platform (SAS/Azure ML), Data Lake (Snowflake), Weather Service, ISO Portal, EMS, DERMS

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
GridAnalytics	Data	QueryHistoricalLoad	Data Lake (Snowflake)	Analytics Platform (Azure ML)	Snowflake SQL	Azure ML SDK (Python)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
GridAnalytics	Weather	FetchWeatherForecast	Weather Service (DTN/The Weather Co)	Analytics Platform (Azure ML)	Weather API (REST)	Azure ML REST API	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
GridAnalytics	Model	TrainLoadForecastModel	Analytics Platform (Azure ML)	Analytics Platform (Azure ML)	Azure ML Internal	Azure ML Internal	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
GridAnalytics	Forecast	SendDAForecast	Analytics Platform (Azure ML)	EMS	Azure ML REST API	EMS API (REST)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
GridAnalytics	ISO	SubmitDASchedule	EMS	ISO Portal (PJM/MISO/CAISO)	EMS API (ICCP)	ISO API (ICCP)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
GridAnalytics	DER	AnalyzeDERPenetration	Analytics Platform (Azure ML)	DERMS	Azure ML REST API	DERMS API (REST)	Batch (Real-time)	Medium	Periodic near-real-time batch with data validation and transformation needs
GridAnalytics	Archive	ArchiveForecastResults	Analytics Platform (Azure ML)	Data Lake (Snowflake)	Azure ML SDK	Snowflake SQL	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

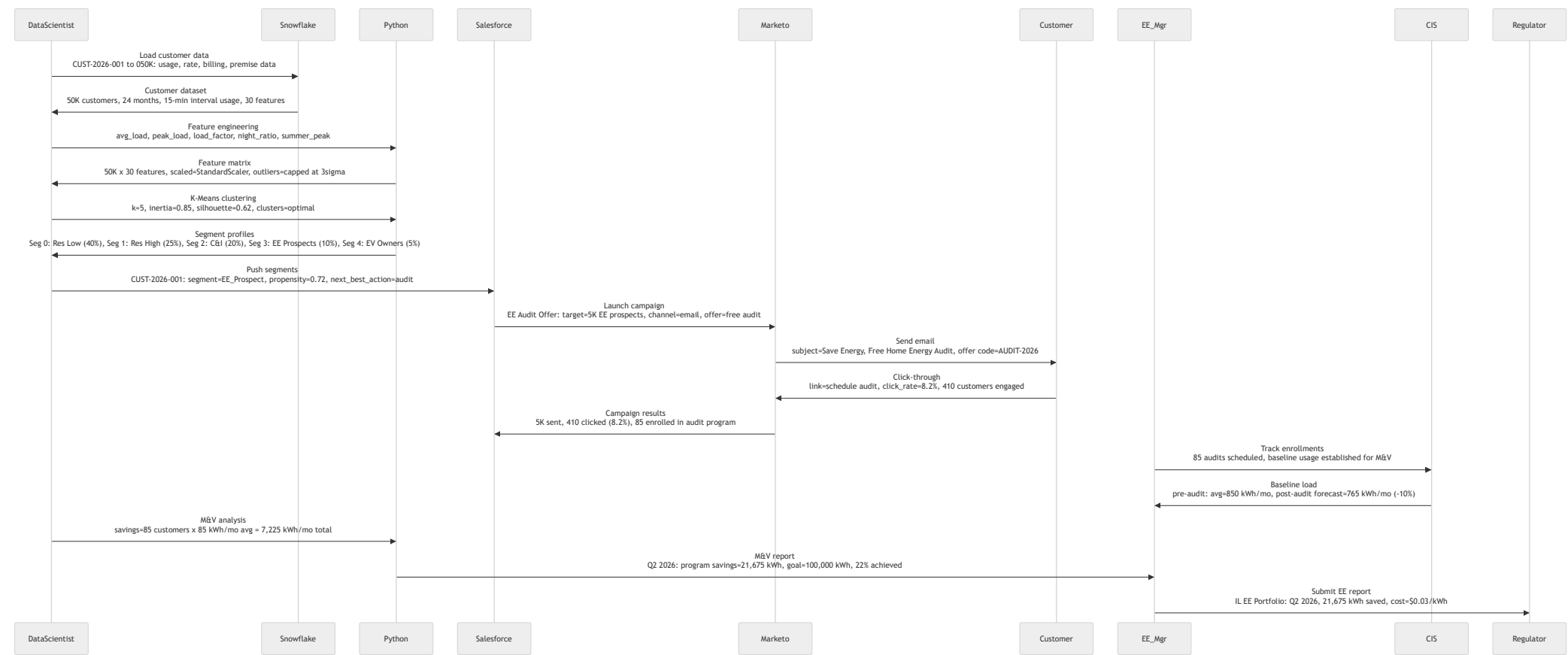
AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-GAF-01	Validate day-ahead load forecast accuracy within ±3% MAPE for system peak	TrainLoadForecastModel	DA forecast MAE <3% of actual peak load	High
AC-GAF-02	Confirm hour-ahead forecast update submitted to ISO at least 30 minutes before delivery hour	SendDAForecast	HA forecast submitted before ISO deadline each hour	High
AC-GAF-03	Verify DER penetration forecast includes solar, wind, and battery scenarios with confidence intervals	AnalyzeDERPenetration	DER forecast includes P50/P90 scenarios and confidence bounds	Medium
AC-GAF-04	Ensure forecast accuracy is tracked and reported monthly with model retraining trigger at drift >5%	ArchiveForecastResults	Model retraining initiated when monthly MAPE exceeds 5% for 2 months	Medium

15.3 Customer Analytics & Segmentation

Description: Customer data analysis using usage patterns for k-means clustering to create segments (residential, commercial & industrial, energy efficiency prospects). Targeted campaign design, program enrollment tracking, energy savings measurement, and M&V (Measurement and Verification) reporting.

Actors: Data Scientist, Marketing Analyst, EE Program Manager
Systems: Analytics Platform (Python/R), Data Lake (Snowflake), CRM (Salesforce), CIS, MDM, Campaign Mgmt (Marketo)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
CustAnalytics	Data	LoadCustomerUsage	Data Lake (Snowflake)	Analytics Platform (Python/R)	Snowflake SQL	Python JDBC	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
CustAnalytics	Cluster	RunKMeansSegmentation	Analytics Platform (Python/R)	Analytics Platform (Python/R)	Internal	Internal	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
CustAnalytics	CRM	PushSegmentsToCRM	Analytics Platform (Python/R)	CRM (Salesforce)	Python REST API	Salesforce REST API	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
CustAnalytics	Campaign	LaunchEECampaign	CRM (Salesforce)	Campaign Mgmt (Marketo)	Salesforce REST API	Marketo REST API	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
CustAnalytics	Enroll	TrackProgramEnrollment	CRM (Salesforce)	CIS	Salesforce REST API	CIS REST API	Event-driven	Simple	Simple event notification with fire-and-forget delivery semantics
CustAnalytics	M&V	CalculateEnergySavings	Analytics Platform (Python/R)	CIS	Python REST API	CIS REST API	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
CustAnalytics	Report	GenerateEEReport	Analytics Platform (Python/R)	Regulatory Portal	Python REST API	Reg Portal (REST/SOAP)	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-CAN-01	Validate segmentation silhouette score above 0.5 indicating well-separated clusters	RunKMeansSegmentation	Silhouette score >=0.5 for customer segmentation model	Medium
AC-CAN-02	Confirm campaign targeting accuracy: at least 70% of EE prospects enroll or engage	LaunchEECampaign	EE prospect engagement rate >=70% across all campaigns	Medium
AC-CAN-03	Verify M&V savings calculation uses approved IPMVP Option C methodology	CalculateEnergySavings	M&V methodology documented and consistent with IPMVP Option C	High
AC-CAN-04	Ensure EE program savings are tracked quarterly and reported to regulator within deadlines	GenerateEEReport	Quarterly EE report submitted within 30 days of quarter end	High

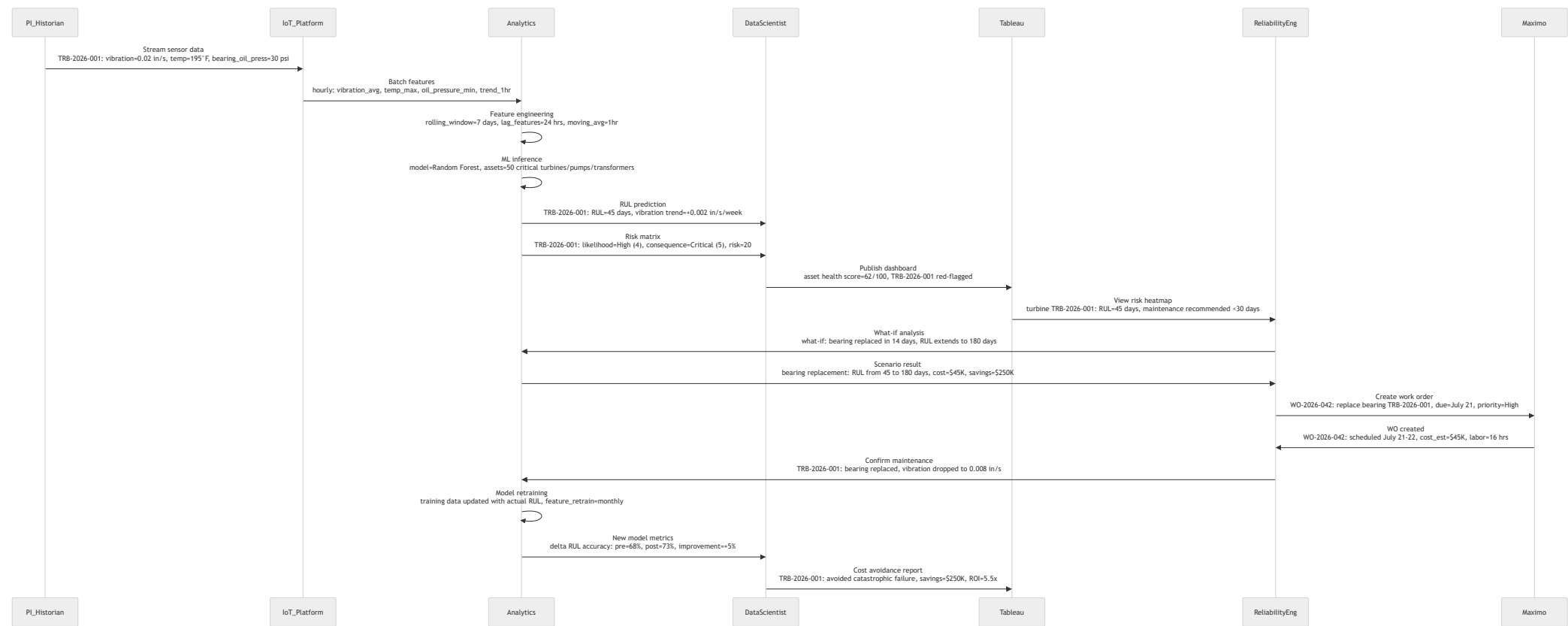
15.4 Asset Health & Predictive Analytics

Description: IoT sensor data collection from critical assets, feature engineering, ML model training (Random Forest/XGBoost) for remaining useful life (RUL) prediction, risk matrix scoring, maintenance recommendation generation, CMMS work order creation, cost avoidance calculation, and model retraining lifecycle.

Actors: Data Scientist, Reliability Engineer, Maintenance Manager

Systems: IoT Platform (C3 AI / AWS IoT), Analytics Platform (Azure ML / DataBricks), EAM (Maximo), SCADA Historian (PI), Visualization (Tableau)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
AssetHealth	Sensors	IngestSensorData	SCADA Historian (PI)	IoT Platform (C3 AI / AWS IoT)	PI API (REST)	IoT Platform API (MQTT)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
AssetHealth	Features	EngineerFeatures	IoT Platform (C3 AI / AWS IoT)	Analytics Platform (Azure ML)	IoT API (REST)	Azure ML SDK	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
AssetHealth	Model	RunPredictiveModel	Analytics Platform (Azure ML / DataBricks)	Analytics Platform (Azure ML / DataBricks)	Internal	Internal	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
AssetHealth	Risk	GenerateRiskMatrix	Analytics Platform (Azure ML / DataBricks)	Visualization (Tableau)	Analytics REST API	Tableau REST API	Batch (Real-time)	Medium	Periodic near-real-time batch with data validation and transformation needs
AssetHealth	WO	CreateMaintenanceWO	Analytics Platform (Azure ML / DataBricks)	EAM (Maximo)	Analytics REST API	Maximo REST API	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
AssetHealth	Retrain	RetrainPredictiveModel	Analytics Platform (Azure ML / DataBricks)	Analytics Platform (Azure ML / DataBricks)	Internal	Internal	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
AssetHealth	Report	ReportCostAvoidance	Visualization (Tableau)	Maintenance Manager Email	Tableau REST API	SMTP Email	Batch (Scheduled)	Simple	Straightforward periodic data transfer with no real-time constraints

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-AHP-01	Validate RUL prediction accuracy within ±15% of actual failure date for critical assets	RunPredictiveModel	RUL prediction within 15% of actual remaining life for training set	High
AC-AHP-02	Confirm risk matrix scoring follows standardized 5x5 likelihood x consequence methodology	GenerateRiskMatrix	All critical assets scored using approved risk matrix method	Medium

AC ID	Description	Related Info Flow	Expected Outcome	Priority
			ology	
AC-AHP-03	Verify predictive model triggers work order creation at RUL threshold >30 days for actionable lead time	CreateMaintenanceWO	WO created when predicted RUL crosses 30-day threshold	High
AC-AHP-04	Ensure model retraining occurs monthly with validation on latest failure events	RetrainPredictiveModel	Model retrained monthly and validated against held-out failure data	Medium

Regulatory, Market & Environmental Compliance - UT-RRP UT-ISO UT-ENV UT-EHS

Manages regulatory reporting (FERC, NERC, PUC, EPA), ISO/RTO wholesale market operations, environmental compliance (emissions, water discharge, waste), and EHS (safety, health, incident management).

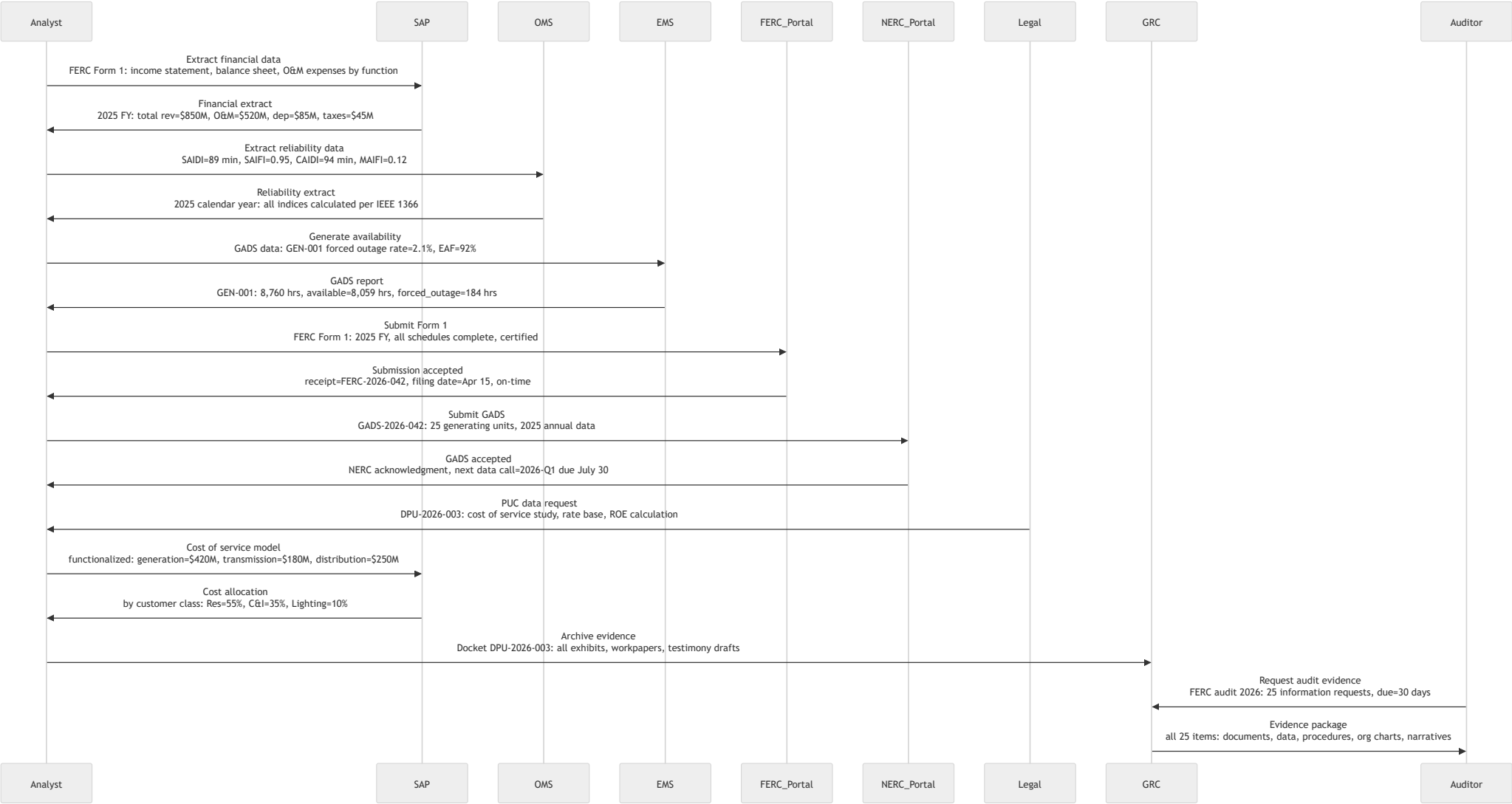
16.1 Regulatory Reporting (FERC/NERC/PUC)

Description: Data collection from financial, operational, and reliability systems, validation, FERC Form 1 preparation, NERC GADS reporting for generator availability, PUC rate case support with cost-of-service studies, audit evidence compilation, submission via regulatory portals, and responses to data requests.

Actors: Regulatory Analyst, Compliance Manager, Legal Counsel, Auditor

Systems: ERP (SAP), OMS, EMS Historian, GRC (Archer), FERC eForms Portal, NERC Compliance Portal

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
RegReporting	Financial	ExtractFERCForm1Data	ERP (SAP)	Regulatory Analyst	SAP JDBC	Analyst Excel/SQL	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
RegReporting	Reliability	ExtractReliabilityData	OMS	Regulatory Analyst	OMS REST API	Analyst REST API	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
RegReporting	GADS	GenerateGADSReport	EMS Historian	NERC Compliance Portal	EMS SQL JDBC	NERC Portal Web	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
RegReporting	FERC	SubmitFERCForm1	Regulatory Analyst	FERC eForms Portal	Analyst Portal UI	FERC eForms API (SOAP)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
RegReporting	PUC	RespondToDataRequest	Regulatory Analyst	GRC (Archer)	Analyst REST API	GRC REST API	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
RegReporting	Evidence	ArchiveAuditEvidence	Regulatory Analyst	GRC (Archer)	Analyst REST API	GRC REST API	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
RegReporting	Audit	SubmitAuditEvidence	GRC (Archer)	Auditor Portal	GRC REST API	Auditor Portal (REST)	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-RRP-01	Validate FERC Form 1 data consistency across all schedules before submission	ExtractFERCForm1Data	Cross-schedule totals balance with zero variance flagged	High
AC-RRP-02	Confirm GADS data submission meets NERC deadline (60 days after quarter end)	GenerateGADSReport	GADS submitted within NERC reporting window	High
AC-RRP-03	Verify PUC data request responses are reviewed by legal counsel before submission	RespondToDataRequest	All PUC responses have legal review stamp in GRC	High
AC-RRP-04	Ensure audit evidence responses are submitted within regulatory deadline	SubmitAuditEvidence	100% of audit data requests responded within 30-day window	High

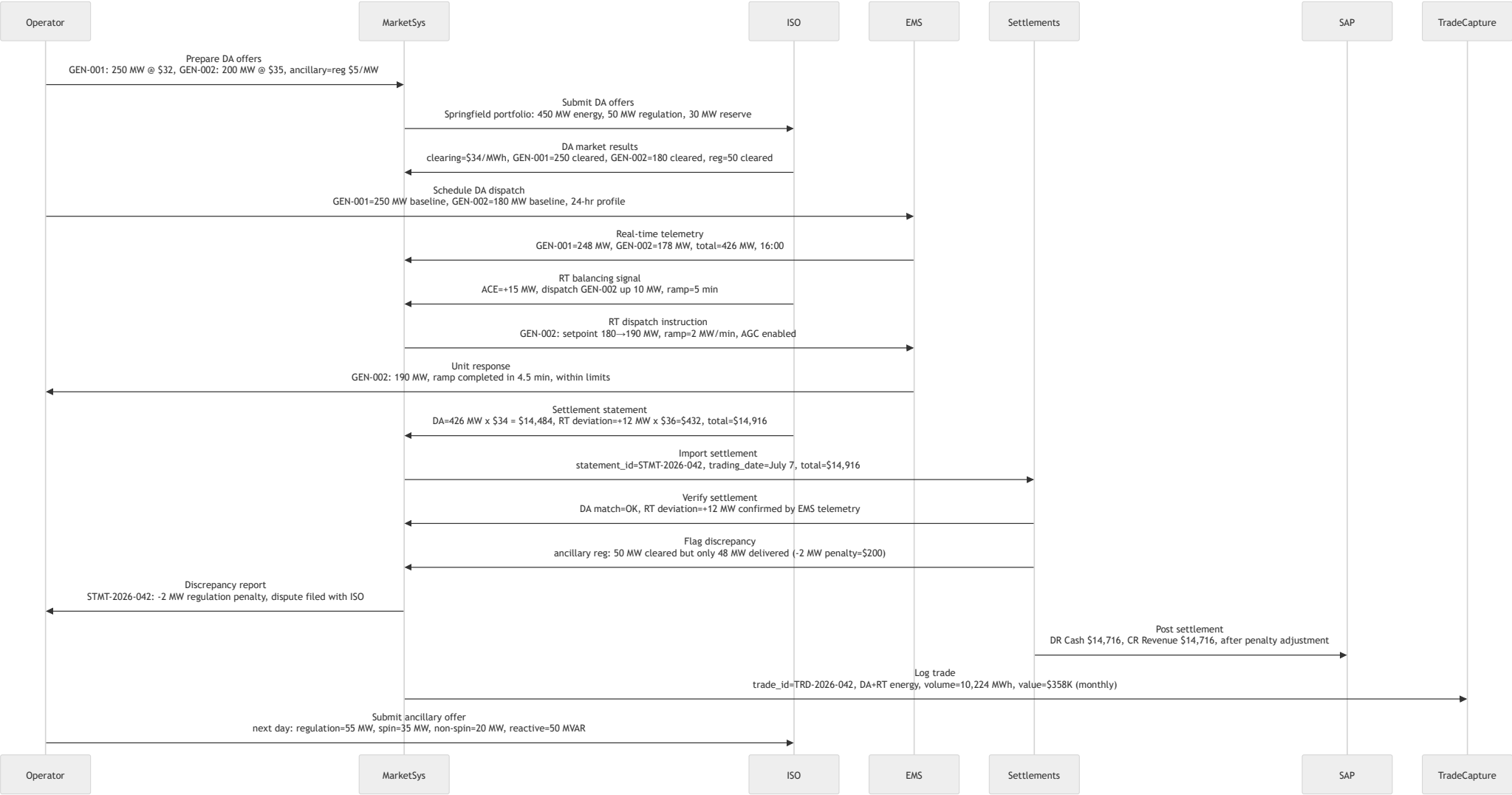
16.2 ISO/RTO Market Operations

Description: Generation offers submitted to ISO, market clearing process, dispatch schedule receipt, real-time balancing and unit commitment, settlement statements reconciliation, discrepancy resolution, and invoicing. Day-ahead and real-time market participation including ancillary services offers (regulation, reserves, reactive).

Actors: Market Operator, Settlements Analyst, Generation Scheduler

Systems: Market System (ABB Promod), EMS, ISO Portal (PJM/MISO/CAISO), ERP (SAP), Trade Capture System

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Market	Offer	SubmitDAOffer	Market System (ABB Promod)	ISO Portal (PJM/MISO/CAISO)	Market Sys API (ICCP)	ISO API (ICCP)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Market	Clear	ReceiveDAClearing	ISO Portal (PJM/MISO/CAISO)	Market System (ABB Promod)	ISO API (ICCP)	Market Sys API (ICCP)	Event-driven	High	Mission-critical event processing requiring guaranteed delivery, sequencing, and audit trail
Market	Dispatch	SendDispatchInstruction	ISO Portal (PJM/MISO/CAISO)	EMS	ISO API (ICCP)	EMS API (ICCP)	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Market	Settlement	ImportSettlementStatement	ISO Portal (PJM/MISO/CAISO)	Market System (ABB Promod)	ISO API (REST)	Market Sys API (REST)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Market	Reconcile	ReconcileSettlement	Market System (ABB Promod)	ERP (SAP)	Market Sys JDBC	SAP JDBC	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
Market	Trade	LogTradeCapture	Market System (ABB Promod)	Trade Capture System	Market Sys API (REST)	Trade Capture API (REST)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Market	Invoice	PostSettlementInvoice	Market System (ABB Promod)	ERP (SAP)	Market Sys REST API	SAP REST API	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-ISO-01	Validate DA offer submission before ISO deadline (10:00 AM for next day)	SubmitDAOffer	All DA offers submitted before ISO deadline with confirmation receipt	High
AC-ISO-02	Confirm settlement statement reconciliation identifies discrepancies > \$500 within 48 hours	ReconcileSettlement	All material discrepancies flagged and disputed within 48 hrs	High
AC-ISO-03	Verify RT dispatch instructions are executed within ramp time limits per generator capability	SendDispatchInstruction	Generator response to RT dispatch within ramp rate parameters	High
AC-ISO-04	Ensure ancillary services offers are optimized to avoid penalty charges for non-delivery	SubmitDAOffer	Ancillary delivery accuracy > =98% to avoid ISO penalties	Medium

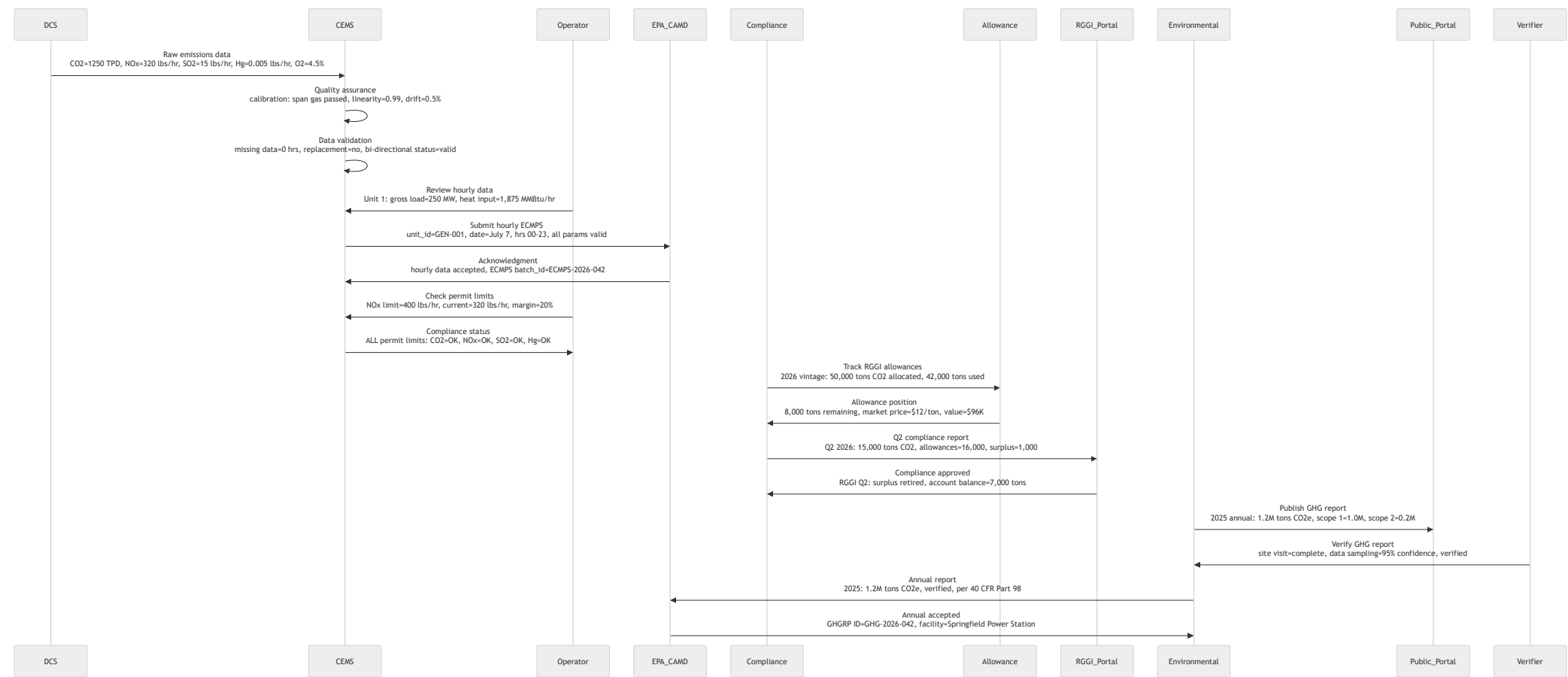
16.3 Environmental Emissions Reporting

Description: Continuous Emissions Monitoring System (CEMS) data quality assurance, EPA CAMD submission for CO2, NOx, SO2, and Hg via ECMPS, permit compliance tracking, allowance management (RGGI/CARB), public GHG report preparation, and third-party verification.

Actors: Environmental Engineer, Compliance Manager, EPA Reviewer

Systems: CEMS Data System, EPA CAMD Portal, DCS (ABB), Emissions Management, Allowance Tracking, Public Portal

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
Emissions	CEMS	SendHourlyEmissions	DCS (ABB)	CEMS Data System	DCS OPC-UA	CEMS Modbus TCP	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Emissions	QA	PerformQualityAssurance	CEMS Data System	CEMS Data System	Internal	Internal	API-led (Real-time)	High	OT/IT boundary crossing with sub-second latency and strict data fidelity requirements
Emissions	EPA	SubmitECMPSCAMD	CEMS Data System	EPA CAMD Portal	CEMS API (REST)	EPA CAMD Web Services (SOAP)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Emissions	Permit	VerifyPermitCompliance	CEMS Data System	Emissions Management	CEMS REST API	Emissions Mgmt REST API	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
Emissions	Allowance	TrackRGGIAllowances	Allowance Tracking System	RGGI/CARB Portal	Allowance API (REST)	RGGI Portal API (SOAP)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling
Emissions	GHG	SubmitGHGReport	Emissions Management	EPA CAMD Portal	Emissions REST API	EPA CAMD Web Services	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
Emissions	Verification	CompleteThirdPartyVerification	Third-Party Verifier	Emissions Management	Verifier Portal API	Emissions API (REST)	Batch (Scheduled)	Medium	Scheduled batch transfer requiring field mapping, validation, and exception handling

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-ENV-01	Validate CEMS data completeness >95% with missing data substitution per 40 CFR Part 75	PerformQualityAssurance	No data gaps >2 hrs without valid substitute data	High
AC-ENV-02	Confirm EPA CAMD hourly submission is within 30-day reporting window	SubmitECMPSCAMD	All hourly data submitted by 30th day of following month	High
AC-ENV-03	Verify emission limits are never exceeded: NOx <400 lbs/hr, SO2 <50 lbs/hr, Hg <0.01 lbs/hr	VerifyPermitCompliance	Zero exceedance events for permitted emission limits	High
AC-ENV-04	Ensure annual GHG report is verified by accredited third party before public submission	SubmitGHGReport	Third-party verification statement included with annual GHG submission	Medium

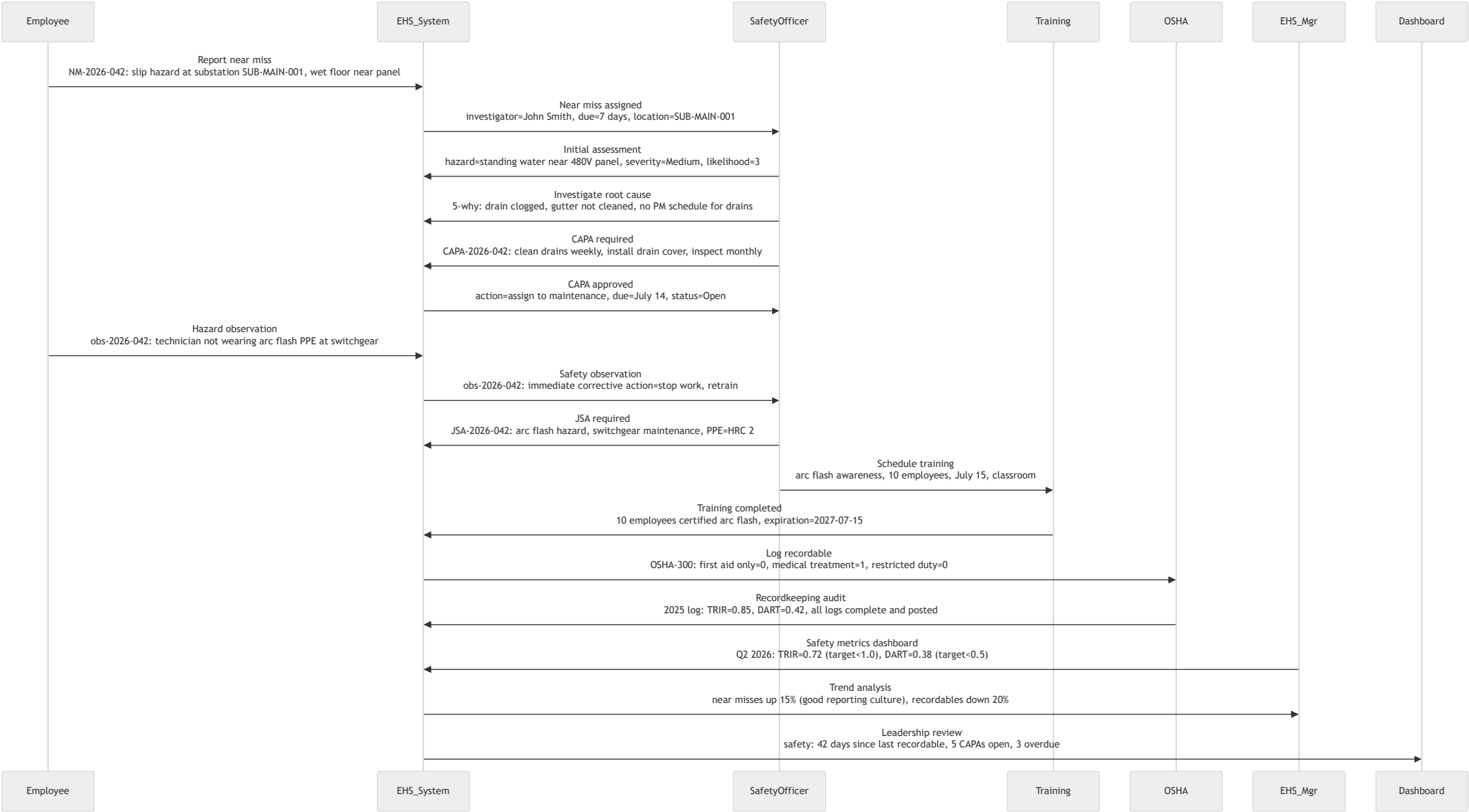
16.4 EHS Management & Safety

Description: Incident reporting, investigation with root cause analysis, corrective action tracking, safety observations, hazard identification and JSA (Job Safety Analysis), training management, safety audits, OSHA recordkeeping (300 log), injury metrics (DART, TRIR), safety meetings, and leadership dashboard review.

Actors: Safety Officer, Employee, EHS Manager, OSHA Representative

Systems: EHS System (Intelx/Enablon), HRIS (SuccessFactors), Training System, OSHA Portal, IoT Safety Sensors

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity	Complexity Reason
EHS	Incident	ReportNearMiss	Employee Mobile App	EHS System (Intelex/Enablon)	Mobile API (OAuth2)	EHS REST API	API-led (Real-time)	Simple	Direct API integration with minimal transformation and single-system lookup
EHS	Investigation	PerformRootCauseAnalysis	EHS System (Intelex/Enablon)	EHS System (Intelex/Enablon)	Internal	Internal	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
EHS	CAPA	TrackCorrectiveAction	EHS System (Intelex/Enablon)	EHS System (Intelex/Enablon)	Internal	Internal	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation
EHS	Training	ScheduleSafetyTraining	EHS System (Intelex/Enablon)	Training System (LMS)	EHS REST API	LMS SCORM	Event-driven	Simple	Simple event notification with fire-and-forget delivery semantics
EHS	OSHA	SubmitOSHA300Log	EHS System (Intelex/Enablon)	OSHA Portal	EHS REST API	OSHA Portal API (SOAP)	Batch (Scheduled)	High	Large-volume scheduled processing with data integrity validation and reconciliation
EHS	Dashboard	GenerateSafetyDashboard	EHS System (Intelex/Enablon)	HRIS (SuccessFactors)	EHS SQL JDBC	SuccessFactors REST API	Batch (Real-time)	Medium	Periodic near-real-time batch with data validation and transformation needs
EHS	IoT	MonitorIoTSafetySensors	IoT Safety Sensors (gas/arc flash)	EHS System (Intelex/Enablon)	Sensor MQTT	EHS API (MQTT)	API-led (Real-time)	Medium	API integration requiring moderate transformation, error handling, and data reconciliation

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-EHS-01	Validate all safety incidents (near miss, first aid, recordable) are reported within 24 hours	ReportNearMiss	100% of incidents reported in EHS system within 24 hrs of occurrence	High
AC-EHS-02	Confirm root cause investigation completed within 7 days for all recordable incidents	PerformRootCauseAnalysis	RCA report approved in EHS system within 7-day target	High
AC-EHS-03	Verify CAPA closure rate above 90% within original due date	TrackCorrectiveAction	At least 90% of CAPAs closed on or before original due date	Medium
AC-EHS-04	Ensure OSHA 300 log is posted annually by February 1 and retained for 5 years	SubmitOSHA300Log	OSHA 300 log posted in workplace and submitted per regulatory requirement	High